



Get Started with Databricks for Data Warehousing

Databricks Academy
August 2025

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Agenda

Databricks Overview	Time	Lecture	Demo	Lab
Databricks Data Intelligence Platform	8 mins	✓		
Databricks Workspace Walkthrough	15 mins		✓	



Agenda

Using Databricks for Data Warehousing	Time	Lecture	Demo	Lab
Introduction to Data Warehousing with Databricks	10 mins	✓		
Databricks SQL Warehouses	5 mins	✓		
Delta Lake Overview	8 mins	✓		
Using Delta Lake features with Databricks SQL	5 mins		✓	



Agenda

Data Ingestion and Transformation	Time	Lecture	Demo	Lab
Delta Lake UniForm	5 mins	✓		
Data Ingestion Techniques Overview	5 mins	✓		
Data Ingestion Techniques	10 mins		✓	
Data Transformation	8 mins	✓		
Exploring Data Transformation in Databricks	10 mins		✓	



Agenda

Data Orchestration and Querying Capabilities	Time	Lecture	Demo	Lab
Orchestration in Databricks	5 mins	✓		
Setting Up and Managing Serverless Lakeflow Jobs	15 mins		✓	
Databricks Querying Capabilities	5 mins	✓		



Agenda

Data Presentation	Time	Lecture	Demo	Lab
Introduction to AI/BI	5 mins	✓		
Creating a Dashboard in Databricks	5 mins		✓	
Creating Genie Spaces	5 mins		✓	
Data Warehousing Comprehensive Lab	30 mins			✓



Lab Exercise Environment



Technical Details

- Your lab environment is provided by Vocareum.
- It will open in a new tab.
- It has been configured with the permissions and resources required to accomplish the tasks outlined in the lab exercise.
- Third party cookies must be enabled in your browser for Vocareum's user experience to work properly.
- Make sure to enable pop ups!



Course Learning Objectives

- Describe the available compute options for workloads performed on the Databricks Data Intelligence Platform.
- List the products and features Databricks offers for different data-centric needs within the Databricks Platform.
- Navigate the Databricks Workspace UI.
- Use Databricks to complete common data warehousing tasks.
- Explain the purpose of Delta Lake for data warehousing purposes.
- Describe and apply various techniques for ingesting and transforming data in Databricks.
- Explain how Databricks support data orchestration needs within the platform.
- Describe the impact of data warehousing work on Databricks AI/BI needs.
- Identify the AI supported features in the Databricks platform.





Databricks Overview

Get Started with Databricks for Data Warehousing



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Learning Objectives

- List the four key product lines available from Databricks.
- Describe how each product line serves its respective audience.
- Navigate the Databricks Workspace UI to locate key features and functionalities.



Agenda

Databricks Overview	Time	Lecture	Demo	Lab
Databricks Data Intelligence Platform	8 mins	✓		
Databricks Workspace Walkthrough	15 mins		✓	





Databricks Overview

LECTURE

Databricks Data Intelligence Platform



Lakeflow
Ingest, ETL,
Streaming



Databricks
SQL
Data
Warehousing



AI/BI
Business
Intelligence



Mosaic AI
Artificial
Intelligence



Knowledge of your data



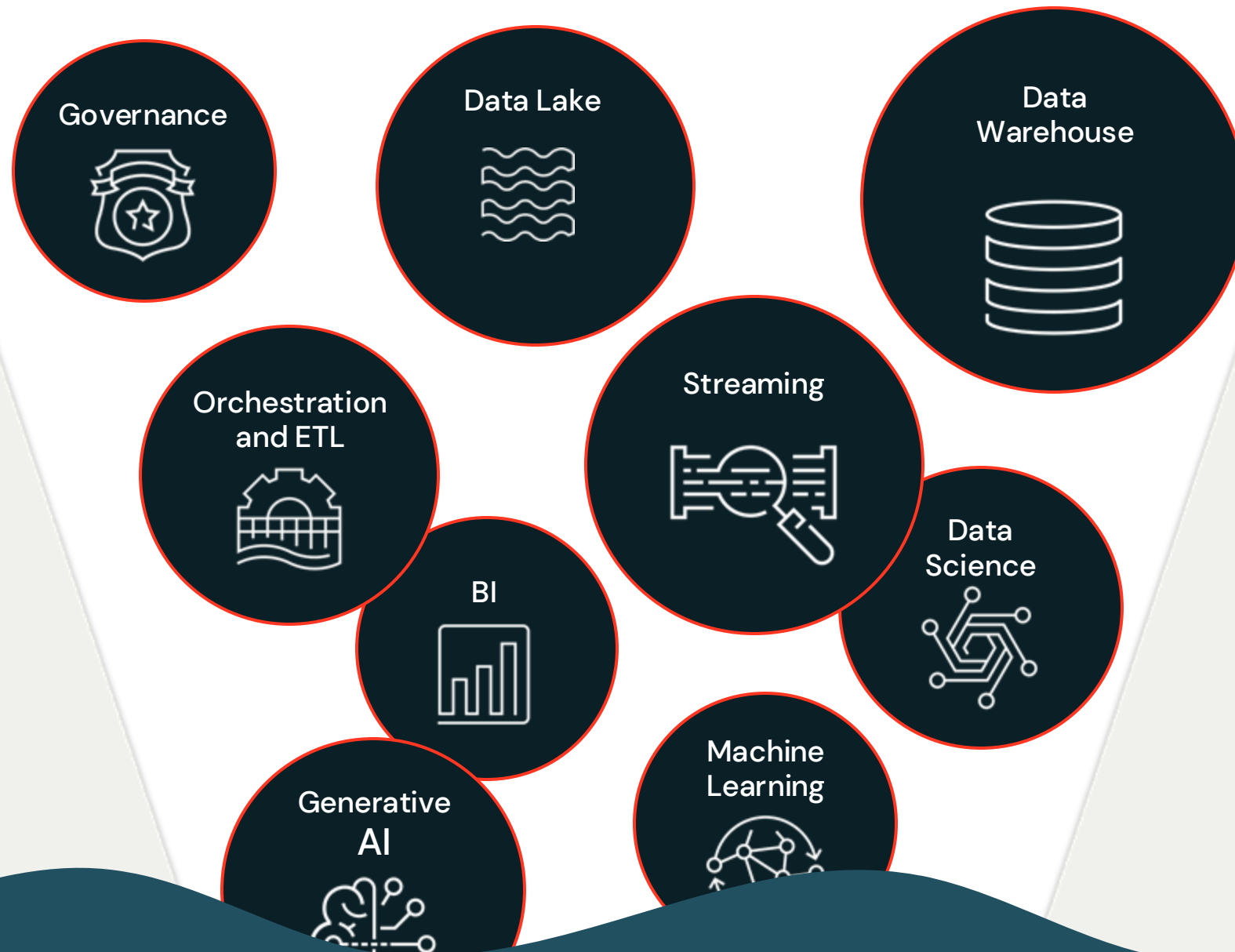
Unity Catalog

**Your data stored in an open, broadly
accessible lakehouse format**

Databricks
simplifies the complexity in your data
estate



Unify and own your data



Adopt an open source Lakehouse format of choice

Break free from lock-in with open formats

Reduce unnecessary costs from multiple copies of data



Iceberg



Delta Lake

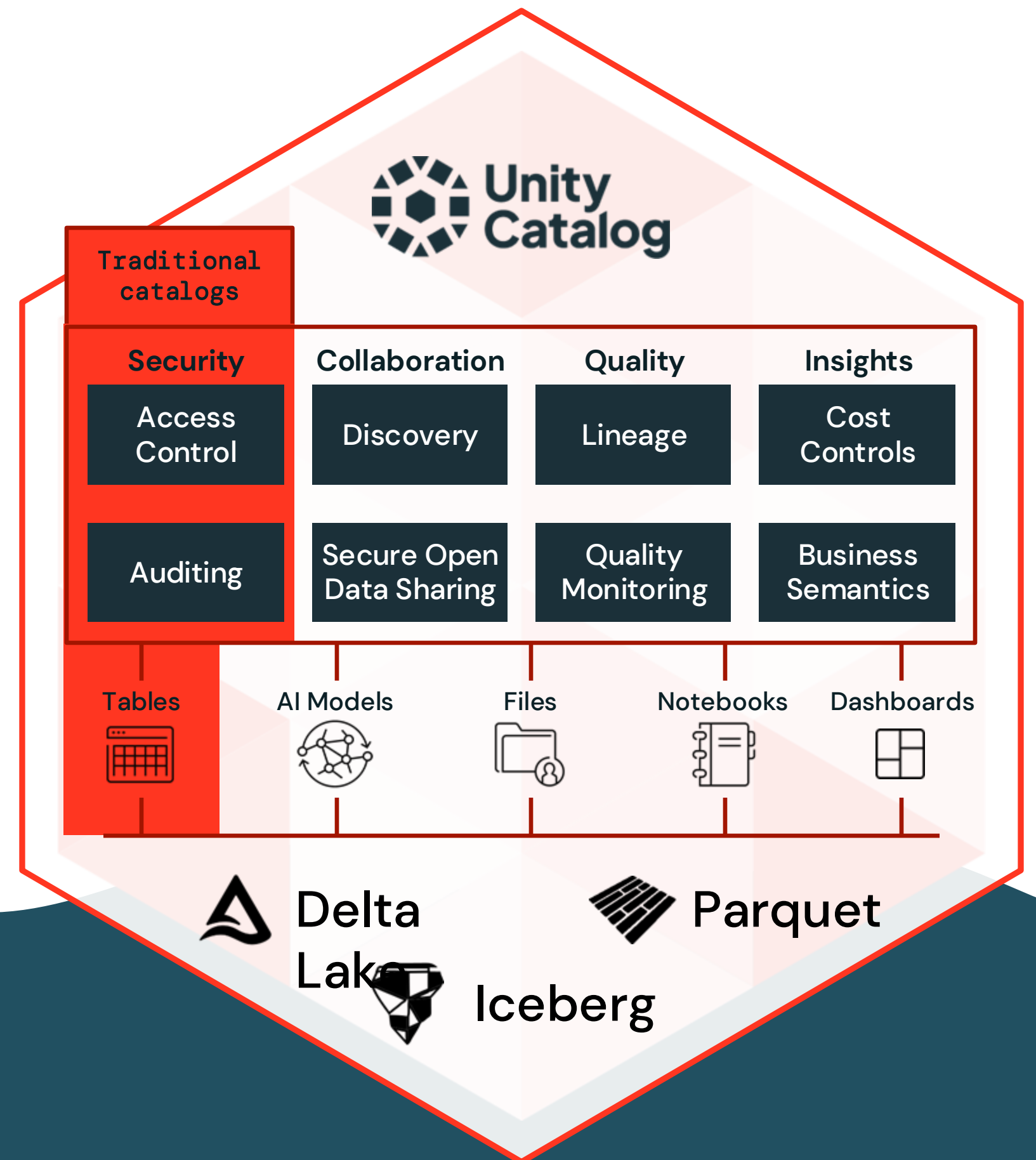


Parquet

Protect and understand your data

Unified capabilities
for every use case

Unified governance
for all assets





Data Intelligence Platform

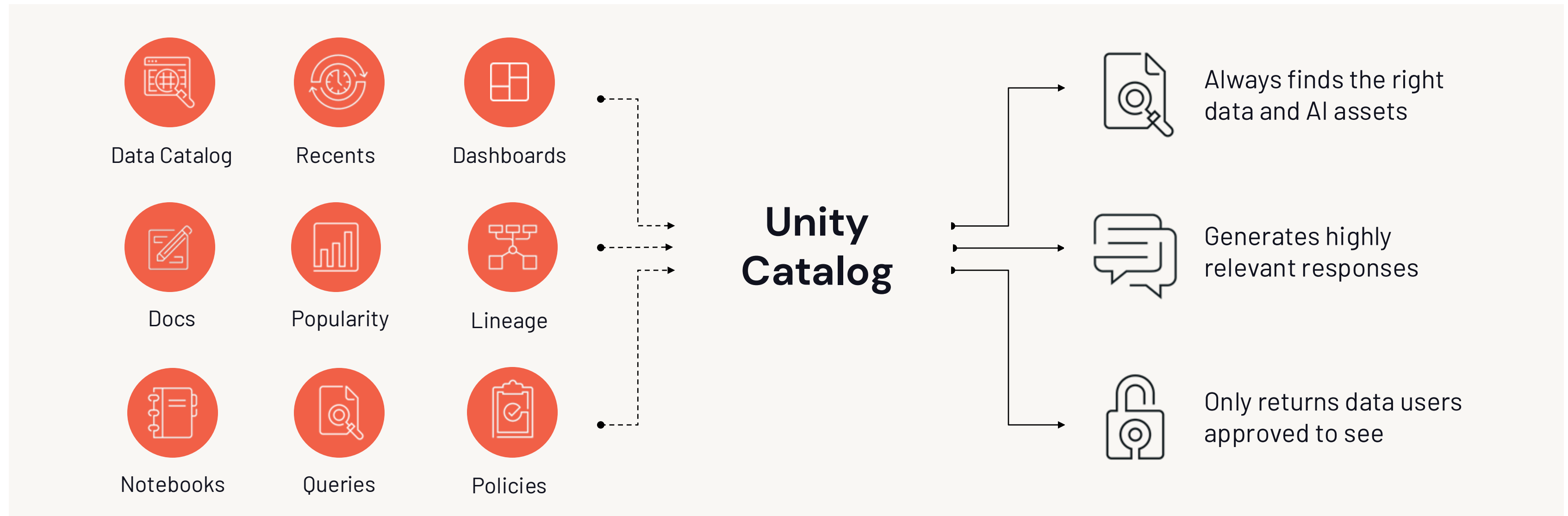
**AI tuned to your
business**



**Unified data
and governance**



Data Intelligence delivers the unique context of your business



All of the signals flowing into Unity Catalog allow AI to learn

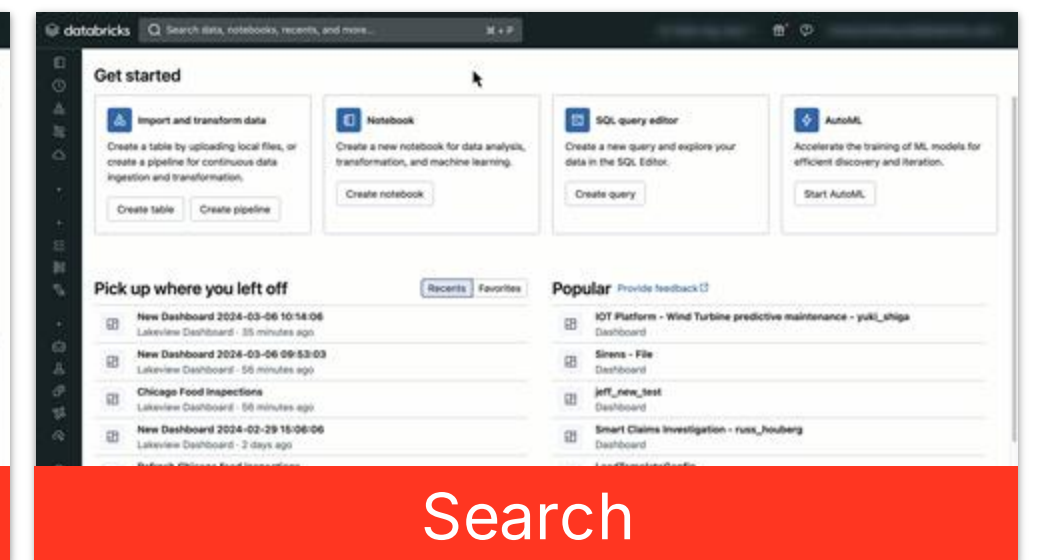
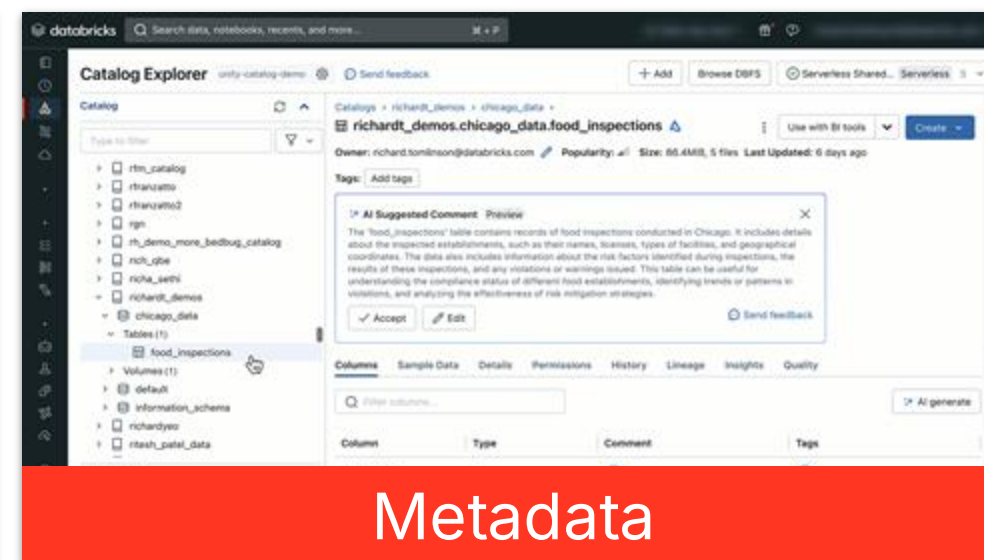
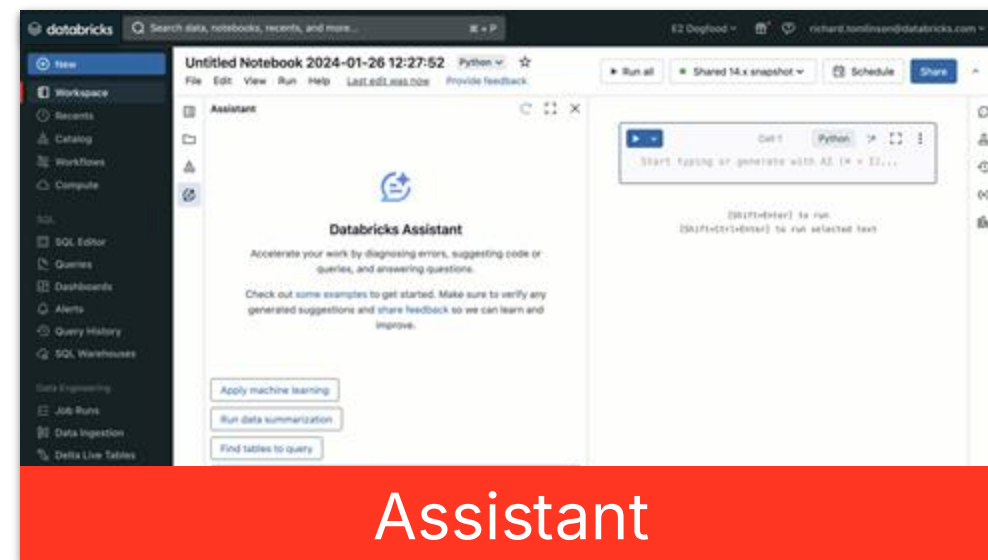
Databricks gains awareness of your unique data and business context



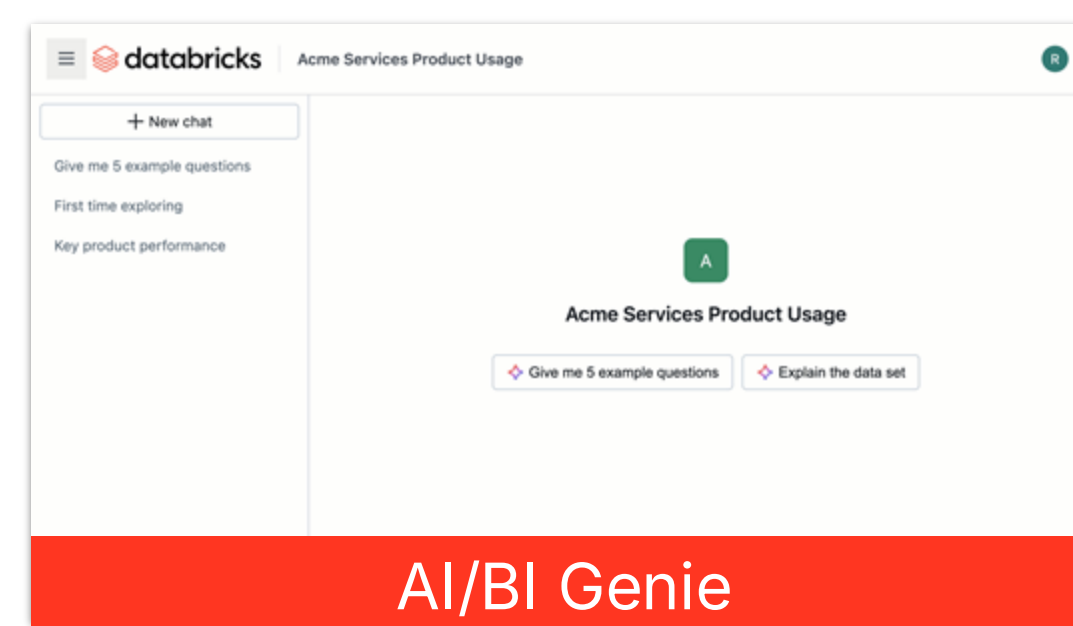
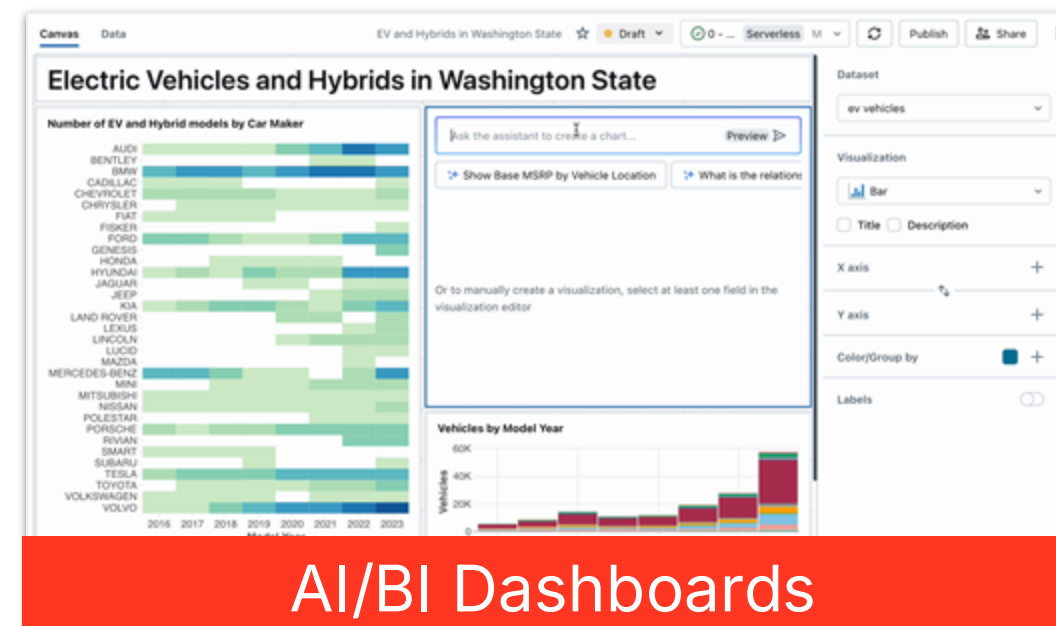
Boosts productivity for all users

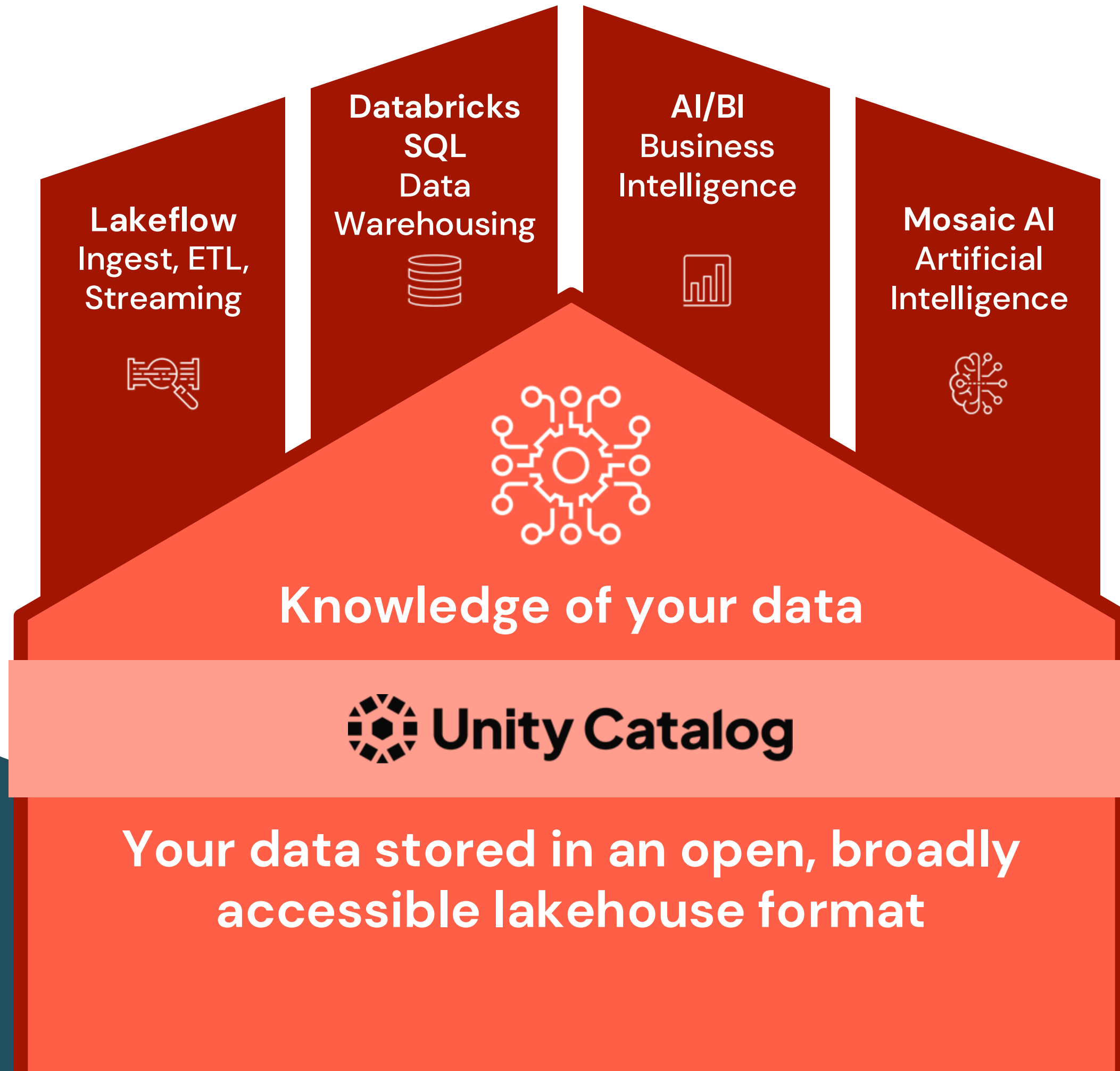
Data Scientists | Data Engineers | SQL Analysts | Business Users

For
Data
Teams



For
Business
Teams





Achieving **data intelligence**
begins with getting data in



Lakeflow

Reliable & automated
dataflow from
systems-of-record

Lakeflow
Ingest, ETL,
Streaming



Databricks
SQL
Data
Warehousing



AI/BI
Business
Intelligence



Mosaic AI
Artificial
Intelligence



Knowledge of your data

 **Unity Catalog**

**Your data stored in an open, broadly
accessible lakehouse format**

Get the right data without the hassle



Connect



**Easily connect key
data sources**

ORACLE
DATABASE



PostgreSQL

MySQL



workday

Google
Analytics

servicenow



ORACLE
NETSUITE

SharePoint

Dynamics 365

DLT



**Reliable data
pipelines made easy**

Jobs



**Unified orchestration
for analytics and AI**

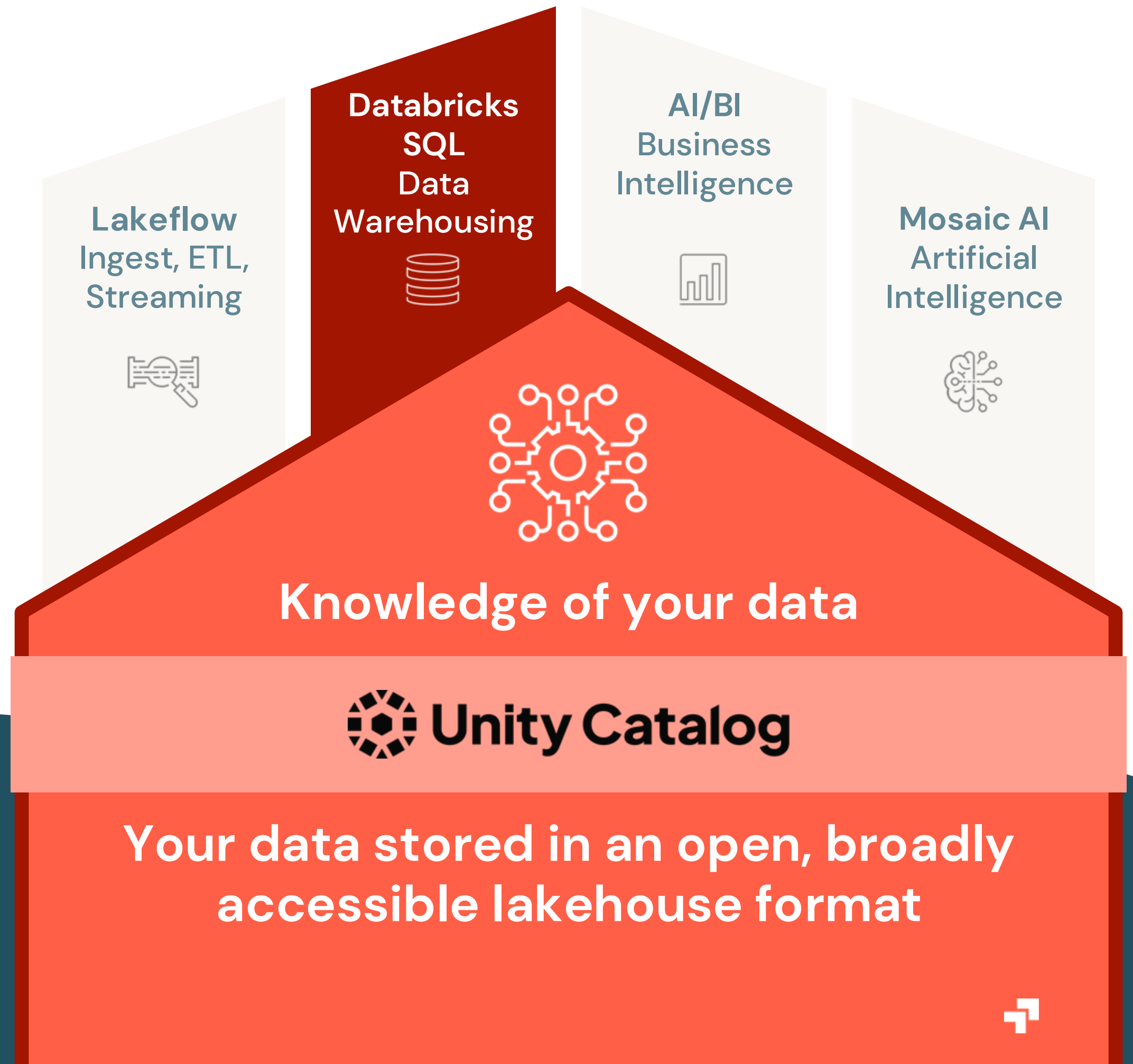


What is **data intelligence** for analytics and BI?



Databricks SQL

The intelligent enterprise
data warehouse



Complete data warehousing capabilities

Lower migration cost

Serverless Warehouses	Views/Temp Views	ANSI SQL	Data Quality Monitoring	Monitor Permissions
1P/3P Connectors	SQL Scripting	SQL UDFs	Table ACLs	Billing System Tables
CDC Support	H3 GeoSpatial	Session Vars	Lakehouse Federation	Internet On/Off
Streaming Tables	Spatial SQL (ST_)	SQL Alerts	HMS Federation	Queries System Table
Materialized Views	Foreign/Primary Keys	Dashboards	Table Lineage	Warehouses Sys Table
Autoloader for Ingest	Lateral Col Alias	Dashboard Sharing	Entity Relation Diagram	WH Events Sys Table
Scheduled Workflows	Named Arguments	Publish to PowerBI	Row/Col Security	Query Duration Limits
1p/3p Orchestration	HyperLogLog	Publish to Tableau	OAuth	Foreign/Primary Keys
SQL Editor	Array Functions	Cloud Fetch	ABAC	Query History
Python/Go Connectors	Identifier Clause	Python UDFs	Marketplace	Query Profiling
SQL Rest API	Variant Data Type	Row Level Concurrency	100K+ User Support	Warehouse Monitoring

Foundational Functionality

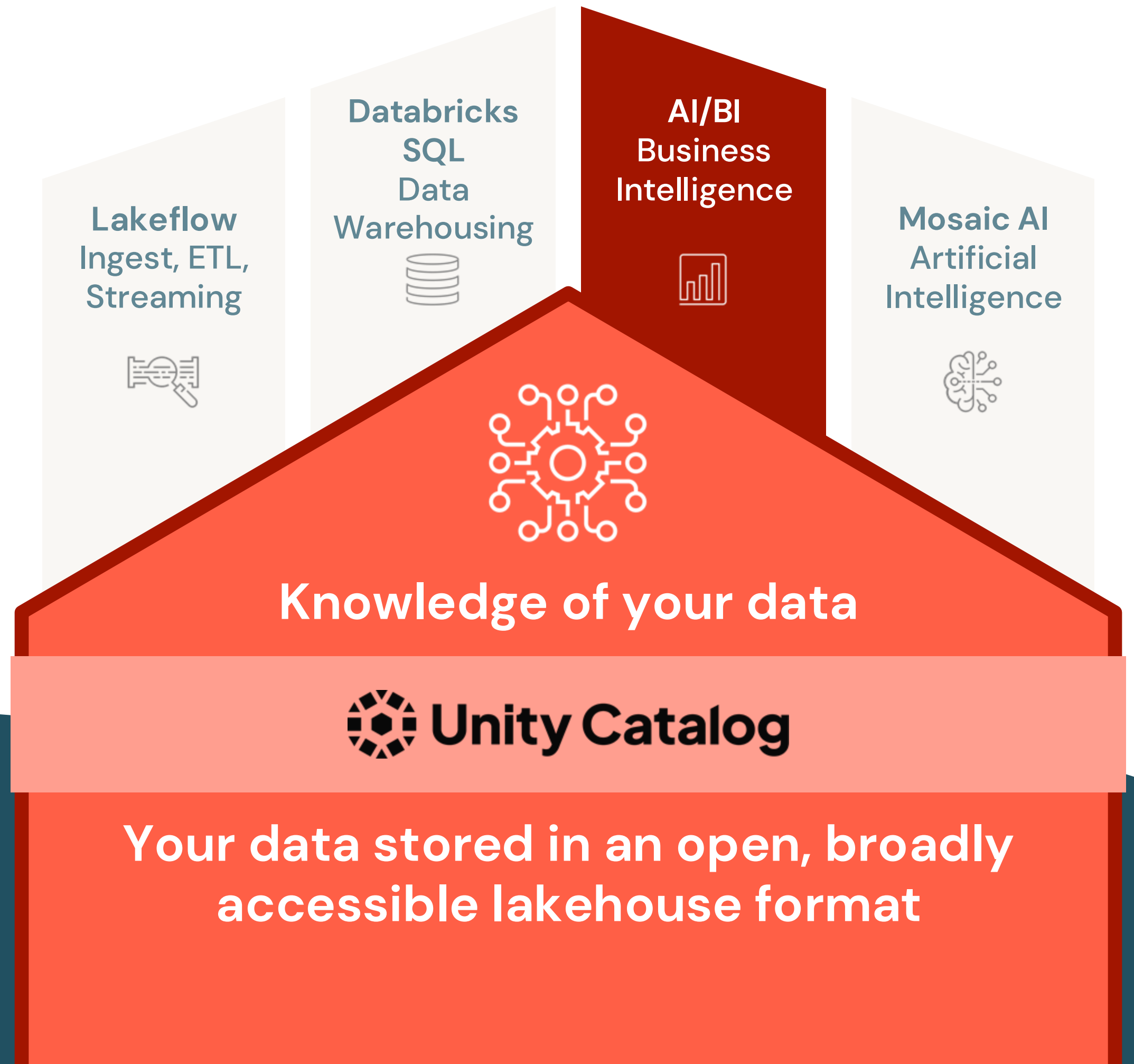
Governance and Administration





AI/BI

Intelligent analytics
for real-world data



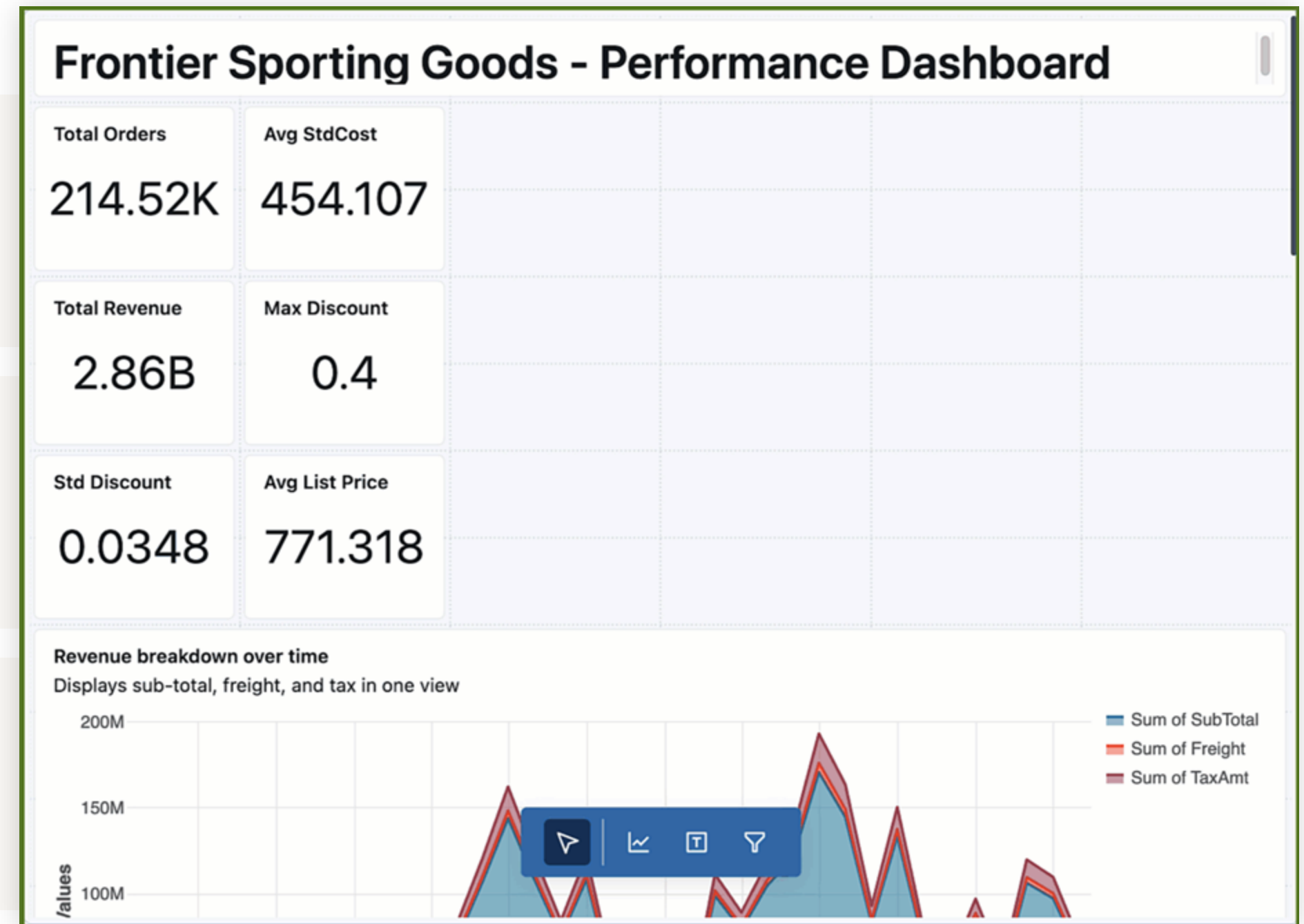
Dashboards to analyze and visualize data

Get useful results with AI-powered understanding

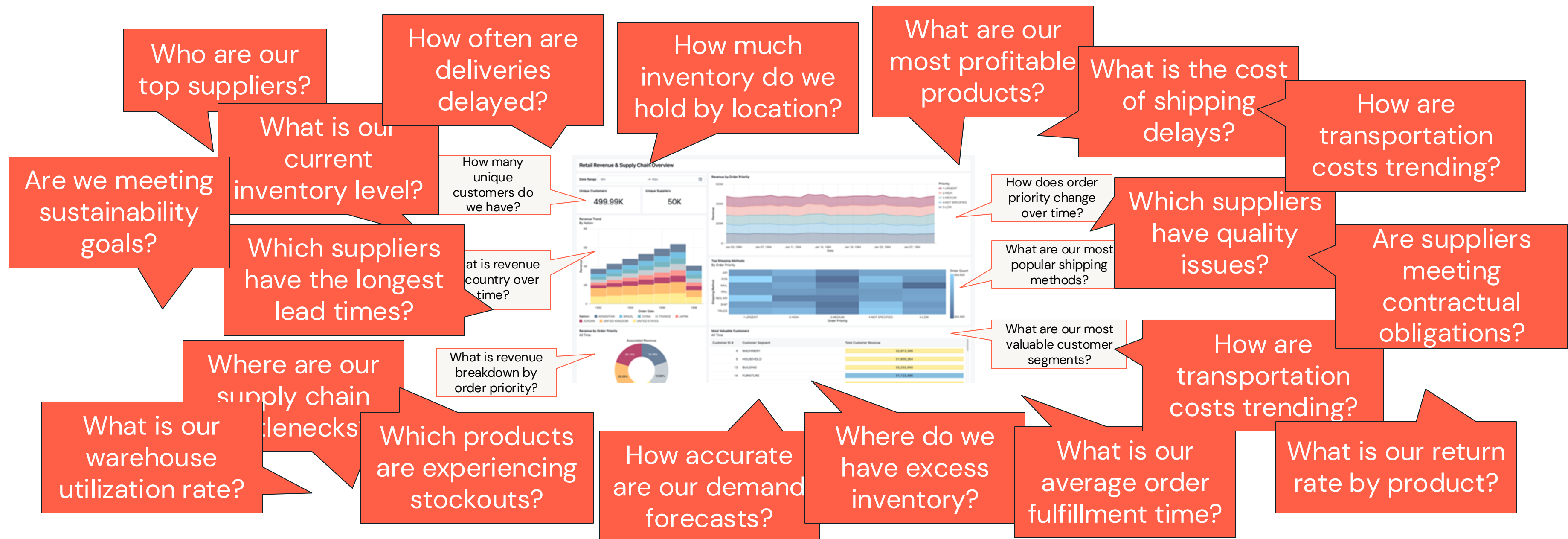
Scale BI to everyone with no additional cost

Blazing fast results

Secure by design with Unity

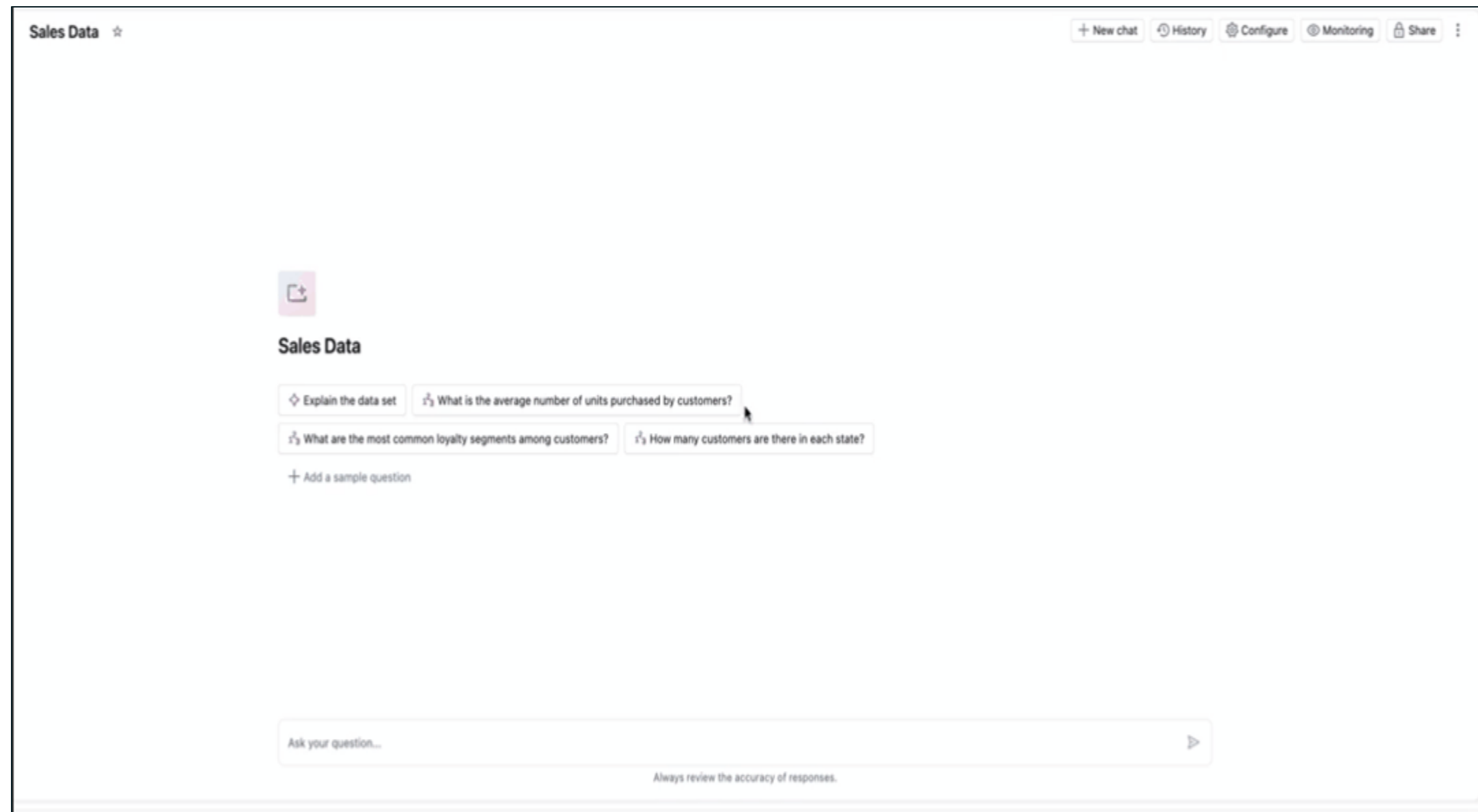


But, you can't anticipate every question users will ask



Genie lets you go beyond dashboards

Talk to your data to explore any question



Query your enterprise data directly with open Q&A

Maintain context between sessions to follow-up on research

Get responses in natural language, data tables, and visualizations

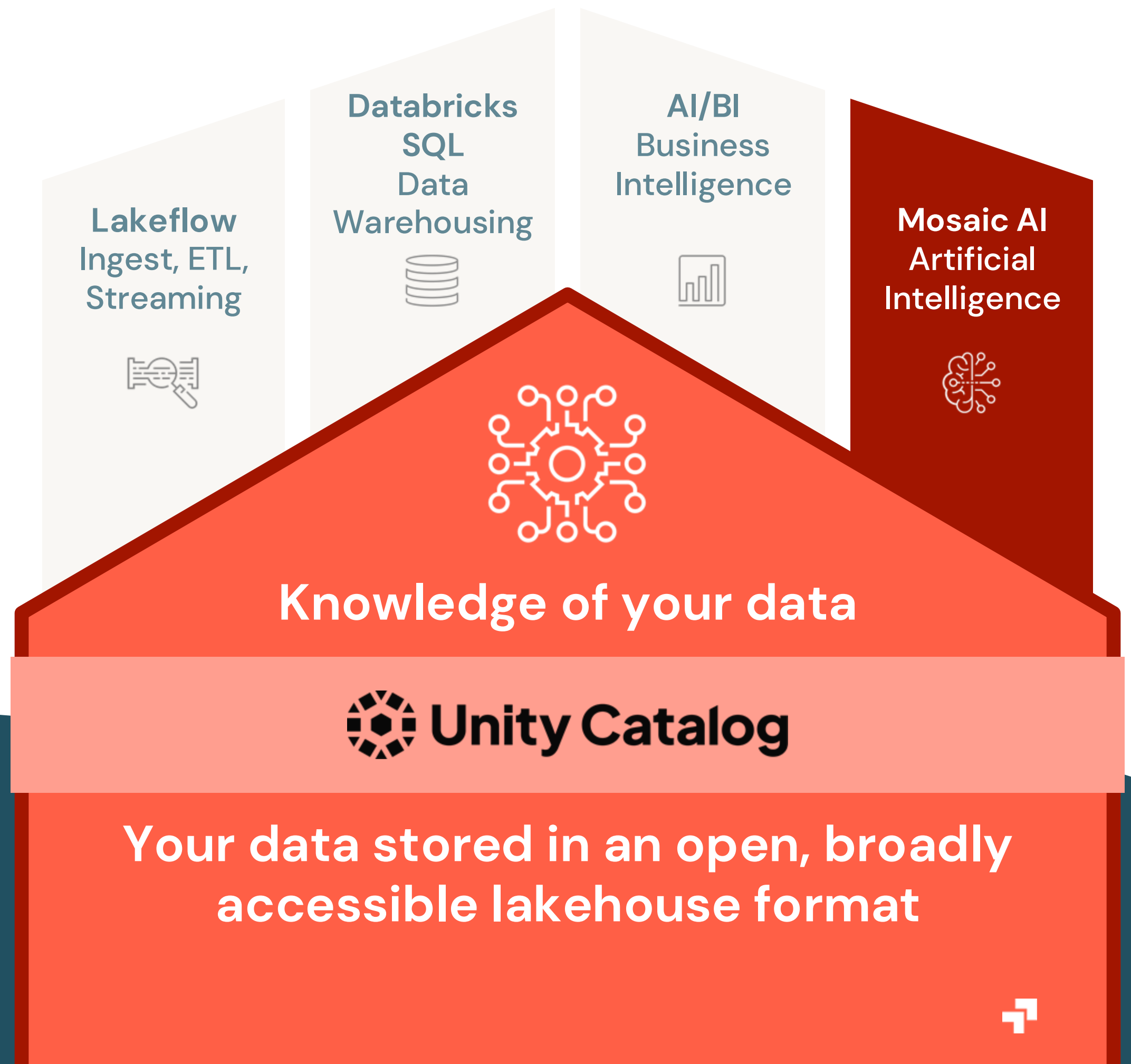


What is **data intelligence** for AI applications?



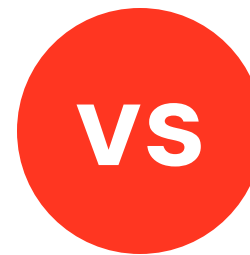
Mosaic AI

Create domain-specific
agentic applications



General Intelligence

Large models
trained on the entire web
leveraging scaling laws



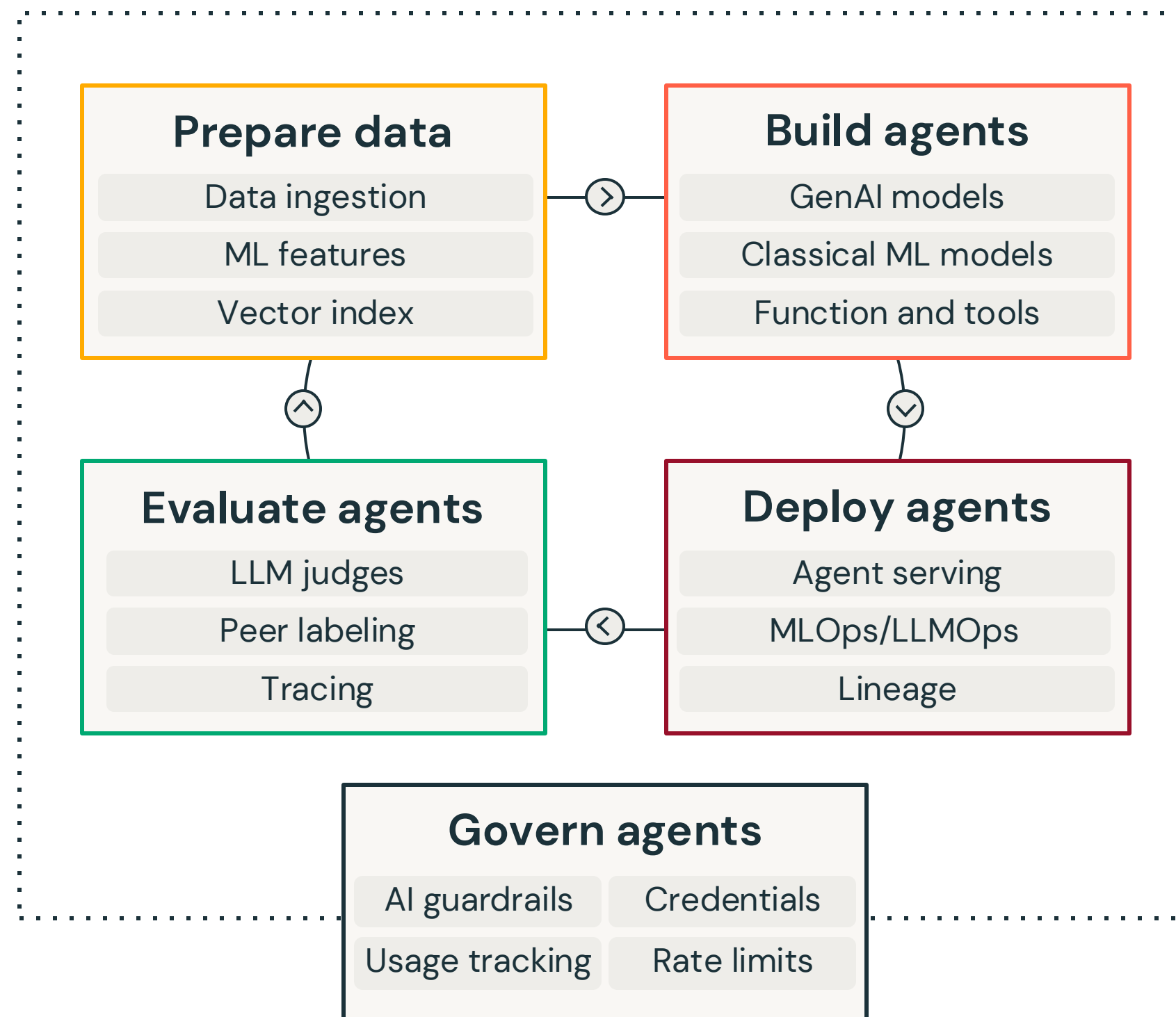
Data Intelligence

AI agents that reason
on your data and solve
domain-specific problems



Get accurate, domain-specific results

Mosaic AI is the complete agent platform



Agents that reason across every enterprise system

Support for all existing and future AI models

Build trust with guardrails, evaluation, and monitoring

o, Apache Iceberg, Iceberg, and the Apache



Lakeflow
Ingest, ETL,
Streaming



Databricks
SQL
Data
Warehousing



AI/BI
Business
Intelligence



Mosaic AI
Artificial
Intelligence



Knowledge of your data



Unity Catalog

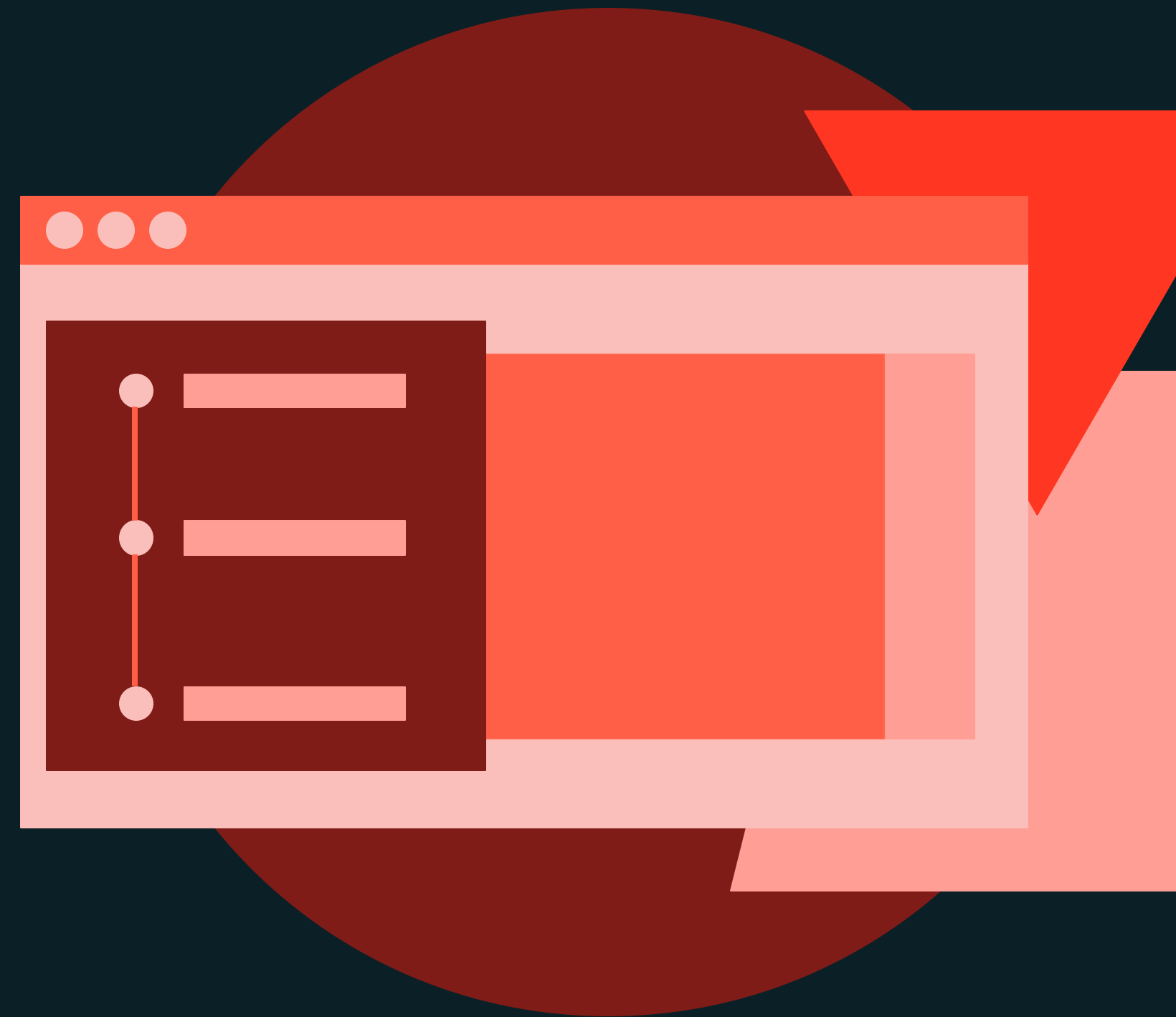
**Your data stored in an open, broadly
accessible lakehouse format**



Databricks Overview

DEMONSTRATION

Databricks Workspace Walkthrough



Follow-along Instructions

Estimated Time: 15 minutes

For this demonstration, the instructor will take you on a guided tour of the platform. If you have access to the Vocareum lab environment, feel free to follow-along to explore the workspace and get oriented with where future labs will be completed.





Using Databricks for Data Warehousing

Get Started with Databricks for Data Warehousing



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Objectives

- Use Databricks to complete common data warehousing tasks.
- Explain the purpose of Delta Lake for data warehousing purposes.



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Introduction to Data Warehousing with Databricks	10 mins	✓		
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Delta Lake Overview	8 mins	✓		
Using Delta Lake features with Databricks SQL	5 mins		✓	





Using Databricks for Data Warehousing

LECTURE

Introduction to Data Warehousing with Databricks

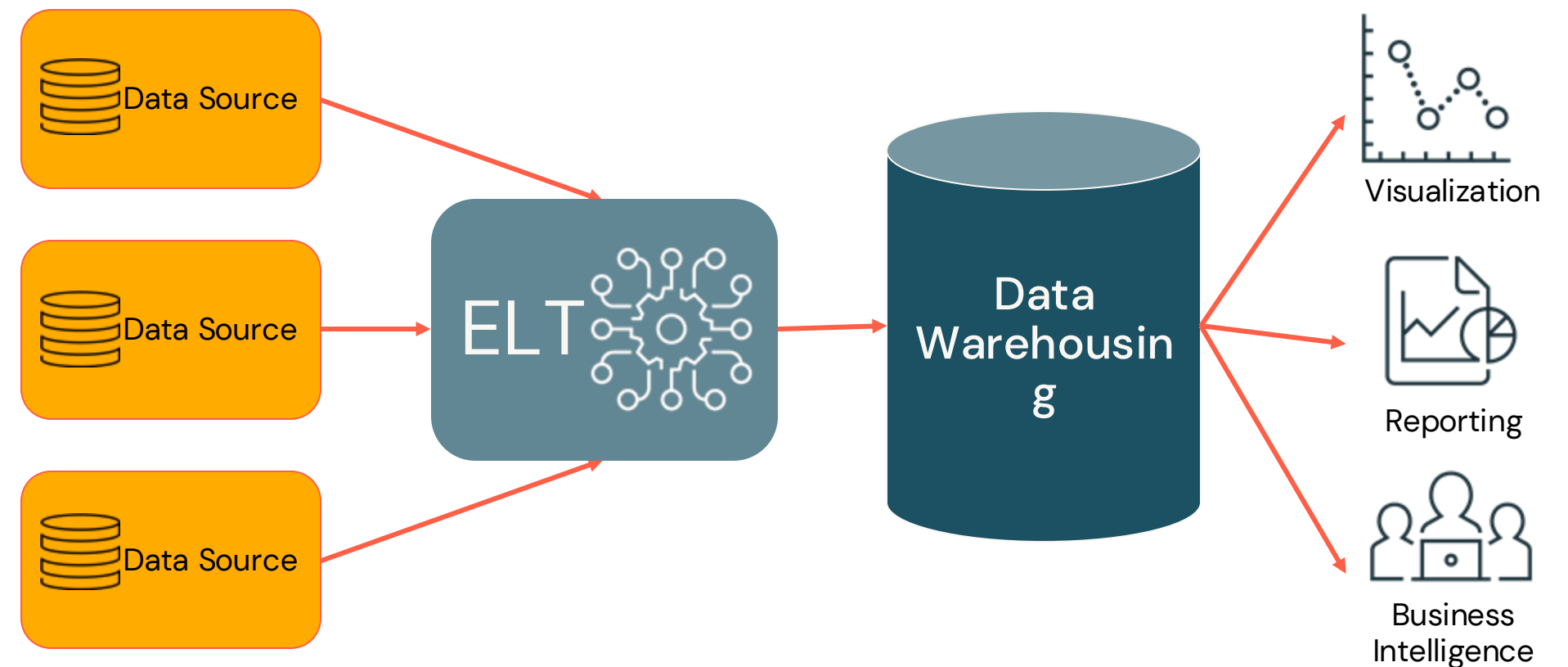


Introduction to Data Warehousing

A Centralized Solution for Reporting, Analytics, and Insights

A data warehouse serves as the foundation for reporting, analytics, and BI, helping organizations make informed decisions.

- **Centralized Storage:** Integrates data from multiple sources (databases, logs, etc.).
- **ETL Process:** Data is extracted, cleaned, and organized for analysis.
- **Optimized for Insights:** Powers visualization, reporting, and BI tools.



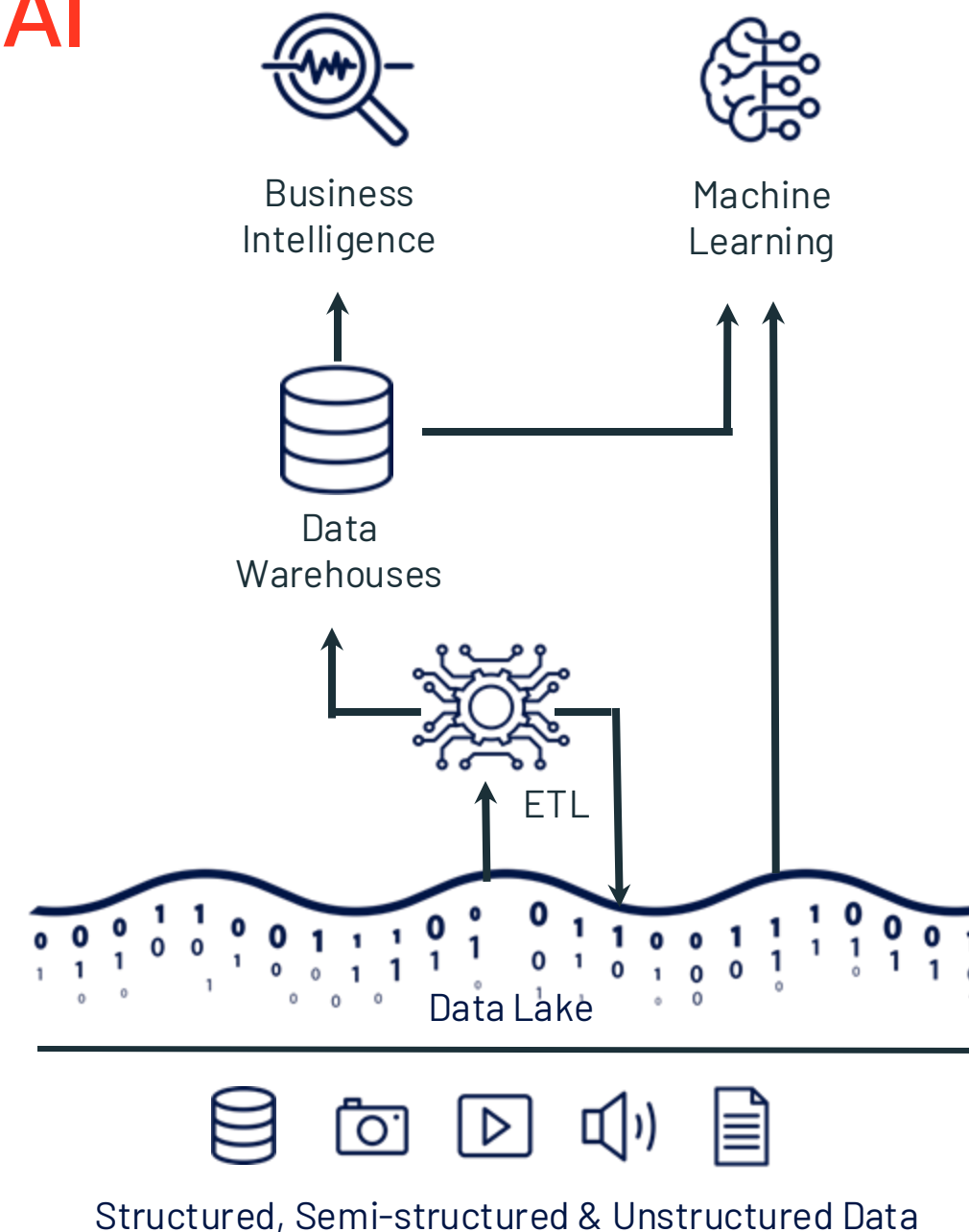
Data warehousing professionals ensure the ETL process runs smoothly and data is centralized, reliable, and optimized for reporting and insights.



Why Data Warehousing Matters

Unified data powers analytics, reporting, and AI

- Consolidates Data from Disparate Sources
- Improves Data Quality and Consistency
- Enables Reporting, Analytics, and BI
- Supports Better Decision-Making



Unified data enables faster insights and better decisions, driving organizational success.



Challenges in Data Warehousing

Data warehousing can be complex and expensive.



Users need specialized technical skills



Complex infrastructure



Data silos and fragmented systems



Issues with data governance and compliance



Slow performance at scale



Inconsistent data quality and access

- Users must have specialized knowledge to operate and manage traditional data warehouses.
- Talent is hard to obtain and retain as the industry is highly competitive.
- Expensive to train up these skills to have them lost to other companies.



Challenges in Data Warehousing

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Slow performance at scale



Inconsistent data quality and access

- Redundant tooling within the data warehouse make it difficult to achieve consistency
- Systems and processes become inefficient over time and need to be frequently adjusted to maintain performance requirements
- Maintenance difficulties arise with easily outdated foundational components



Challenges in Data Warehousing

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Slow performance at scale



Inconsistent data quality and access

- Data is spread across multiple systems in the ecosystem, fragmenting management
- Difficulty in consolidating insights and identifying accurate and clean data
- Delayed decision-making as processes are lengthy and heavily manual



Challenges in Data Warehousing

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Data silos and fragmented systems



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Slow performance at scale



Inconsistent data quality and access

- Lack of centralized access controls in traditional platforms
- Challenges in ensuring compliance with regulations (GDPR, HIPAA)
- Limited capabilities to track data lineage and challenges with tackling quality issues



Challenges in Data Warehousing

Data warehousing can be complex and expensive.



Users need specialized technical skills



Complex infrastructure



Data silos and fragmented systems



Issues with data governance and compliance



Slow performance at scale



Inconsistent data quality and access

- Runaway increase in cost for cloud compute as the data warehouses scale
- Degraded performance as the amount of data increases from small initial testing
- Lack of access in traditional on-premise data warehouses to the latest technologies that impact performance capabilities



Challenges in Data Warehousing

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Complex infrastructure



Data silos and fragmented systems



Issues with data governance and compliance



Slow performance at scale



Inconsistent data quality and access

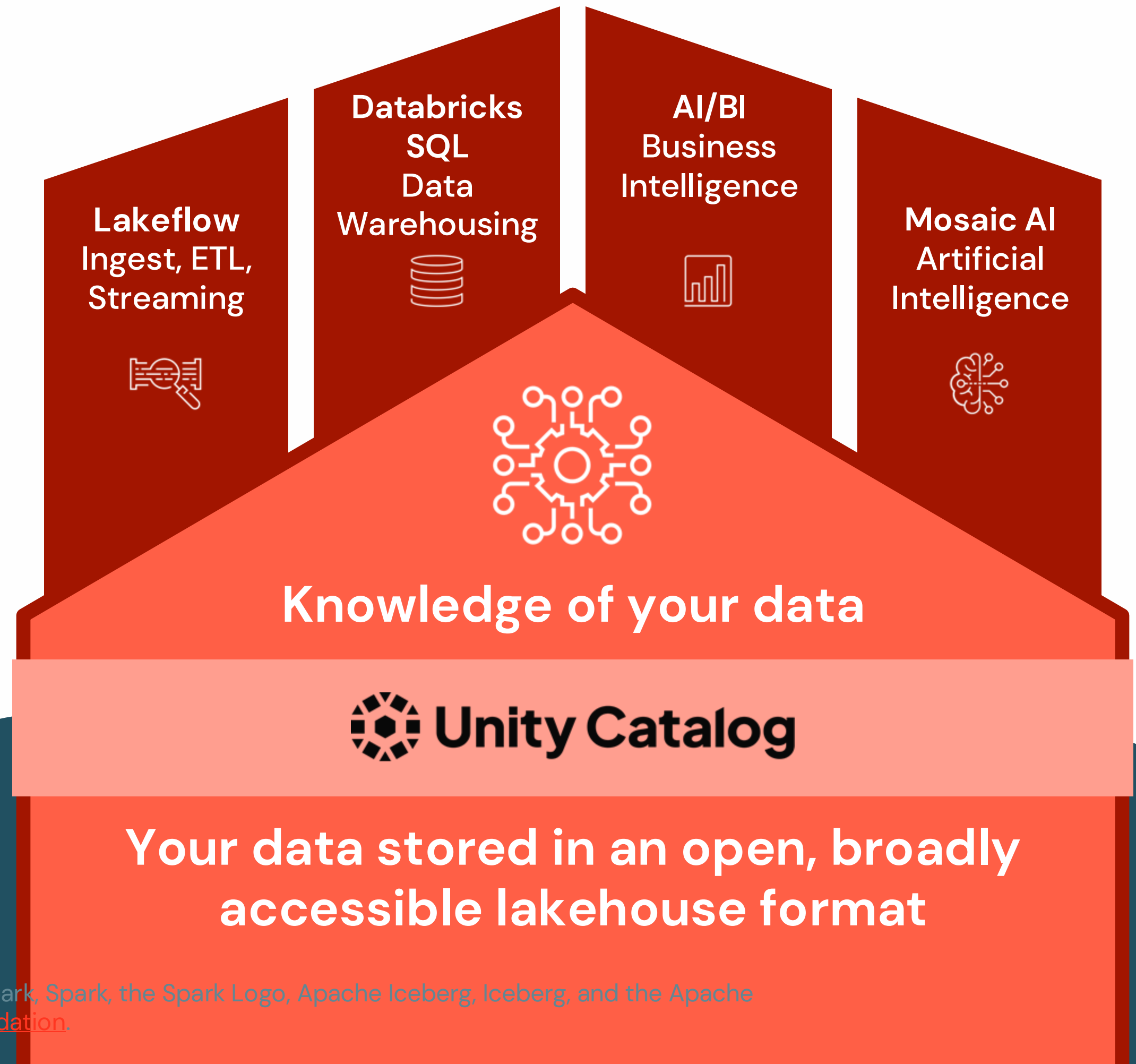
- Inconsistent, incomplete, or outdated data persists within the environment
- Limited access for business users leads to manual overhead to process requests for access
- Complex processes requiring manual oversight to clean and validate data persist regardless of the volume of data in the warehouse



**With Databricks, data warehousing
becomes simpler, faster, and smarter
– enabling you to overcome
traditional challenges.**



Databricks Data Intelligence Platform



What Makes Databricks Different?

Databricks Data Warehousing: Simplified, Centralized, and Powerful

Unified Data Management:

- Combines structured and semi-structured data using Delta Tables for centralized storage.

Transform with Delta:

- Leverages Databricks' compute for ETL and real-time transformations.

Powered by Unity Catalog:

- Ensures governance, metadata management, and seamless collaboration across teams.

Optimized Outputs:

- Enables data marts, machine learning models, and applications for actionable insights.



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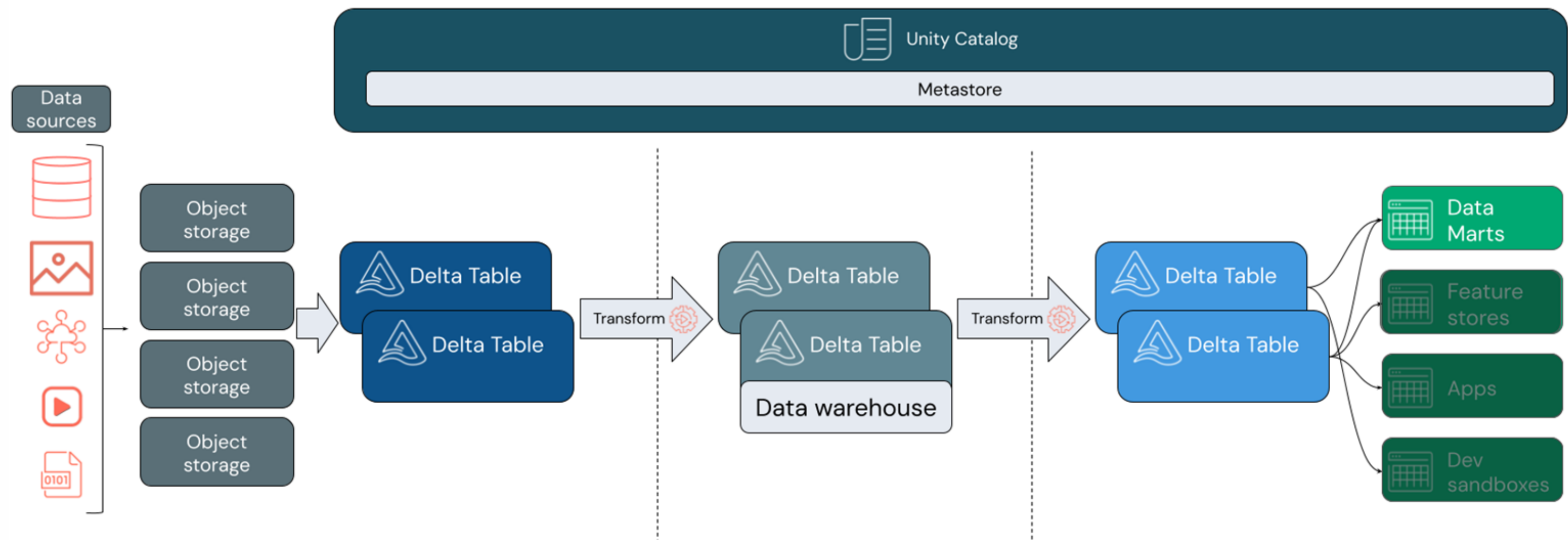
Optimized Outputs:

- Enables data marts, machine learning models, and applications for actionable insights.



Databricks Approach to Data Warehousing

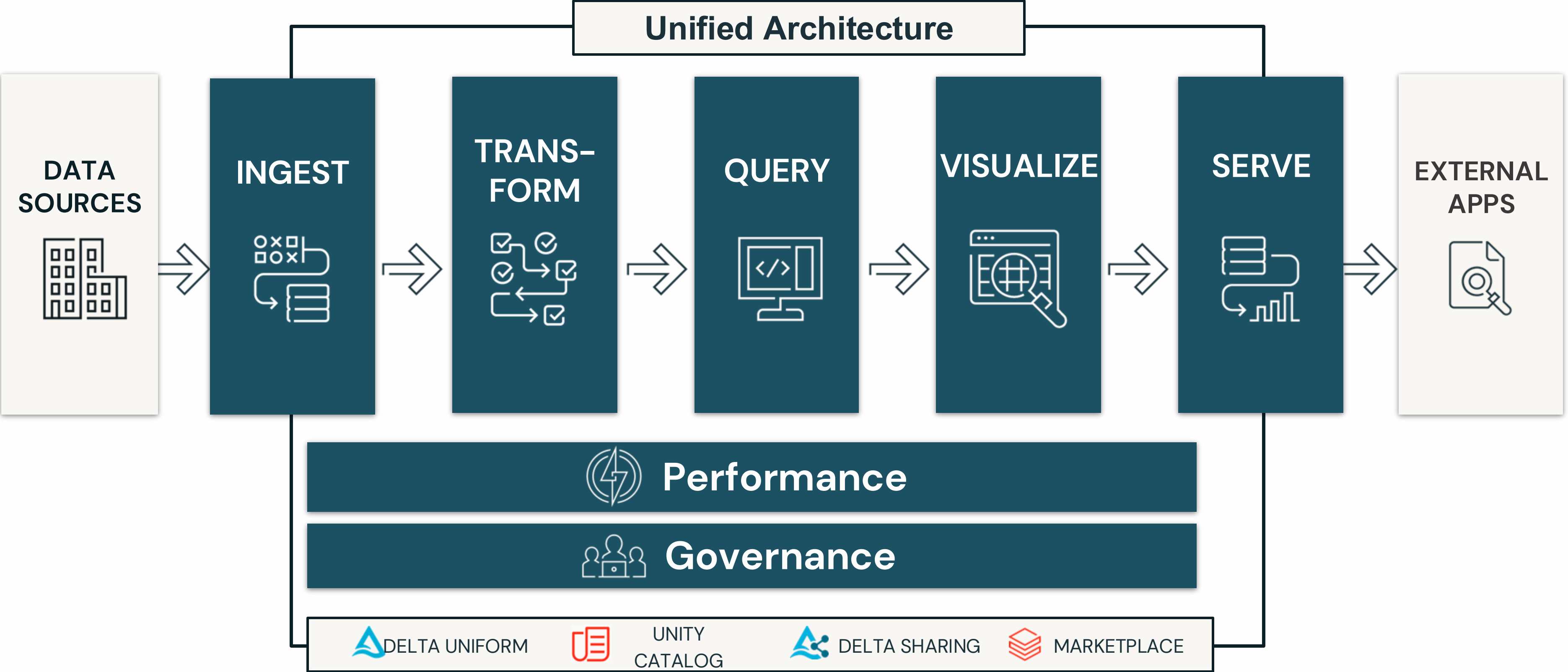
From raw data to insights: Leveraging Delta Tables and Unity Catalog.



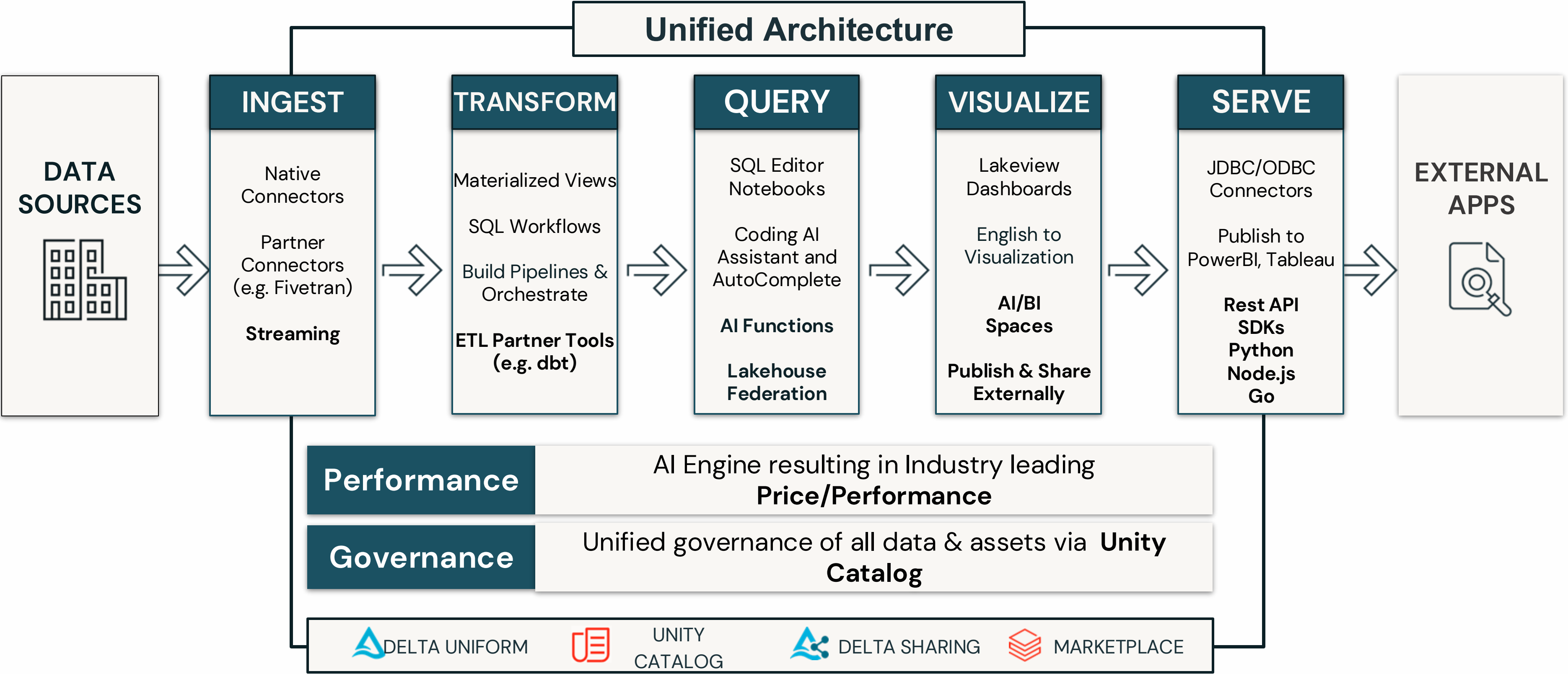
Databricks combines unified storage, real-time transformations, and governance to enable smarter data-driven insights.



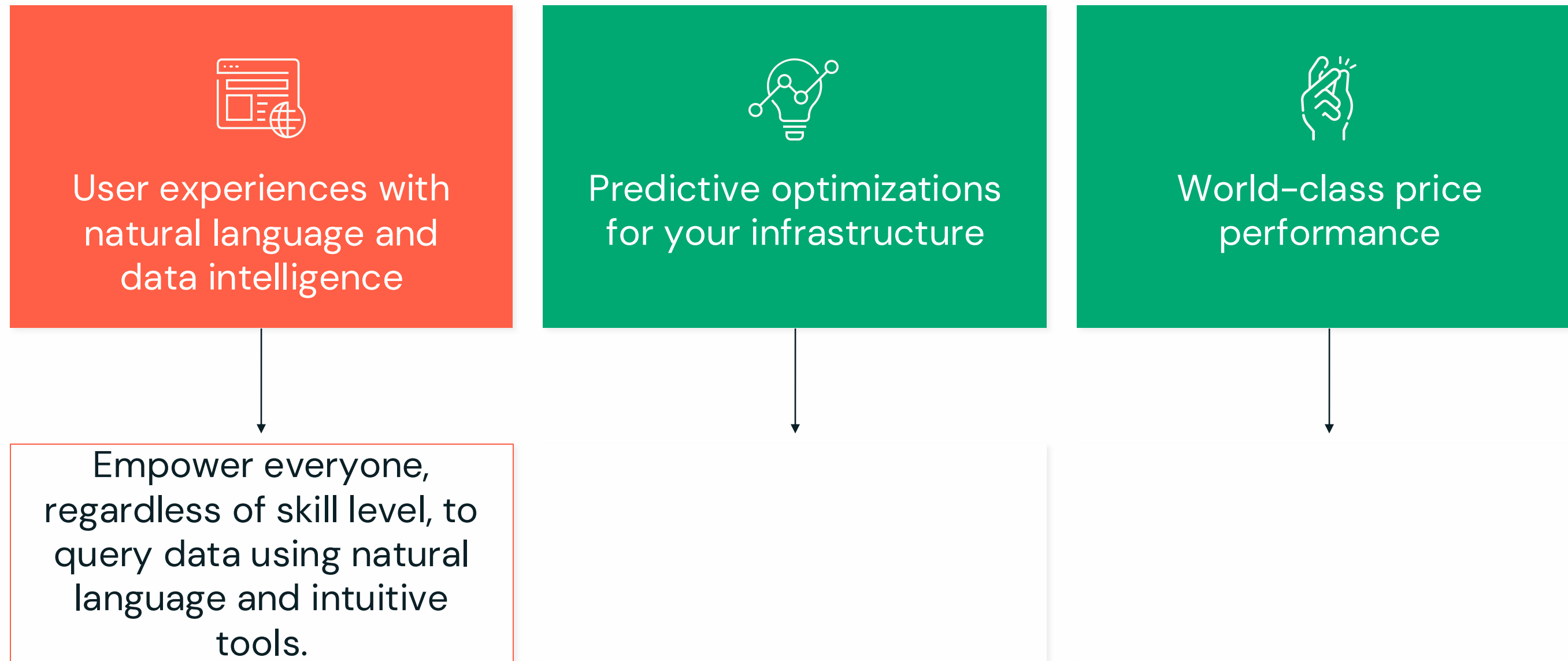
Complete Data Warehousing Solution



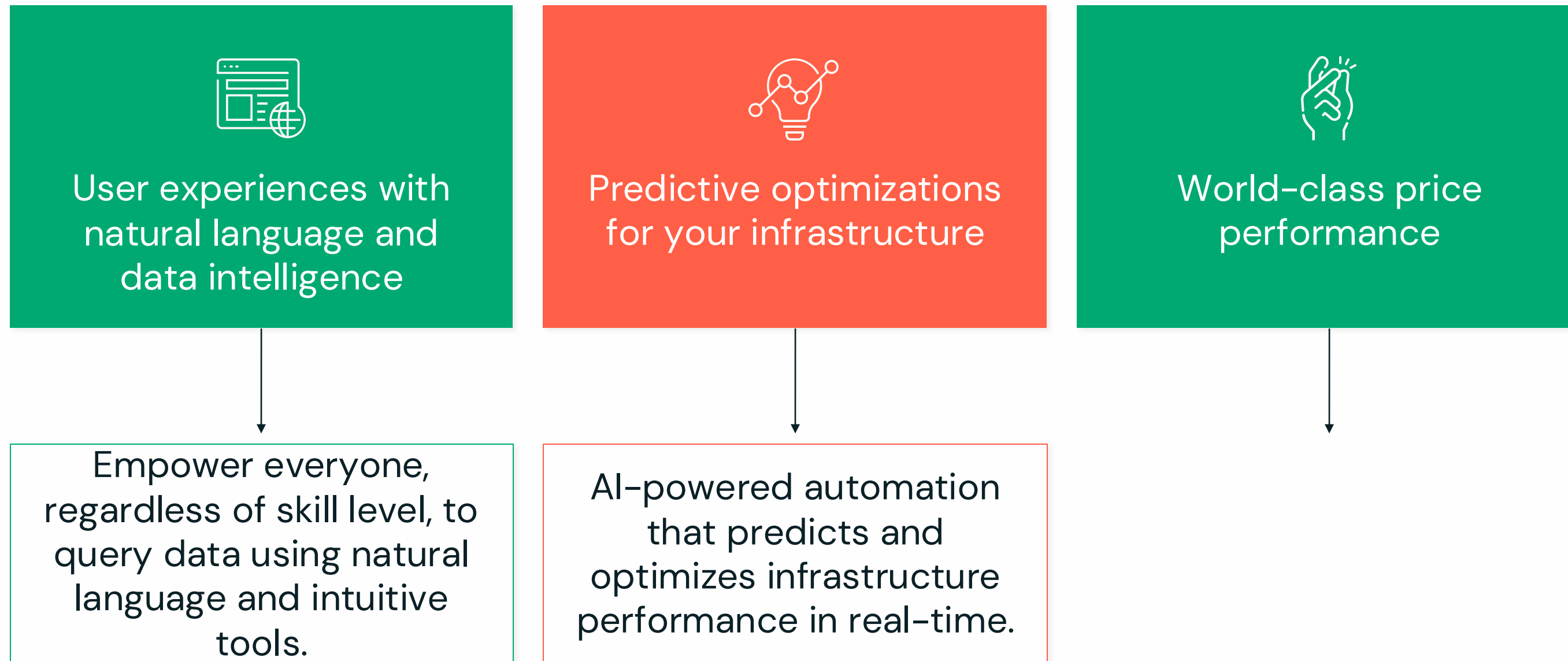
Complete Data Warehousing Solution



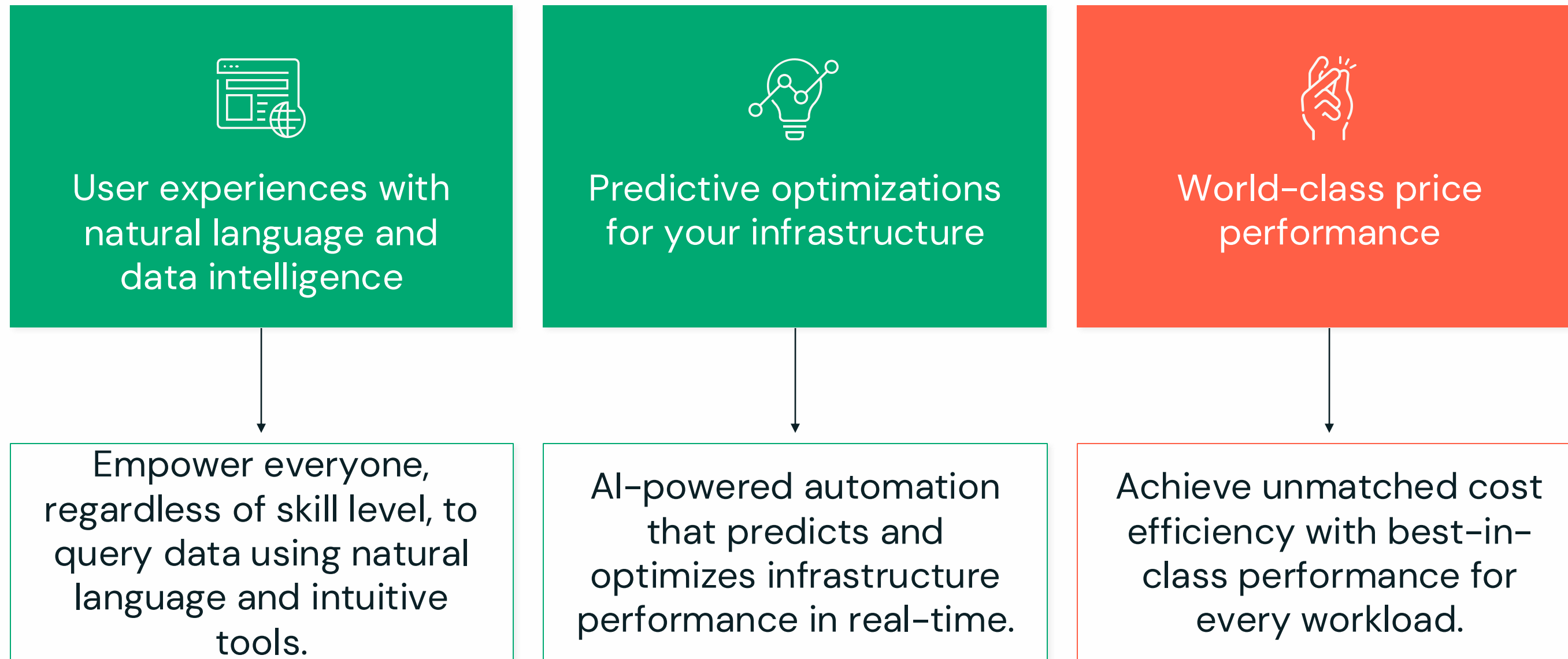
The Future of Data Warehousing: Smarter, Faster, and AI-Powered



The Future of Data Warehousing: Smarter, Faster, and AI-Powered

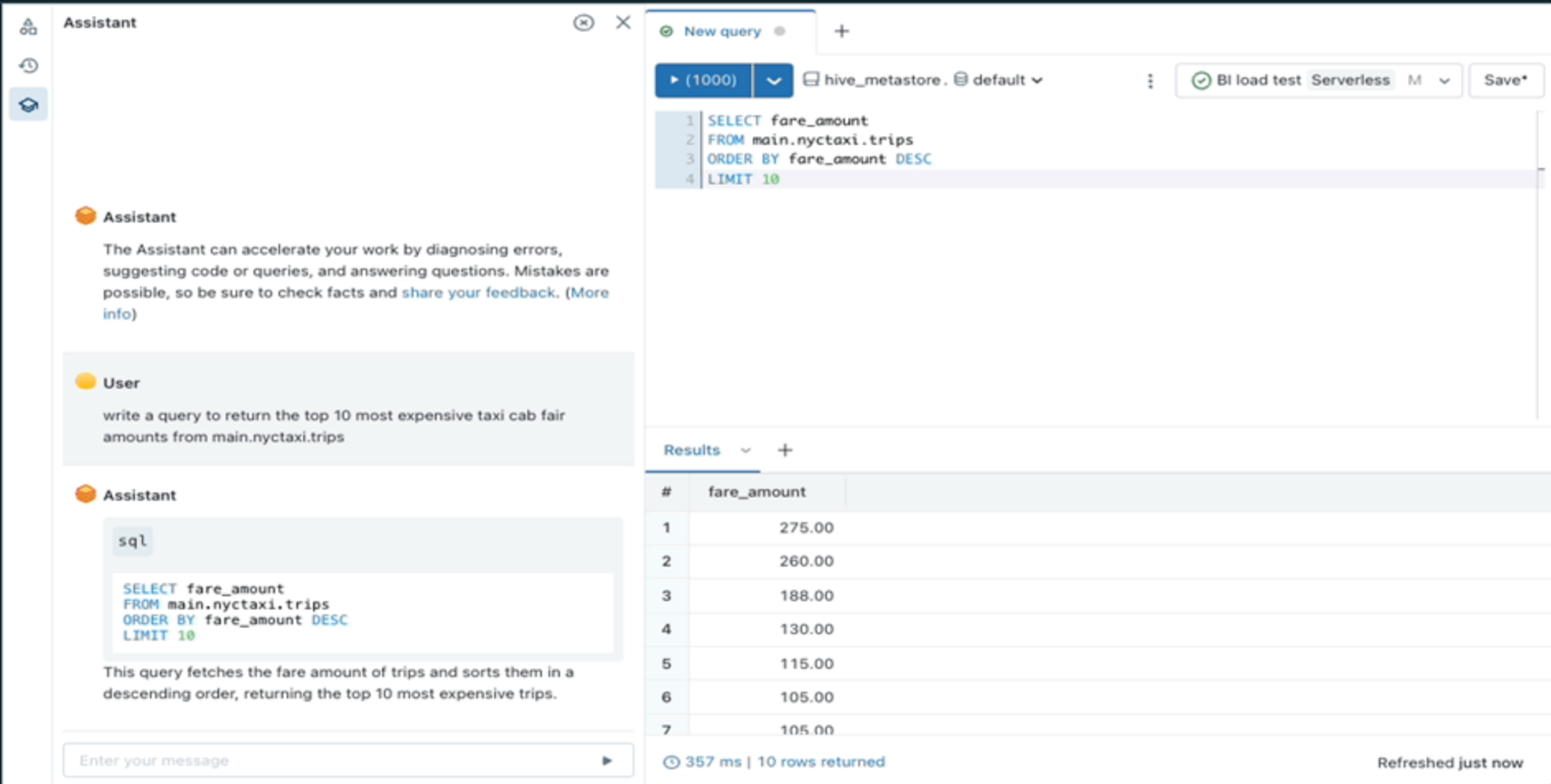


The Future of Data Warehousing: Smarter, Faster, and AI-Powered



Tools for Intelligent Data Warehousing

Powered by Intelligence Engine



The screenshot displays the Unified SQL Editor interface. On the left, a chat window titled 'Assistant' shows a conversation. The Assistant explains its role in accelerating work by diagnosing errors and suggesting code. A User message asks for a query to return the top 10 most expensive taxi cab fare amounts from 'main.nyctaxi.trips'. The Assistant responds with an SQL query: `SELECT fare_amount FROM main.nyctaxi.trips ORDER BY fare_amount DESC LIMIT 10`. Below the chat, there is an input field labeled 'Enter your message'. On the right, the 'New query' editor shows the same SQL query. The results table below the query editor displays the top 10 fare amounts in descending order.

#	fare_amount
1	275.00
2	260.00
3	188.00
4	130.00
5	115.00
6	105.00
7	105.00

357 ms | 10 rows returned Refreshed just now

Unified SQL Editor

Data Science / Engineers



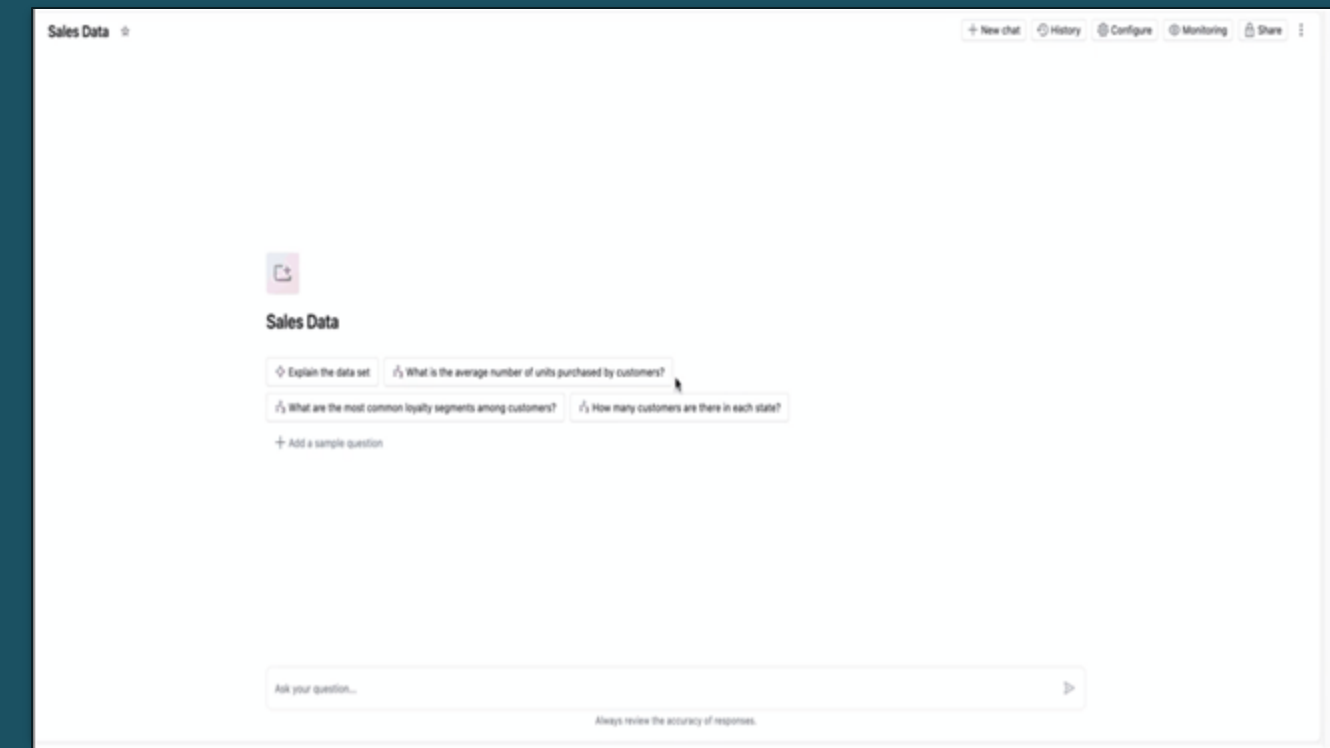
Tools for Intelligent Data Warehousing

Powered by Intelligence Engine



AI/BI Dashboards

Analysts



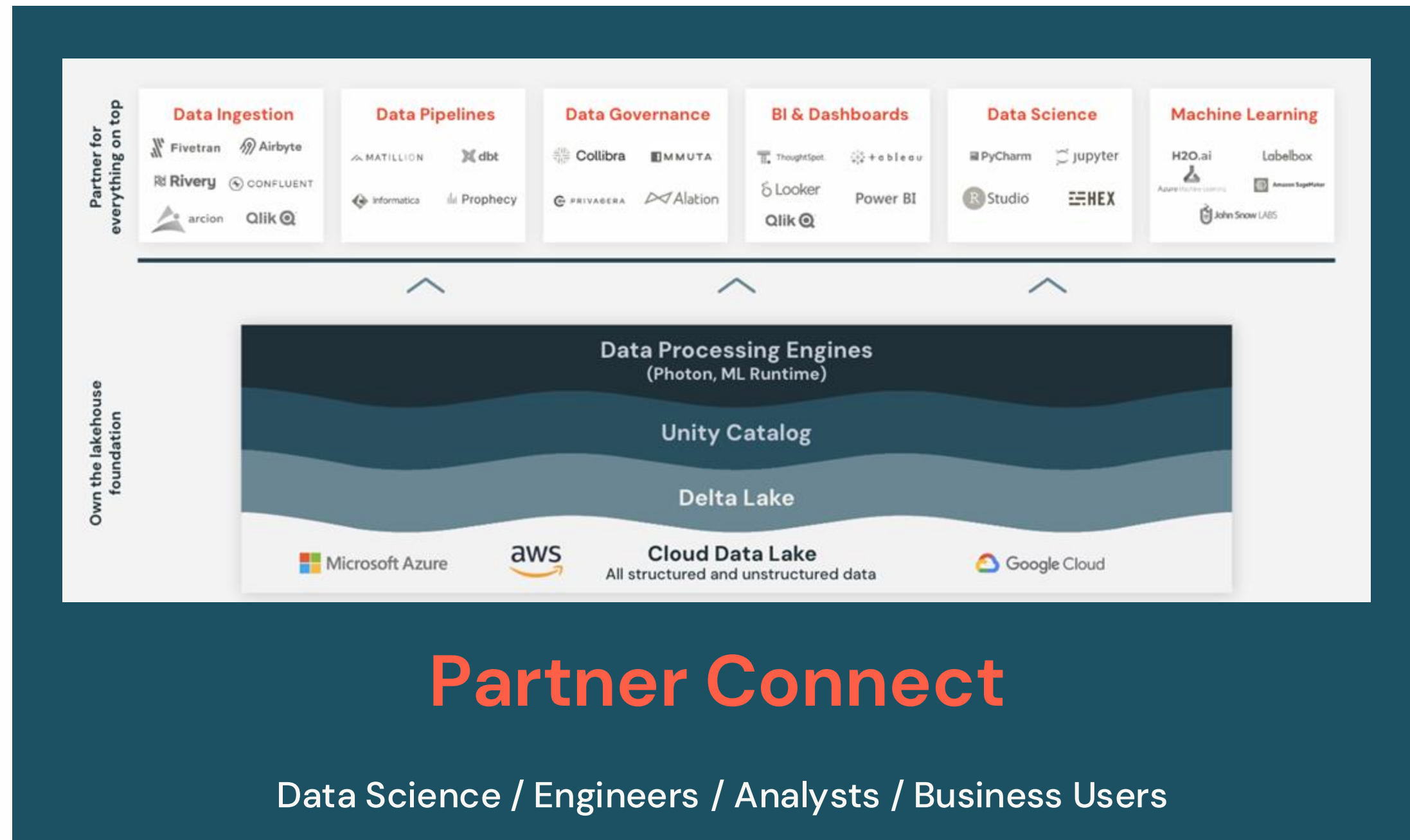
AI/BI Genie

Business Users



Tools for Intelligent Data Warehousing

Powered by Intelligence Engine





Using Databricks for Data Warehousing

LECTURE

Databricks SQL Warehouses



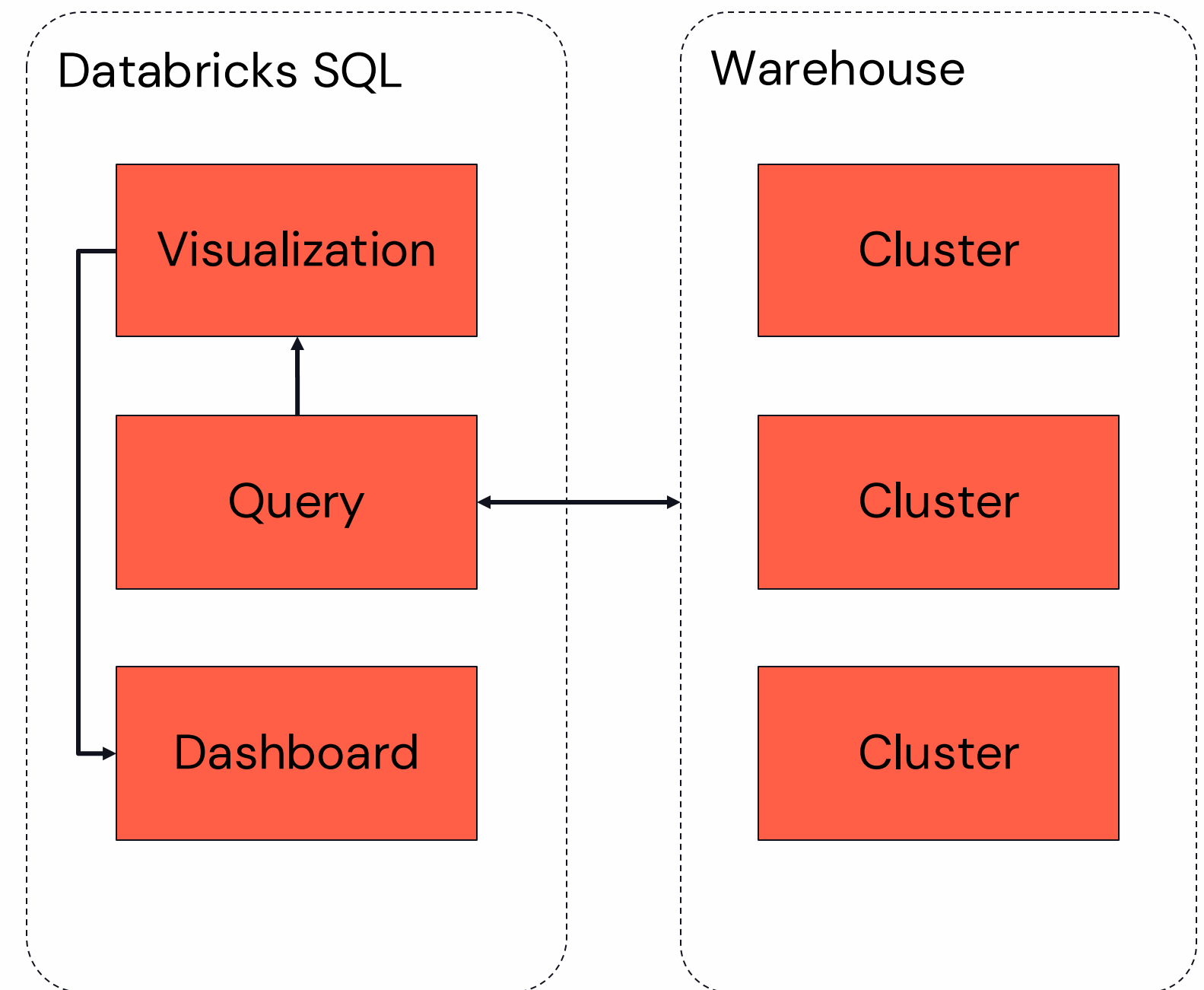
SQL Warehouses

Overview

SQL Warehouse in Databricks is a scalable compute resource optimized for executing SQL queries, analyzing data, creating visualizations, and supporting business intelligence (BI) workflows.

Key Features:

- Scalable compute for distributed SQL workloads.
- Optimized for data queries and BI workflows.
- Supports integration with dashboards and analytics tools.



SQL Warehouses

Cluster Size:

- Refers to the computational power allocated to a SQL Warehouse.
- Expressed in “**t-shirt sizes**” such as **Small**, **Medium**, **Large**, and **X-Large**.
- Each size defines the number of compute nodes and capacity for parallel processing.
- **Example:**
 - **Small:** Suitable for light queries.
 - **X-Large:** Ideal for heavy, high-concurrency workloads.

Cluster size	DBU / h
2X-Small	4 DBU / h
X-Small	6 DBU / h
Small	12 DBU / h
Medium	24 DBU / h
Large	40 DBU / h ✓
X-Large	80 DBU / h
2X-Large	144 DBU / h
3X-Large	272 DBU / h
4X-Large	528 DBU / h



SQL Warehouses

Scaling:

- Ensures SQL Warehouses adapt to varying workloads by dynamically allocating resources.
- **Users can configure:**
 - **Minimum clusters:** Base capacity for consistent performance.
 - **Maximum clusters:** Auto-scale limit to handle peak demand.
- **Benefits:**
 - **Cost Efficiency:** Only pay for resources used.
 - **Performance Optimization:** Prevents resource bottlenecks.

New SQL warehouse

Name: SQL warehouse name

Cluster size: Large 40 DBU / h

Auto stop: ☒ After 10 minutes of inactivity.

Scaling: Min. 1 Max. 1 clusters (40 DBU)

Type: ☒ Pro ☐ Classic

Advanced options

Cancel Create

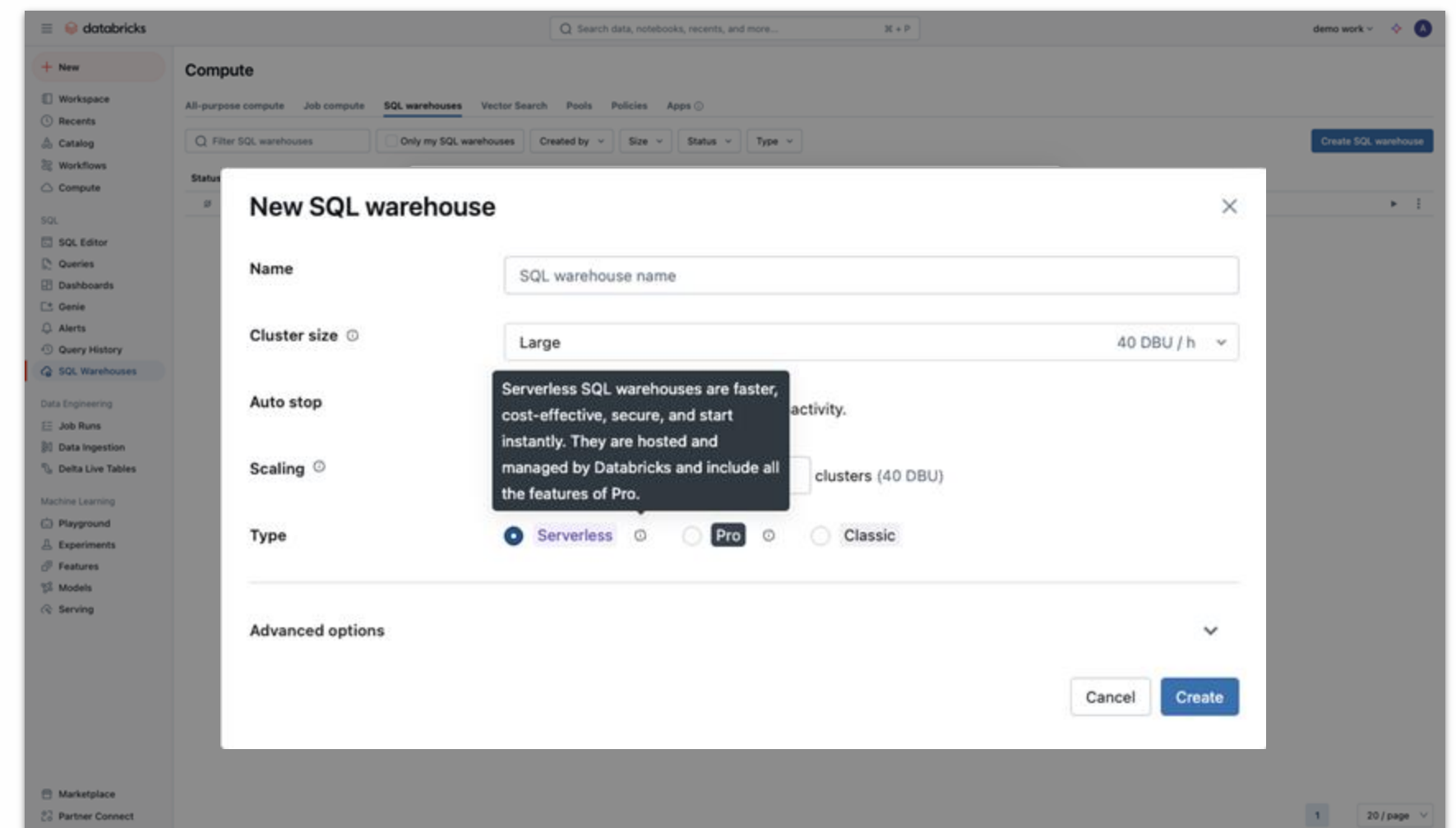
Scale clusters to handle more concurrent users. When clusters recycle each day, maximum will temporarily be exceeded. [Learn more](#)



Serverless Compute for Databricks SQL

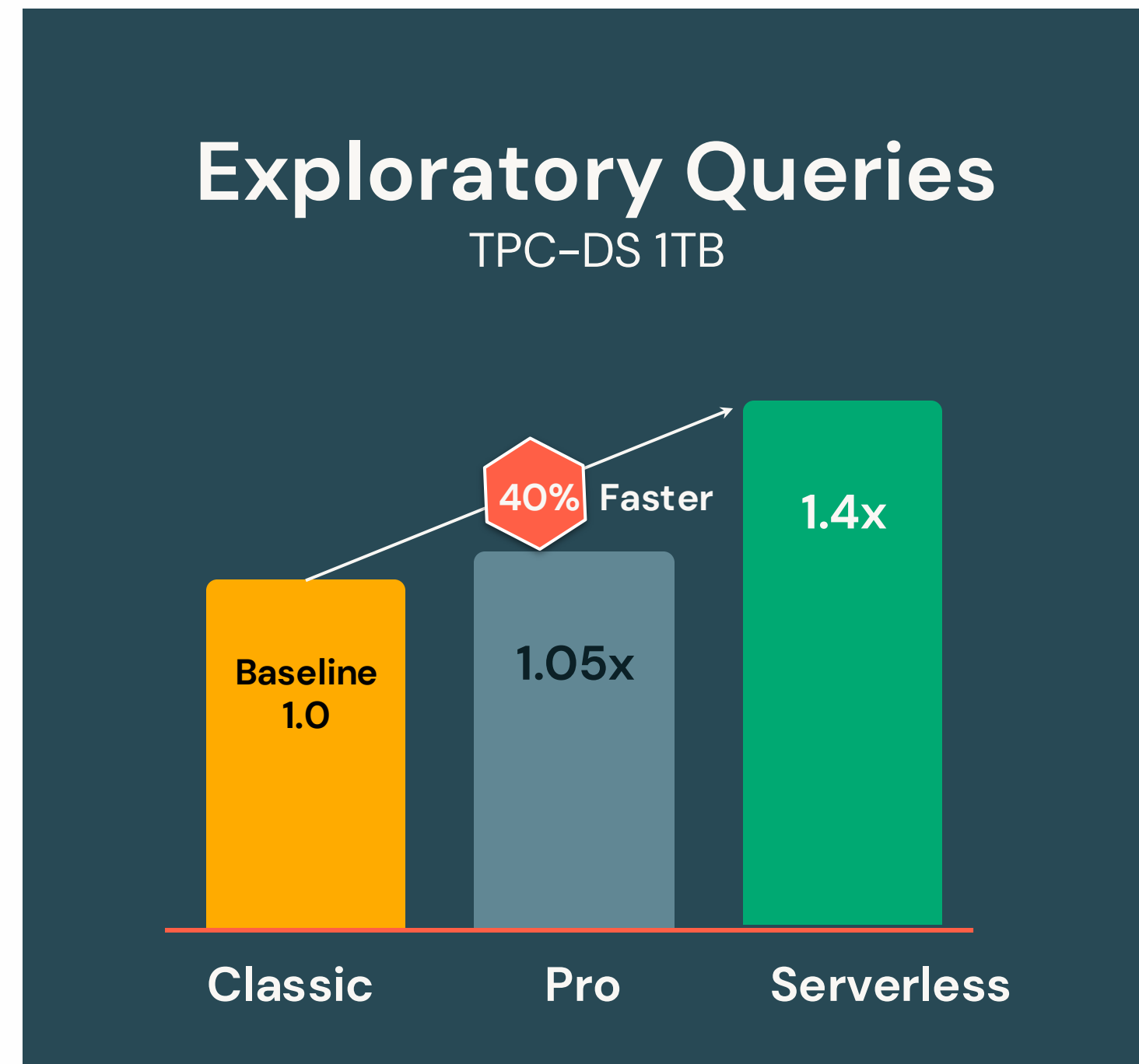
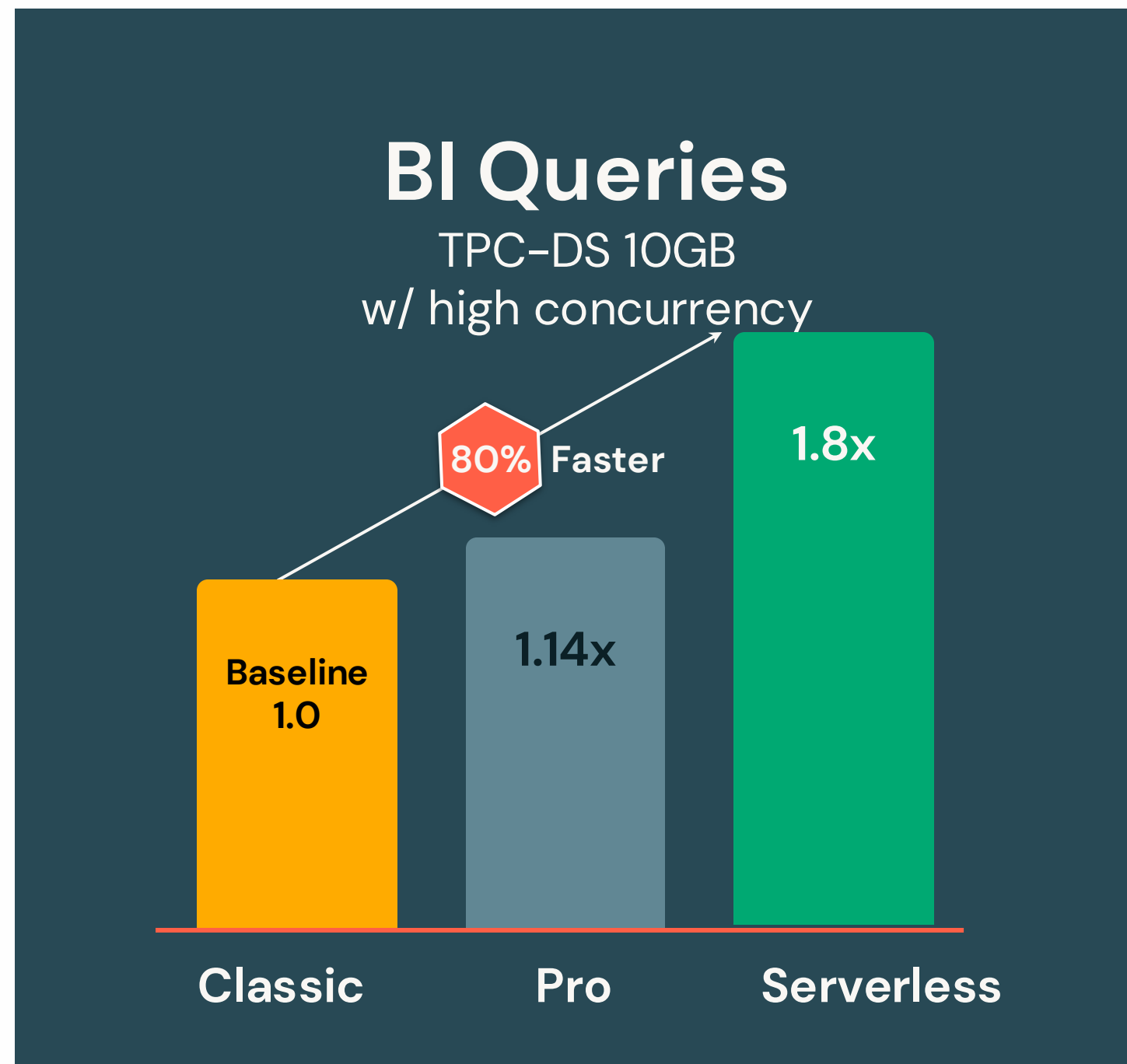
Instant, elastic SQL compute decoupled from storage

- Quickly setup **instant, elastic SQL warehouse** – decoupled from storage – Powered by Photon
- Automatically determines instance types and configuration for best price/perf (up to 12x)
- **High concurrency** built-in, automatic load balancing
- Intelligent **workload management** and faster reads from cloud storage
- Instant startup, greater availability, and **40% average lower overall costs with serverless**



Serverless SQL has the Best Performance!

Let's see how they perform in real-world scenarios:





Using Databricks for Data Warehousing

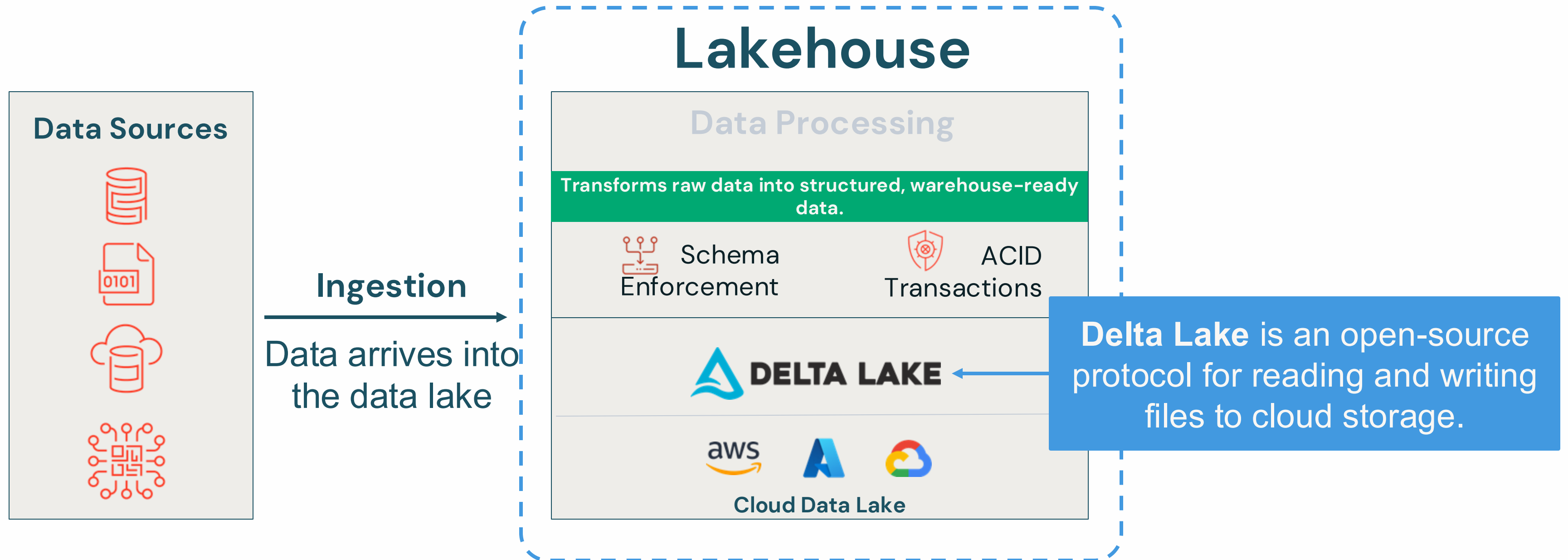
LECTURE

Delta Lake Overview



Delta Lake Overview

Delta Lake: Foundation for Modern Data Warehousing.

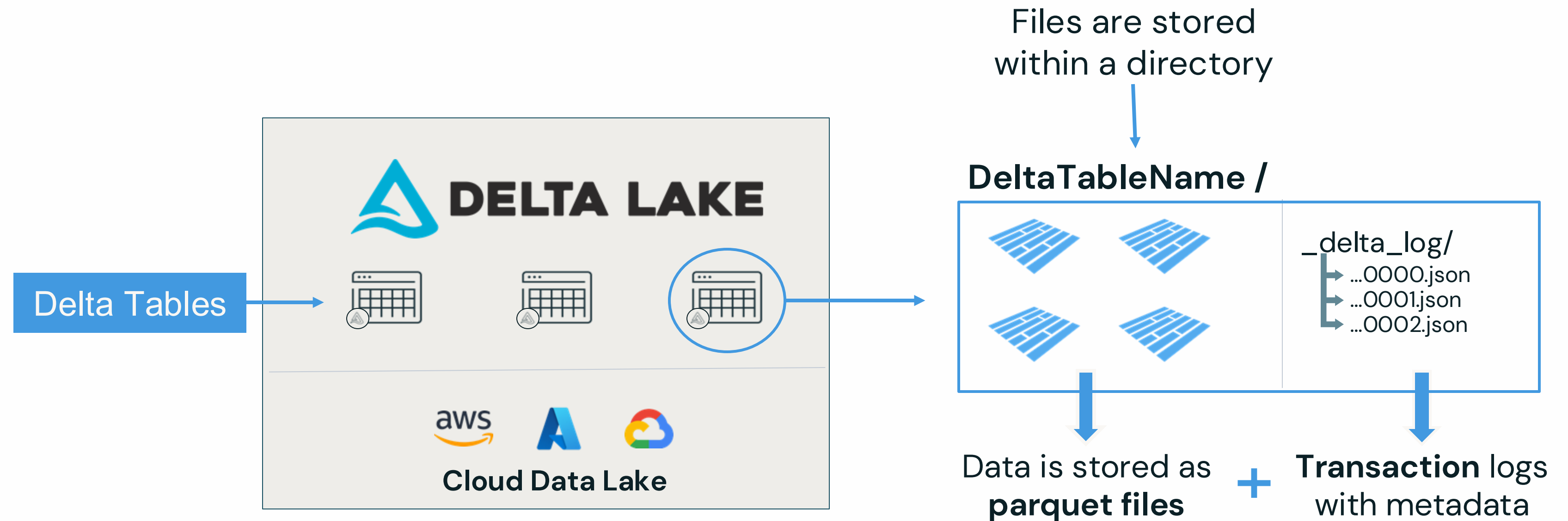


Delta Lake ensures reliability, performance, and scalability for modern data warehousing workflows.



Delta Lake Overview

Delta Tables: The Backbone of Reliable Data Warehousing

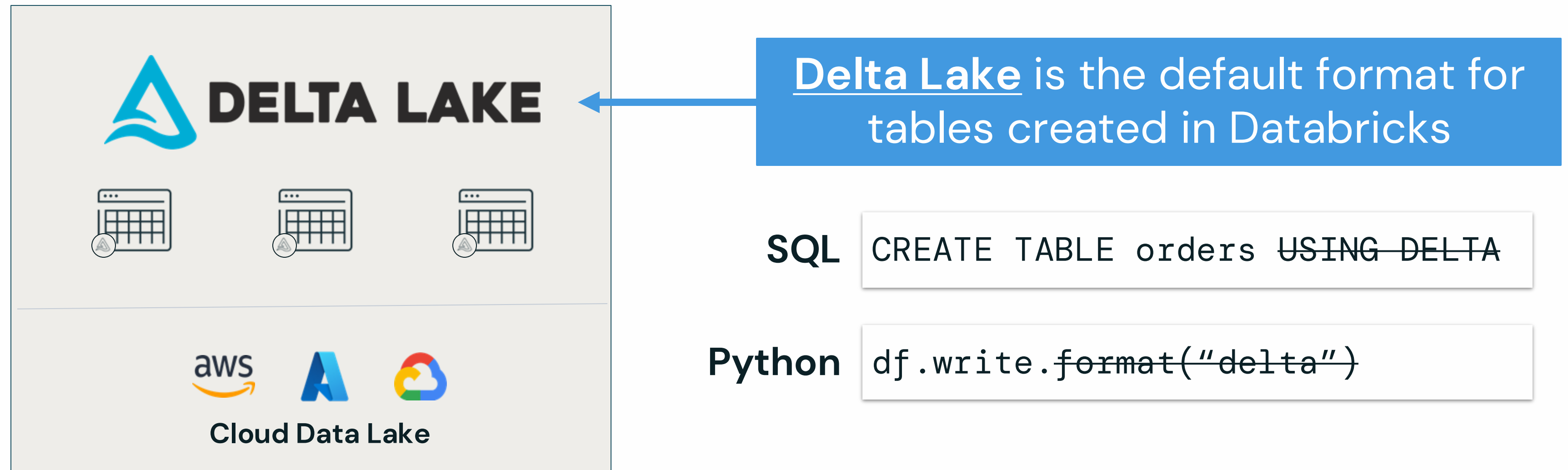


Delta Tables support both batch and streaming data, making them flexible for real-time and historical analytics in a warehouse.



Delta Lake Overview

Delta Lake: Default Format for Databricks Warehousing



Delta Tables power Serverless SQL Warehouses, providing instant scalability and optimized performance for data warehousing workflows.



Delta Lake Overview

Key Features



ACID Transactions



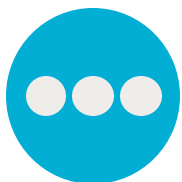
DML Operations



Time Travel



Schema Evolution and
Enforcement



Many more!



Delta Lake Overview

Key Features



ACID Transactions

Delta Lake provides ACID transactional guarantees at the table level to data stored in **cloud-based object storage**.



DML Operations

Atomicity – Entire transaction completes



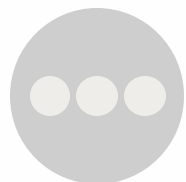
Time Travel

Consistency – Data follows rules or it will be rolled back



Schema Evolution and Enforcement

Isolation – One transaction completed before the start of another



Many more!

Durability – Data is saved in a persistent state once completed



Delta Lake Overview

Key Features



ACID Transactions



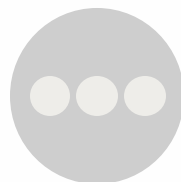
DML Operations



Time Travel



Schema Evolution and
Enforcement



Many more!

Data Manipulation Language (DML)

- **INSERT** – Add new rows
- **UPDATE** – Update existing values
- **DELETE** – Delete rows
- **MERGE** – Selects records from a source table and performs multiple DML operations on a target table



Delta Lake Overview

Key Features



ACID Transactions



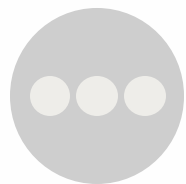
DML Operations



Time Travel



Schema Evolution and
Enforcement



Many more!

Delta Tables keep a **transaction log** for each **version (writes)** of the table.

- Historical Querying
- Snapshot Isolation
- Auditing



Delta Lake Overview

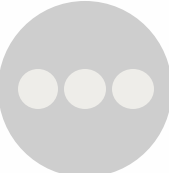
Key Features

 ACID Transactions

 DML Operations

 Time Travel

 Schema Evolution and Enforcement

 Many more!

Schema Evolution

- Automatically **adjusts** the schema of your Delta table as your data changes



Delta Lake Overview

Key Features

 ACID Transactions

 DML Operations

 Time Travel

 Schema Evolution and Enforcement

 Many more!

Schema Evolution

- Automatically **adjusts** the schema of your Delta table as your data changes

Schema Enforcement

- Ensures that any data written to the Delta table **matches** the table's defined schema



Delta Lake Overview

Key Features



ACID Transactions



DML Operations



Time Travel



Schema Evolution and
Enforcement



Many more!

- Unified Batch and Streaming
- Performance
- Scalable Metadata
- Optimization
- **Delta Lake is Open Source!**

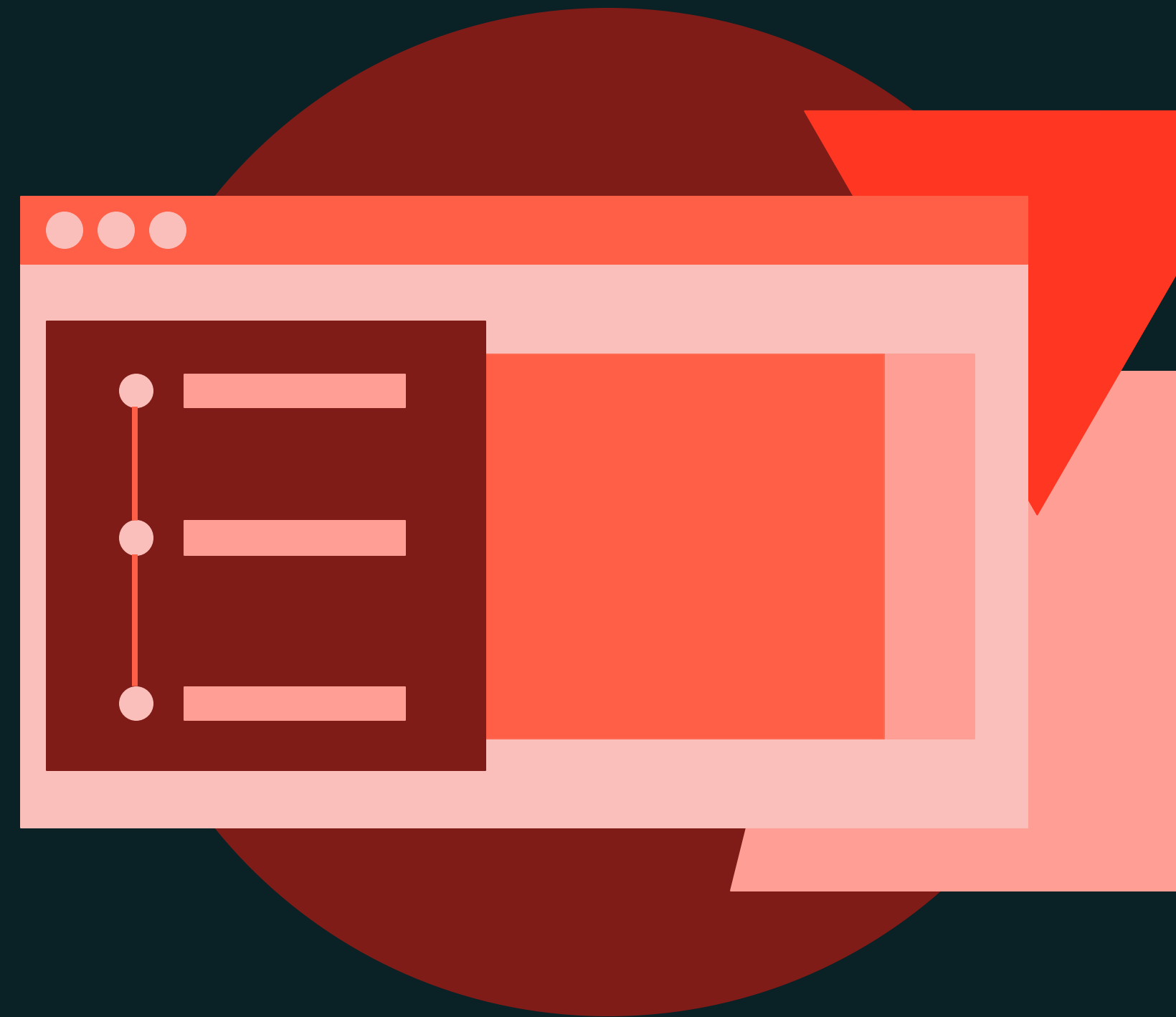




Using Databricks for Data Warehousing

DEMONSTRATION

Using Delta Lake features with Databricks SQL



High Level Steps

- Explore key features of Delta Lake such as Time Travel, Version History, and metadata management.
- Use SQL commands like **DESCRIBE EXTENDED**, **DESCRIBE HISTORY**, **VERSION AS OF**, and **RESTORE TABLE**.
- Understand how Delta Lake's features are beneficial for managing data in data warehousing scenarios.
- Work with the **sales_orders** table in the <schema_name> schema.





Data Ingestion and Transformation

Get Started with Databricks for Data Warehousing



Objectives

- Describe and apply various techniques for ingesting and transforming data in Databricks.



Agenda

Data Ingestion and Transformation	Time	Lecture	Demo	Lab
Delta Lake UniForm	5 mins	✓		
Data Ingestion Techniques Overview	5 mins	✓		
Data Ingestion Techniques	10 mins		✓	
Data Transformation	8 mins	✓		
Exploring Data Transformation in Databricks	10 mins		✓	





Data Ingestion and Transformation

LECTURE

Delta Lake UniForm



Delta Lake with UniForm

Rise of Open Table Formats

Which open table format to pick?



Delta Lake



Hudi



Iceberg

Delta Lake with UniForm simplifies data management by bridging the gap between different table formats, empowering data warehouses with unified access and control.

Introducing Delta Lake with Uniform (Delta Universal Format)



Delta Lake with UniForm

Open Table Formats Overview



Apache Hudi



Delta Lake



Apache Iceberg

Metadata

Used for transactional source of truth, concurrency control, etc.



Metadata



Metadata



Metadata



Parquet

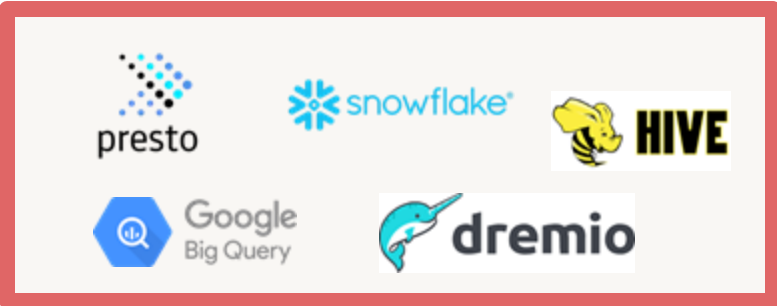


Parquet



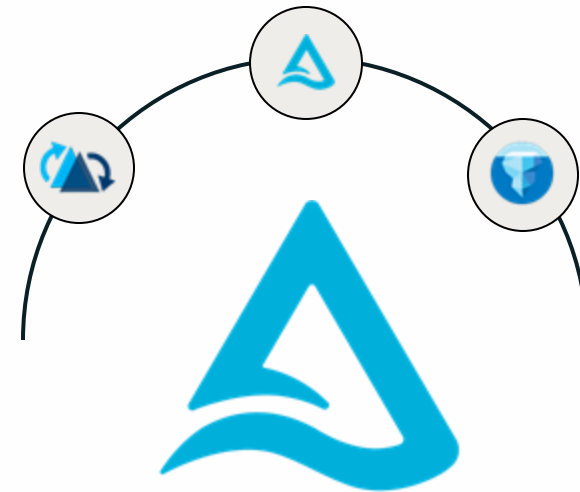
Parquet

Connector Ecosystem



Delta Lake with UniForm

How does it work?



Delta Lake With UniForm



Metadata

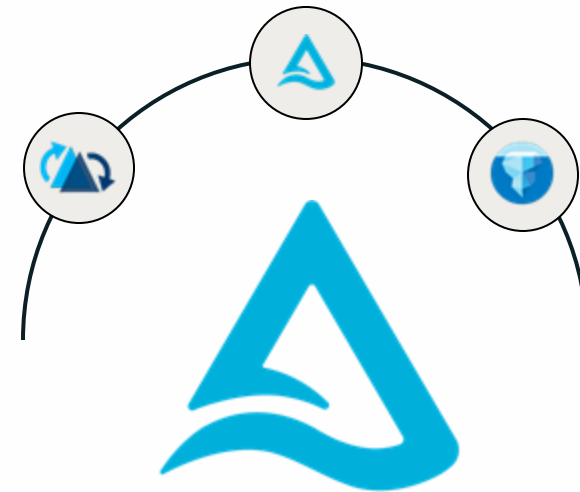
Used for transactional source of truth, concurrency control, etc.

- ✓ Metadata automatically generated to make Delta accessible as Iceberg/Hudi readers



Delta Lake with UniForm

How does it work?



Delta Lake With UniForm

Metadata

Used for transactional source of truth, concurrency control, etc.

Data

All formats use Parquet!

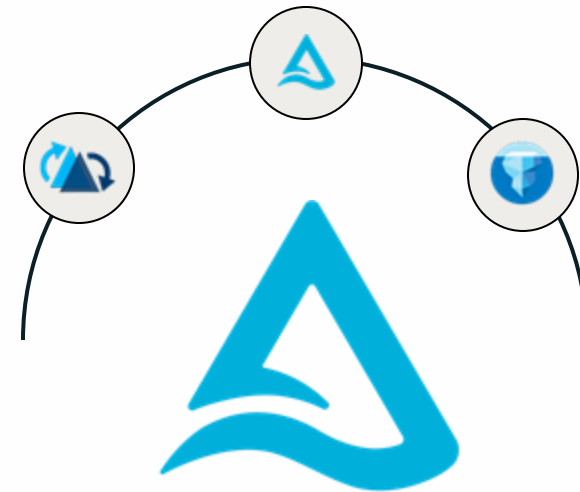


- ✓ Metadata automatically generated to make Delta accessible as Iceberg/Hudi readers
- ✓ Parquet files remain the same



Delta Lake with UniForm

How does it work?



Delta Lake With UniForm

Metadata

Used for transactional source of truth, concurrency control, etc.

Data

All formats use Parquet!

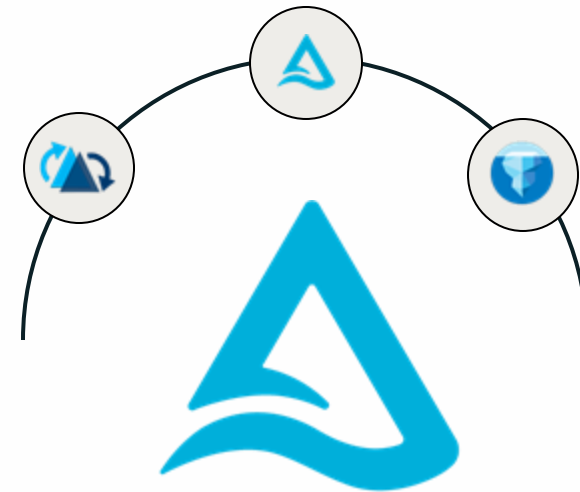


- ✓ Metadata automatically generated to make Delta accessible as Iceberg/Hudi readers
- ✓ Parquet files remain the same
- ✓ **Metadata is co-located with data**



Delta Lake with UniForm

How does it work?



Delta Lake With UniForm



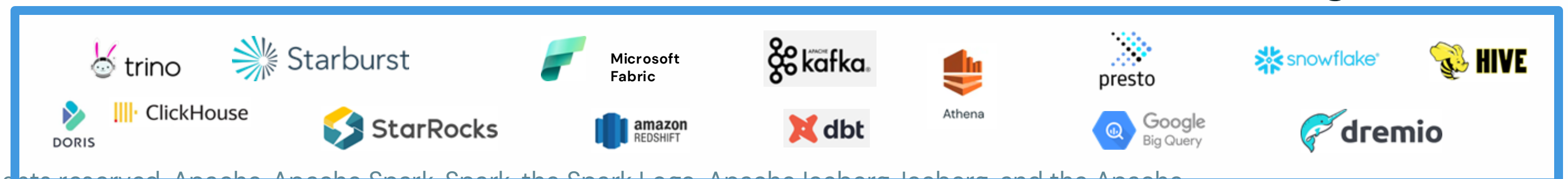
Metadata

Used for transactional source of truth, concurrency control, etc.

Data

All formats use Parquet!

Connector Ecosystem



- ✓ Metadata automatically generated to make Delta accessible as Iceberg/Hudi readers
- ✓ Parquet files remain the same
- ✓ Metadata is co-located with data
- ✓ Data stored in Delta can be read as if it were Iceberg or Hudi





Data Ingestion and Transformation

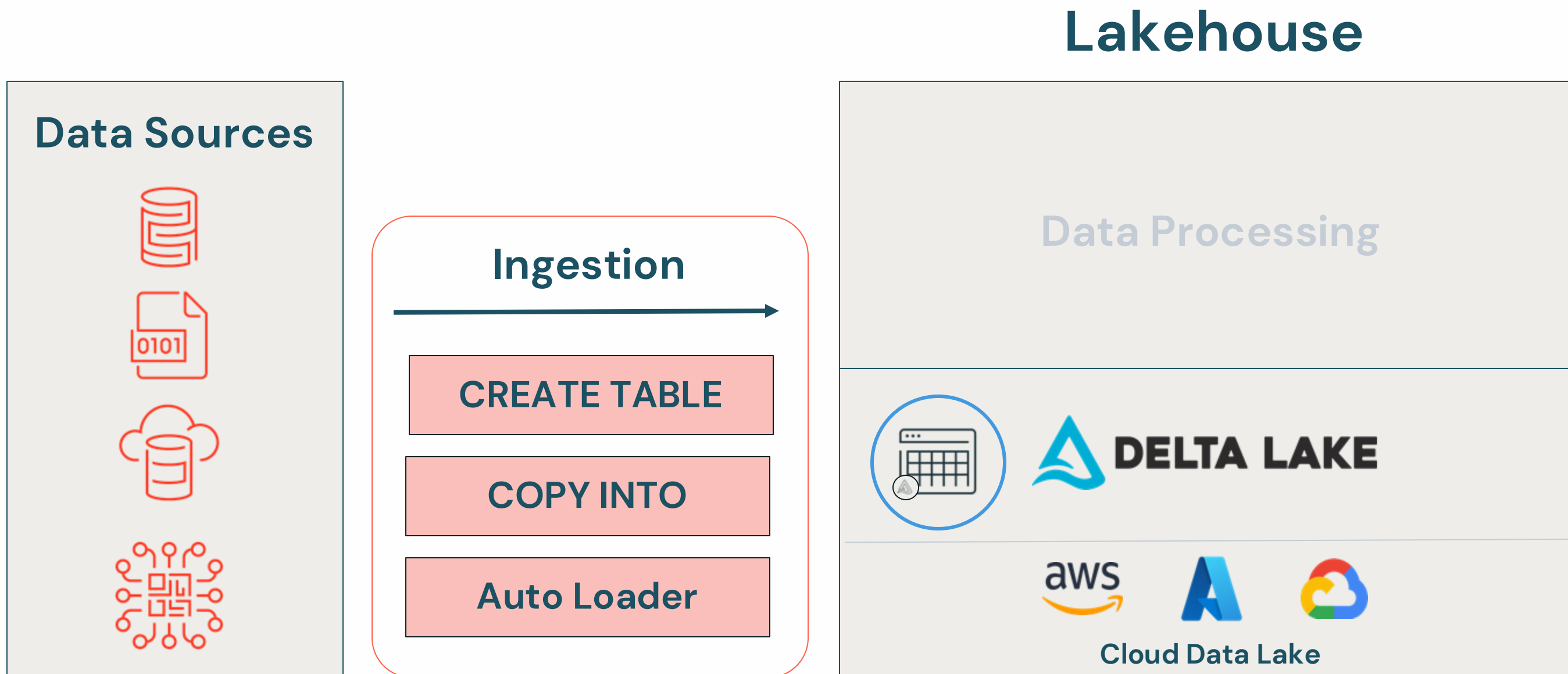
LECTURE

Data Ingestion Techniques Overview



Delta Lake Data Ingestion Techniques

Overview



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

Auto Loader



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

Auto Loader

```
CREATE TABLE mydeltatable  
USING DELTA  
AS  
<query>
```

USING DELTA is optional.

- CREATE TABLE AS (CTAS) creates a table by selecting data from an existing table or data source
- Default table format is **Delta**



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

Auto Loader

- The UPLOAD UI provides a **point-and-click interface** to upload files and create tables
- Allows you to upload CSV, TSV, JSON, Avro, Parquet, or text files
- The UI also enables you to upload files directly to a Unity Catalog volume



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

Auto Loader

```
COPY INTO mydeltatable  
FROM 'your-path'  
FILE_FORMAT = 'format'  
FILE_OPTIONS = ('format-options')
```

- Loads files from a **file location** into a Delta table
- Supports **various file formats** and cloud storage locations
- Automatically can handle **schema changes**.
- **Idempotent** operation—Already-loaded files in the source are skipped



Delta Lake Data Ingestion Techniques

Overview

CREATE TABLE

Upload UI

COPY INTO

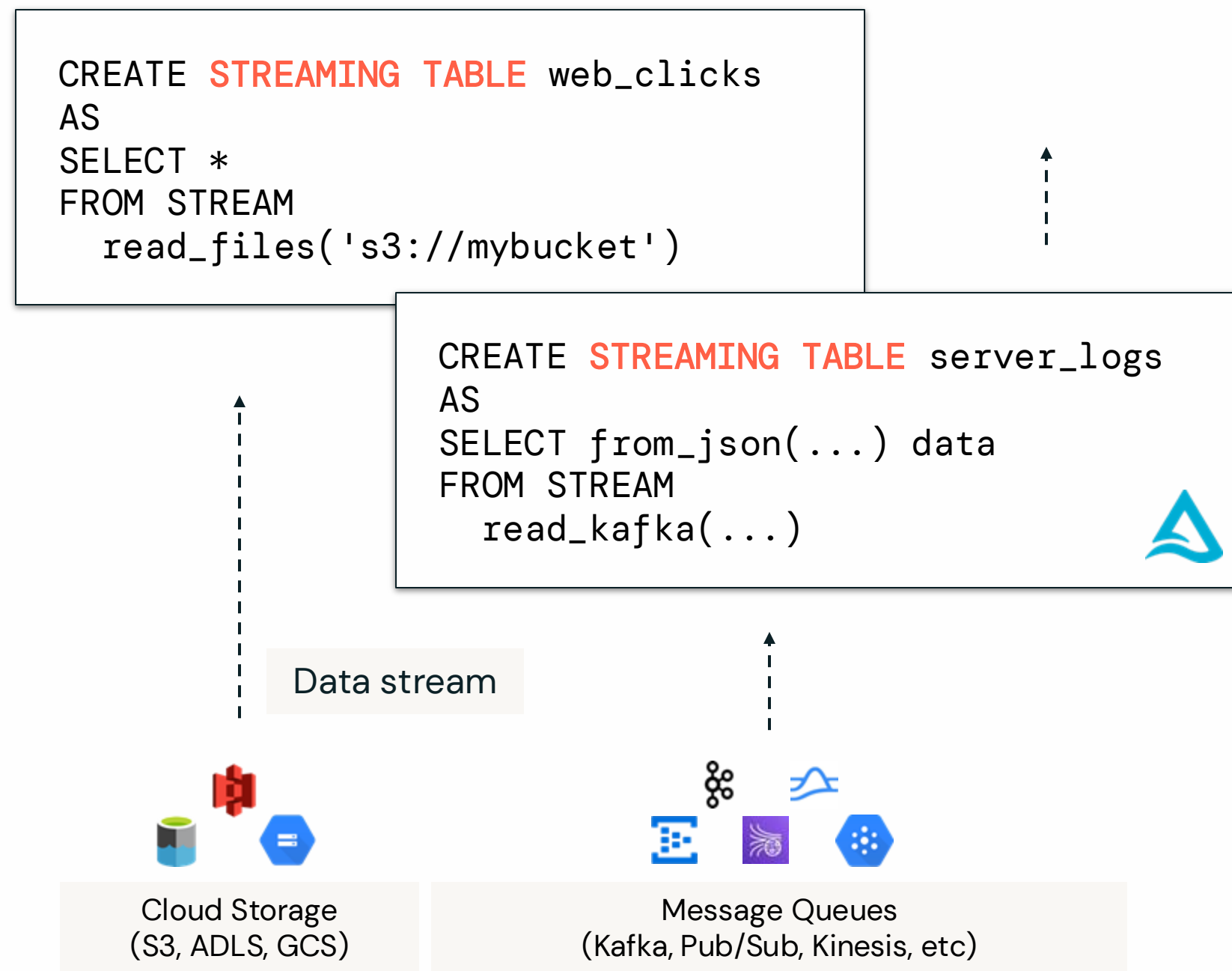
Auto Loader

- Auto Loader **incrementally (streaming) processes** new data files as they arrive in cloud storage
- Automatically **infers schemas** and accommodates **schema changes**
- Includes a **rescue data** column for data that does not adhere to the schema



Streaming Tables in Delta Lake

Real-Time and Incremental Data Processing Made Simple



Benefits:

1. **Unlock real-time use cases.** Ability to support real-time analytics/BI, machine learning and operational use cases with streaming data.
2. **Better scalability.** More efficiently handle high volumes of data via incremental processing vs large batches.
3. **Enable more practitioners.** Simple SQL syntax makes data streaming accessible to all data engineers and analysts.





Databricks LakeFlow Connect

Native ingestion connectors for the Data Intelligence Platform

Efficient ingestion pipelines

Simple setup and maintenance

Unified orchestration, observability, and governance

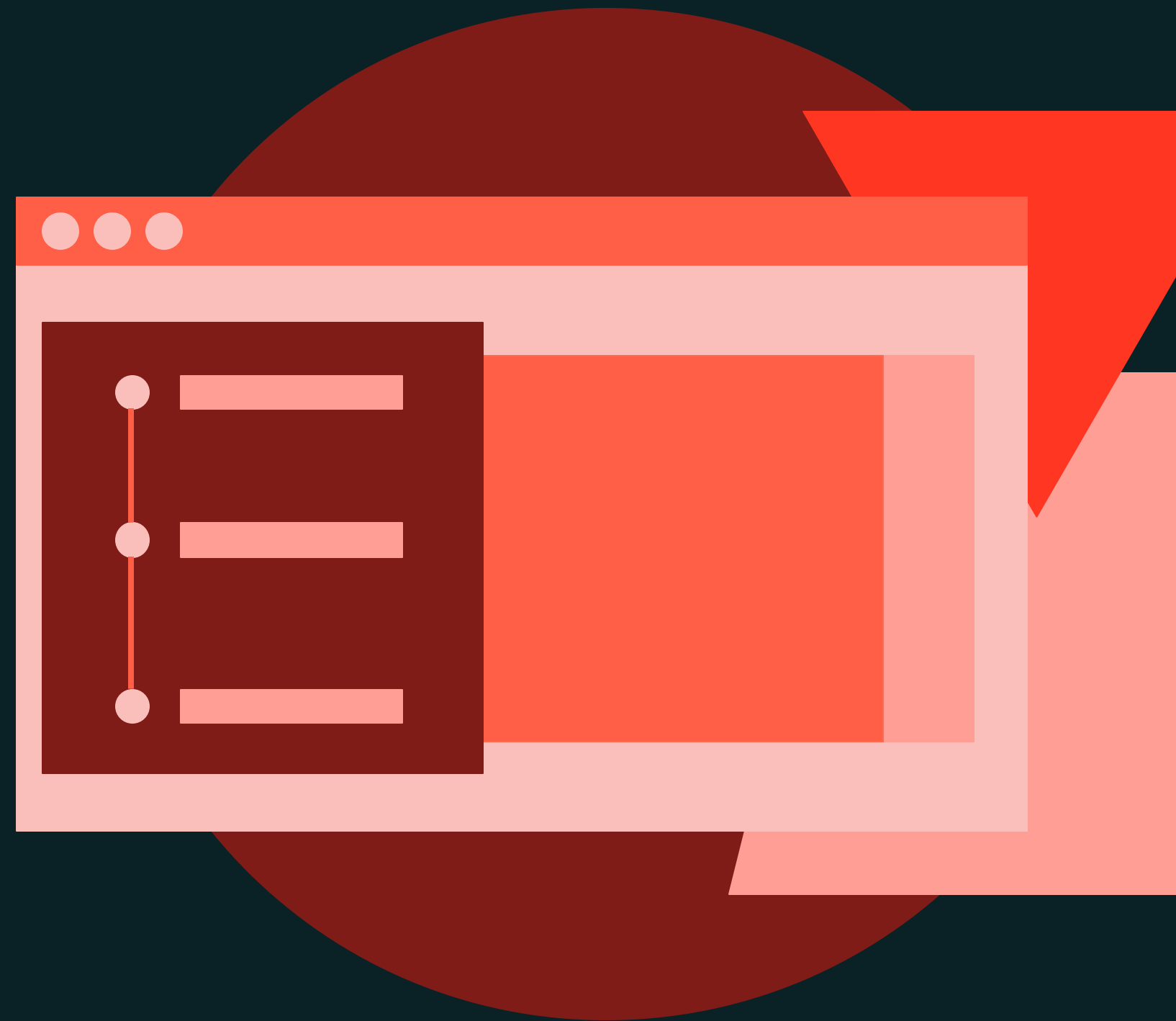




Data Ingestion and Transformation

DEMONSTRATION

Data Ingestion Techniques



High Level Steps

- Ingest data into Delta Lake using CREATE TABLE AS SELECT.
- Incrementally load data using COPY INTO.
- Use the Databricks Upload UI for data ingestion.
- Set up real-time ingestion pipelines using Auto Loader.
- Understand the role of Lakeflow Connect in data ingestion workflows.





Data Ingestion and Transformation

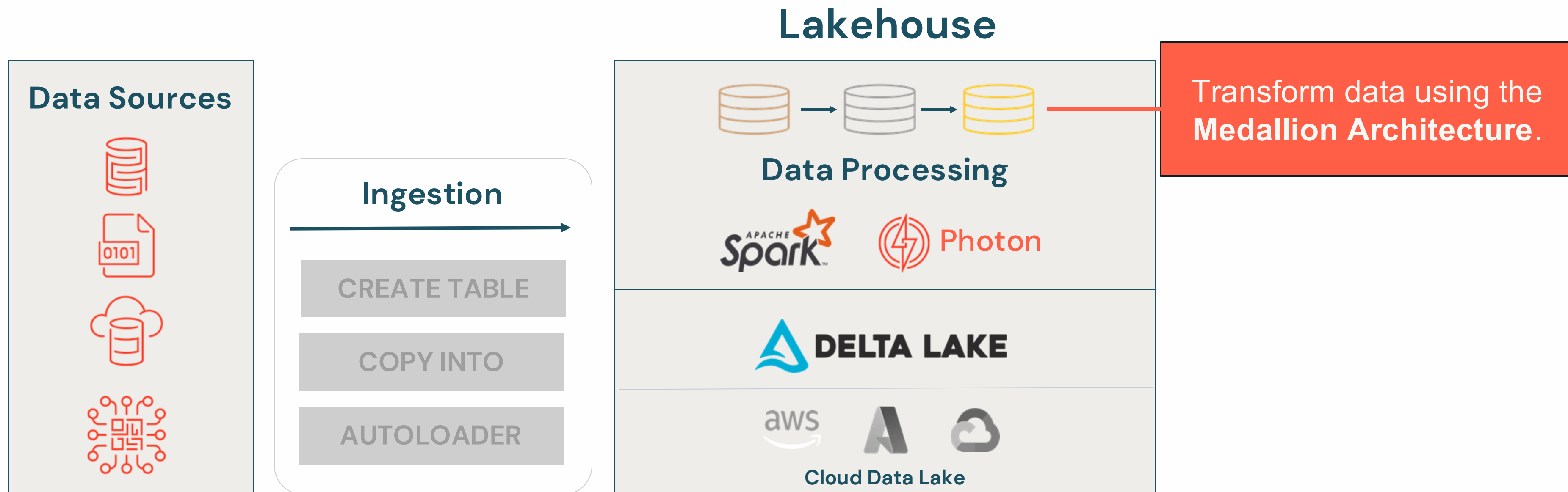
LECTURE

Data Transformation



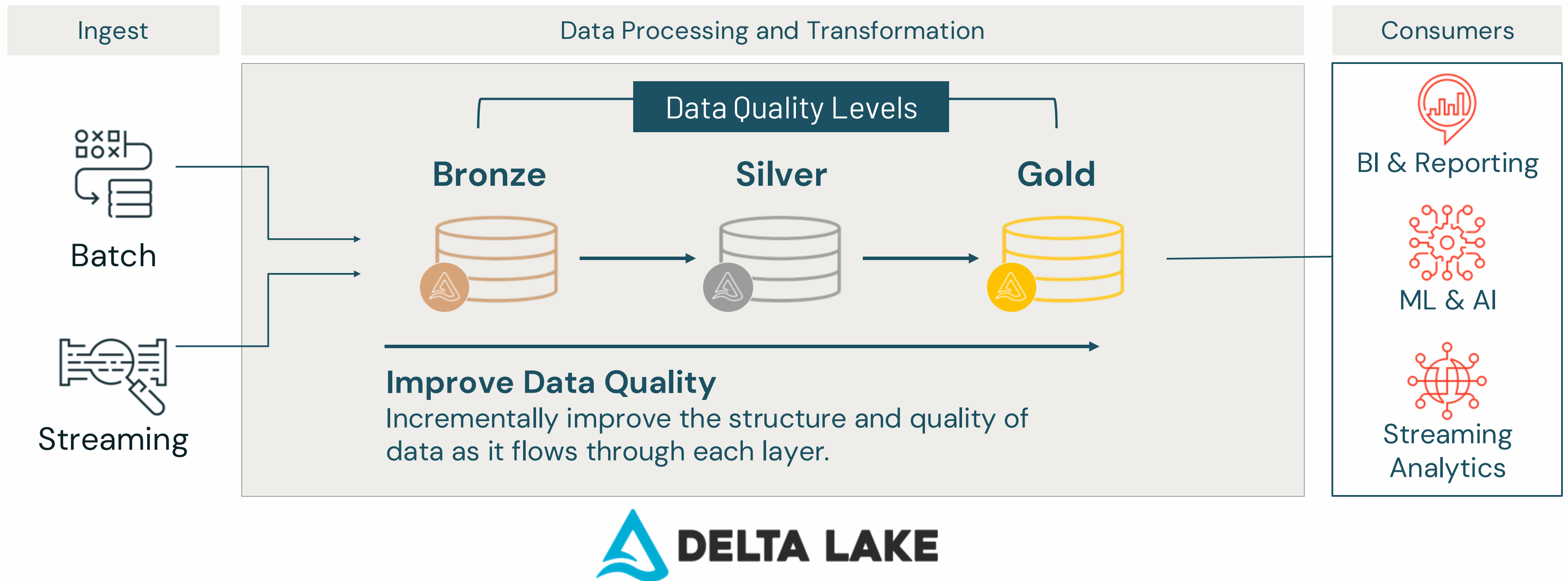
Data Transformation Overview

Medallion Architecture (Multi Hop)



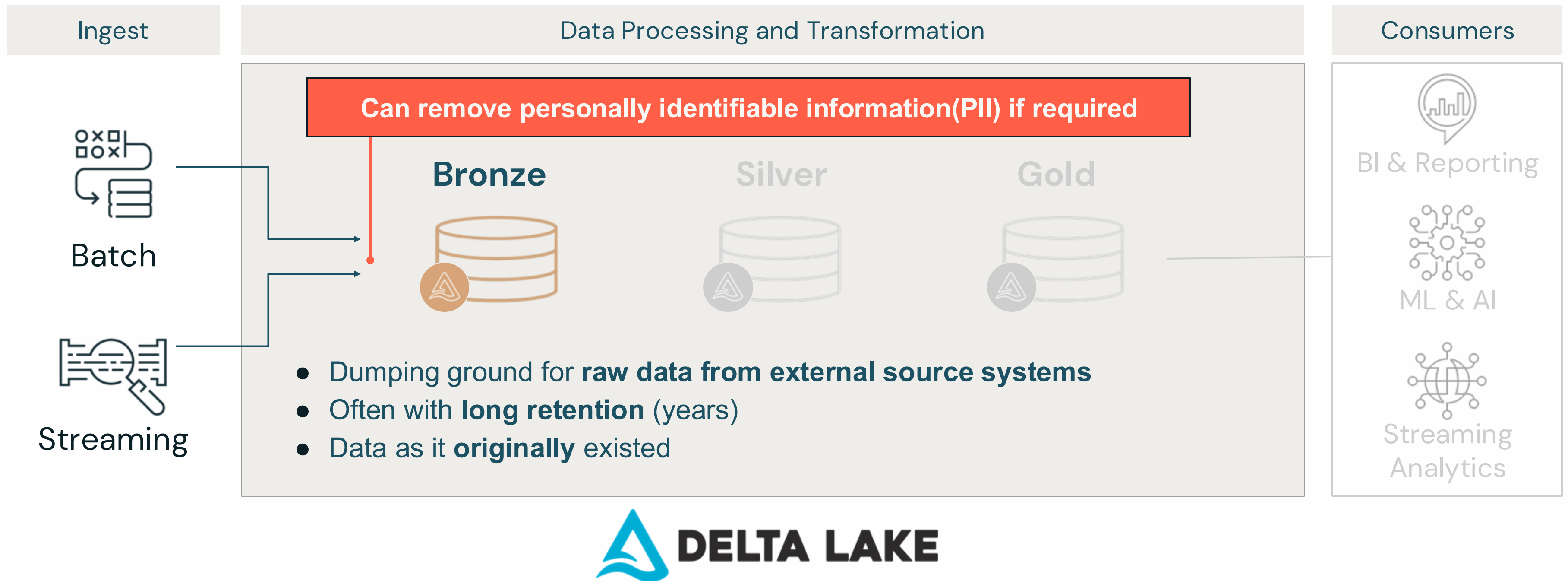
Data Transformation Overview

Medallion Architecture (Multi Hop)



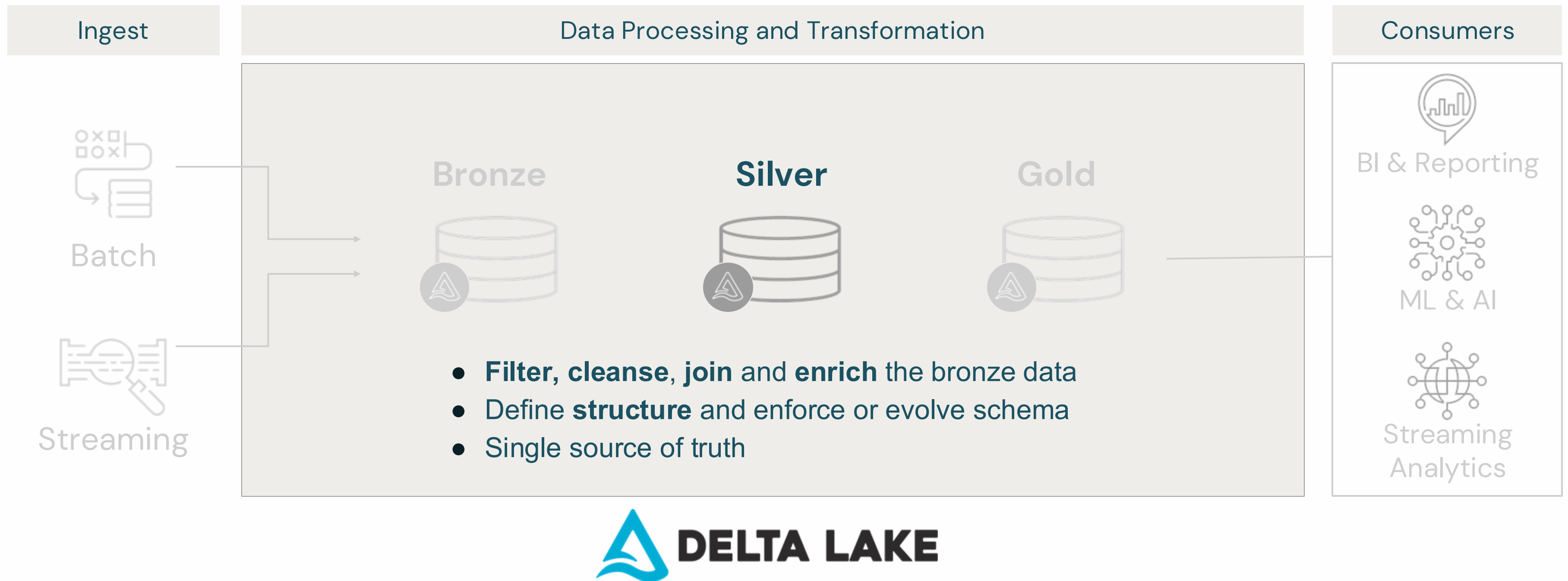
Data Transformation Overview

Medallion Architecture (Multi Hop)



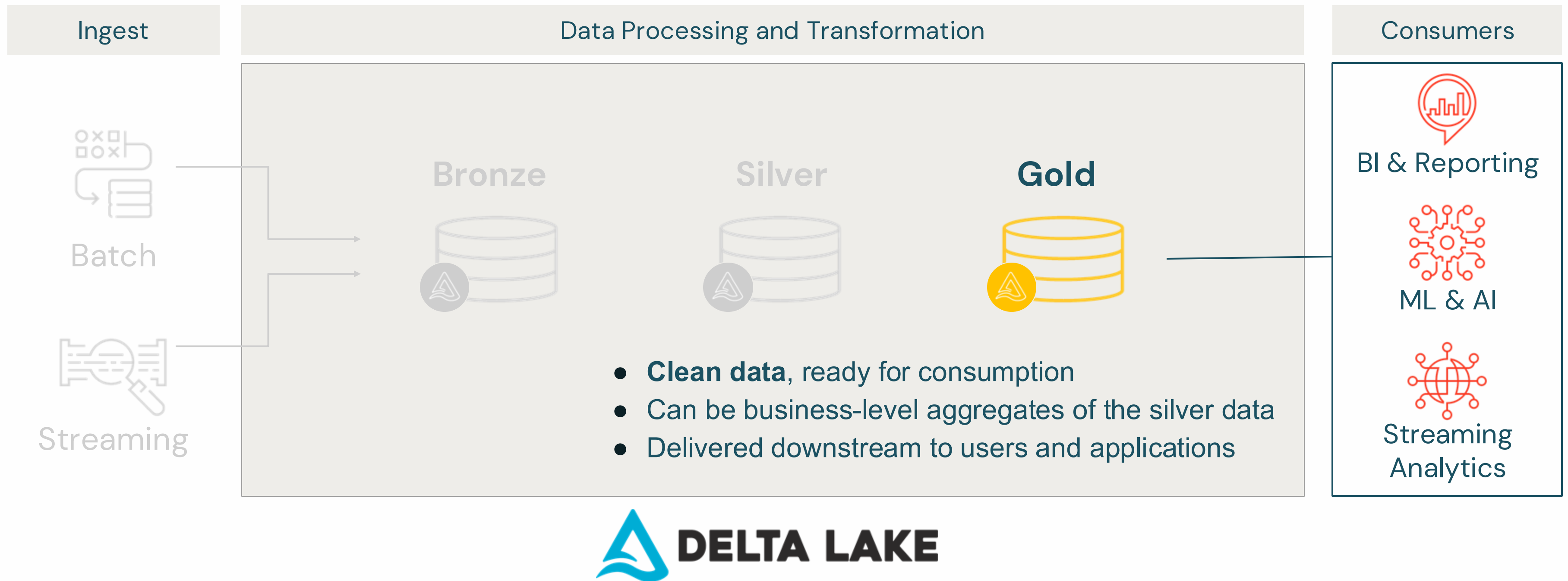
Data Transformation Overview

Medallion Architecture (Multi Hop)



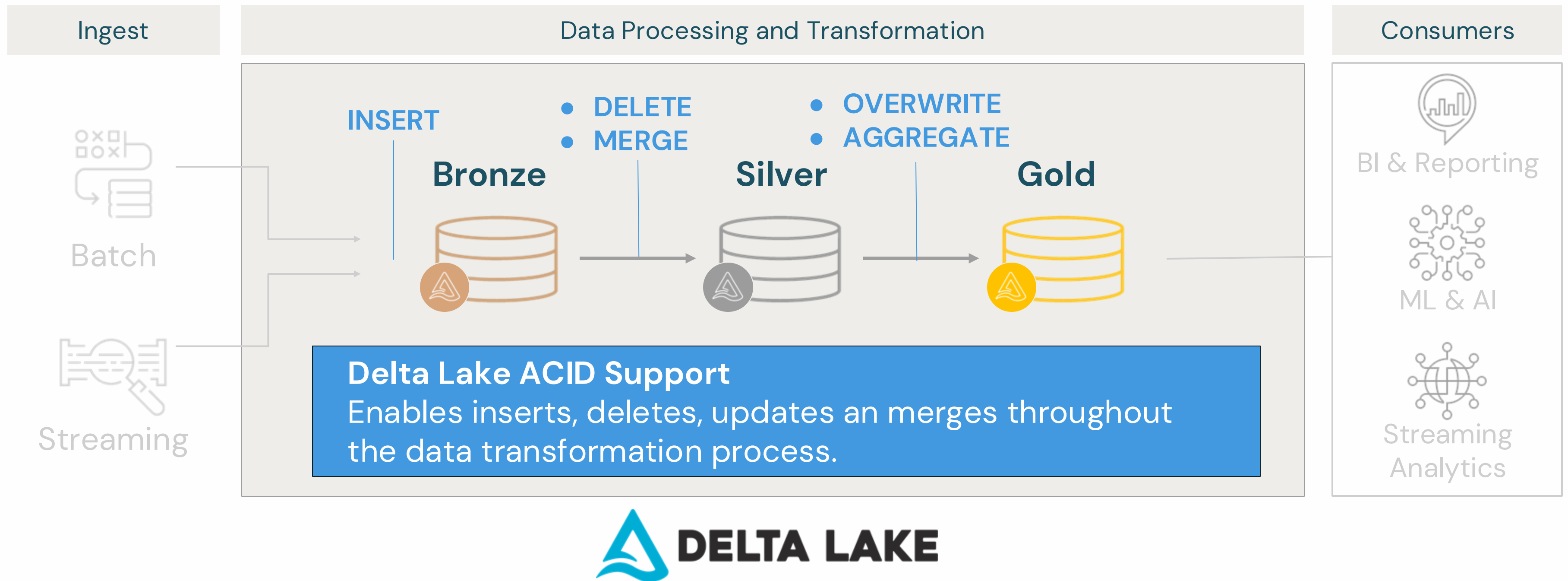
Data Transformation Overview

Medallion Architecture (Multi Hop)



Data Transformation Overview

Medallion Architecture (Multi Hop)

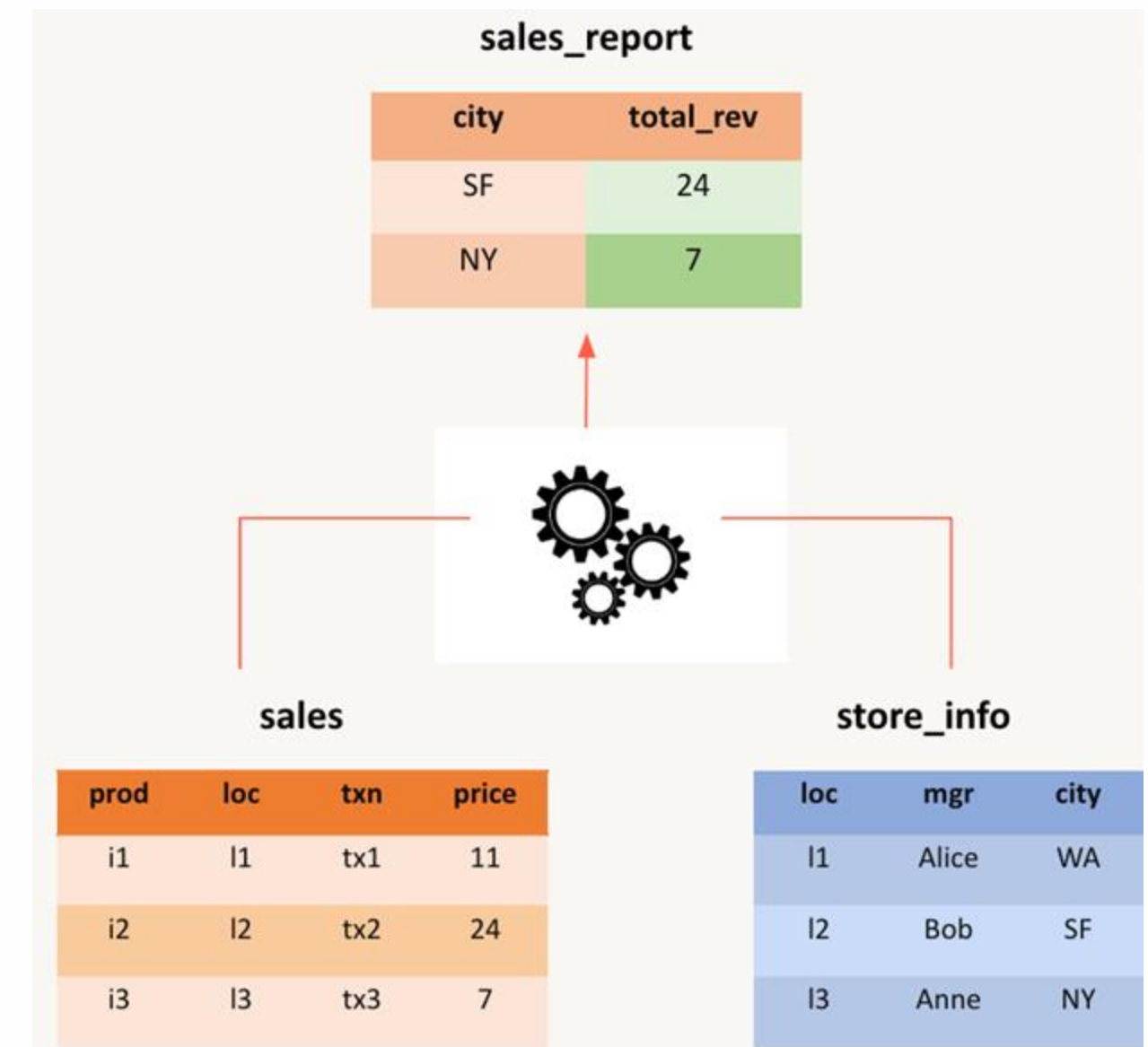


Materialized Views

Speed up queries with pre-computed results

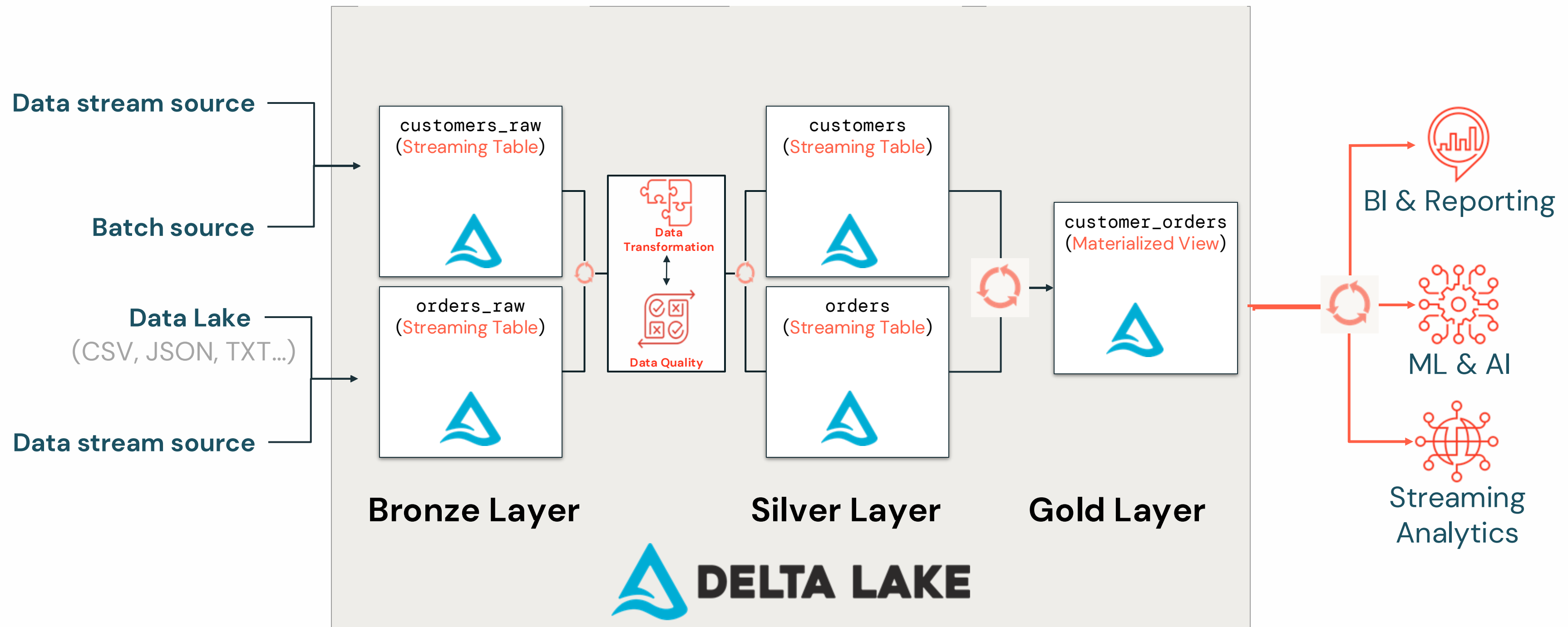
Accelerate end-user queries and reduce infrastructure costs with efficient, incremental computation

- Accelerate **BI dashboards** and **ETL queries**
- **Streaming**: build MVs on top of live tables
- **Easy ELT**: Simplify reporting by cleaning, enriching, denormalizing the base tables
- **Data Sharing & Access Control**: control what info can be seen by internal and external users and organizations



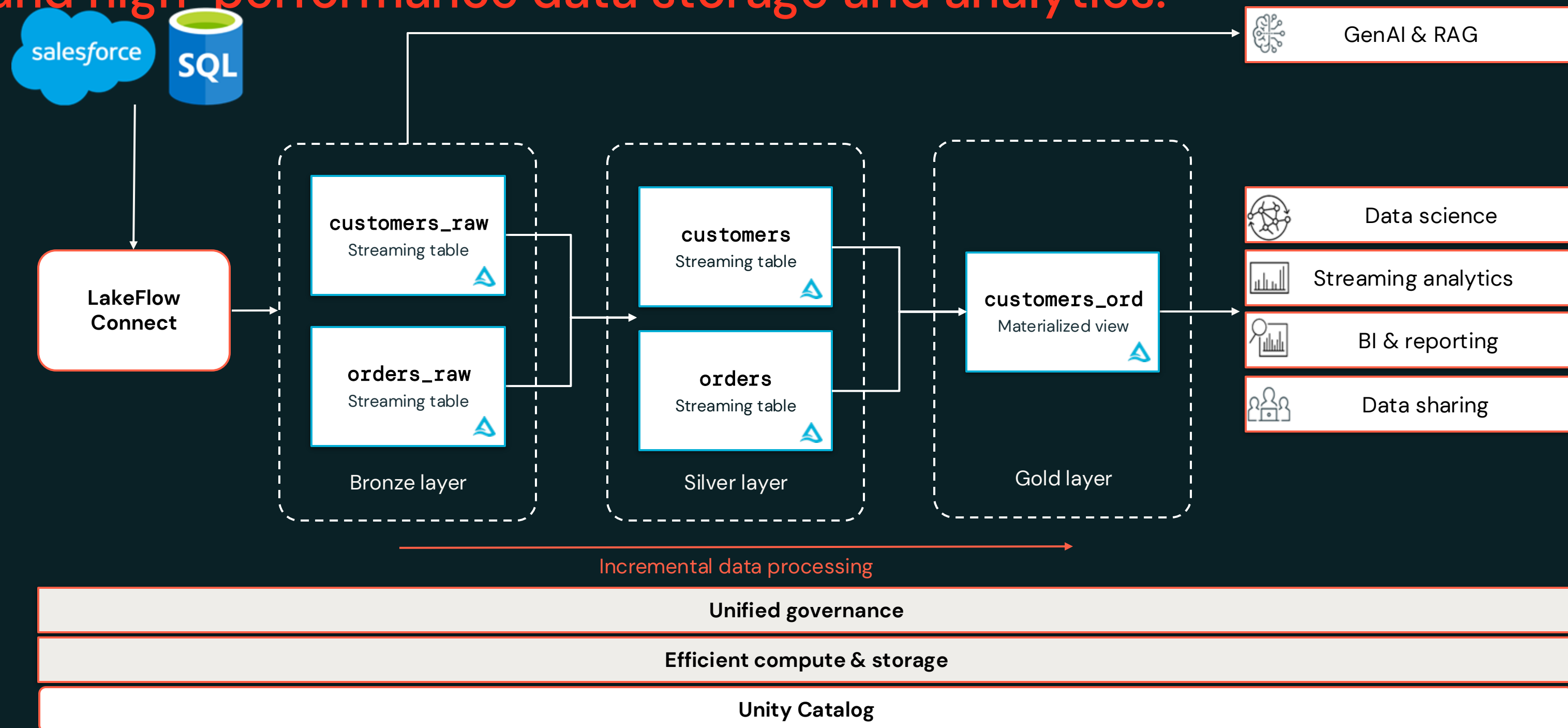
Constructing a Data Transformation Pipeline

Real World Architecture is Typically More Complex



LakeFlow Connect in Action

Medallion Architecture powered by Delta Lake ensures reliable, scalable, and high-performance data storage and analytics.



Overview of Data Modeling with Databricks

Introduction to Data Modeling in Databricks

Data Modeling enables structured, high-quality, and consistent data to drive accurate analysis and informed decision-making.

- **Support for Various Modeling Approaches:**

- Design scalable models using Star Schema, Snowflake Schema, or Data Vault.

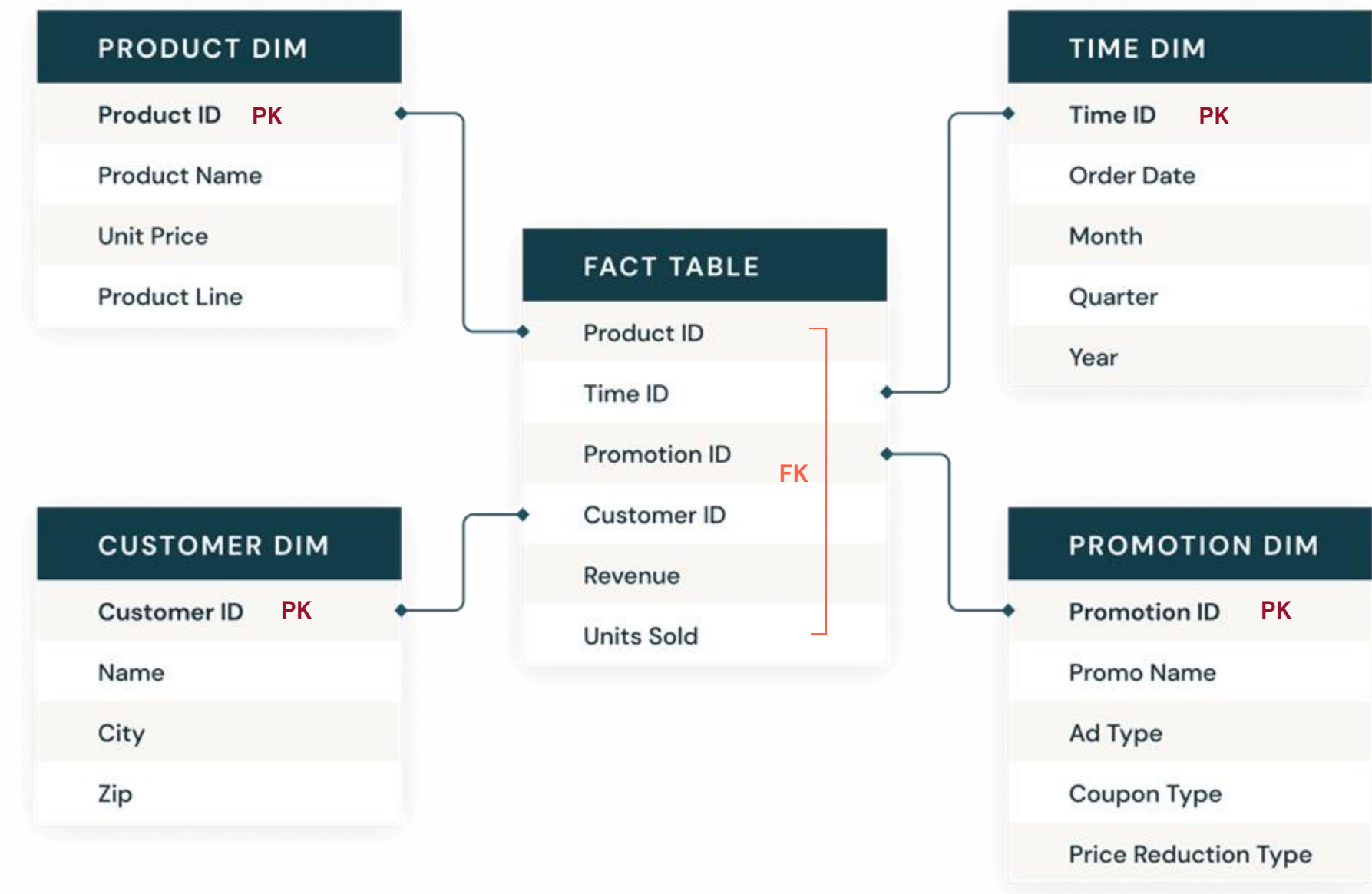
- **Primary Key/Foreign Key Constraints:**

- Ensure referential integrity in Delta tables for reliable relationships.

- **Surrogate Keys:**

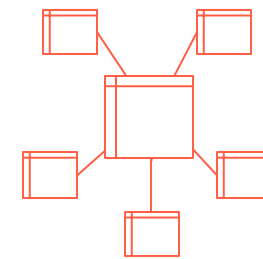
- Enable efficient data management with auto-incrementing IDs.

Star schema



Modern Data Modeling Approaches

Key Modeling Approaches:



Top-Down Approach (Inmon):

- Relational Modeling with normalized data as the core of the Data Warehousing
- Data marts (often dimensional or denormalized models)

Bottom-Up Approach (Kimball):

- Dimensional Modeling
- Denormalized data model, built as a star or snowflake schema: central fact tables surrounded by dimension tables

Data Vault (Linstedt):

The Data Vault model is based on three basic entity types:

- **Hubs** separate core business concepts
- **Links** store relationships between business concepts
- **Satellites** store the attributes of a business concepts or relationships

Unique Databricks Features:

- Seamless integration with Delta Lake for schema management.
- Identity columns for generating Surrogate Keys.

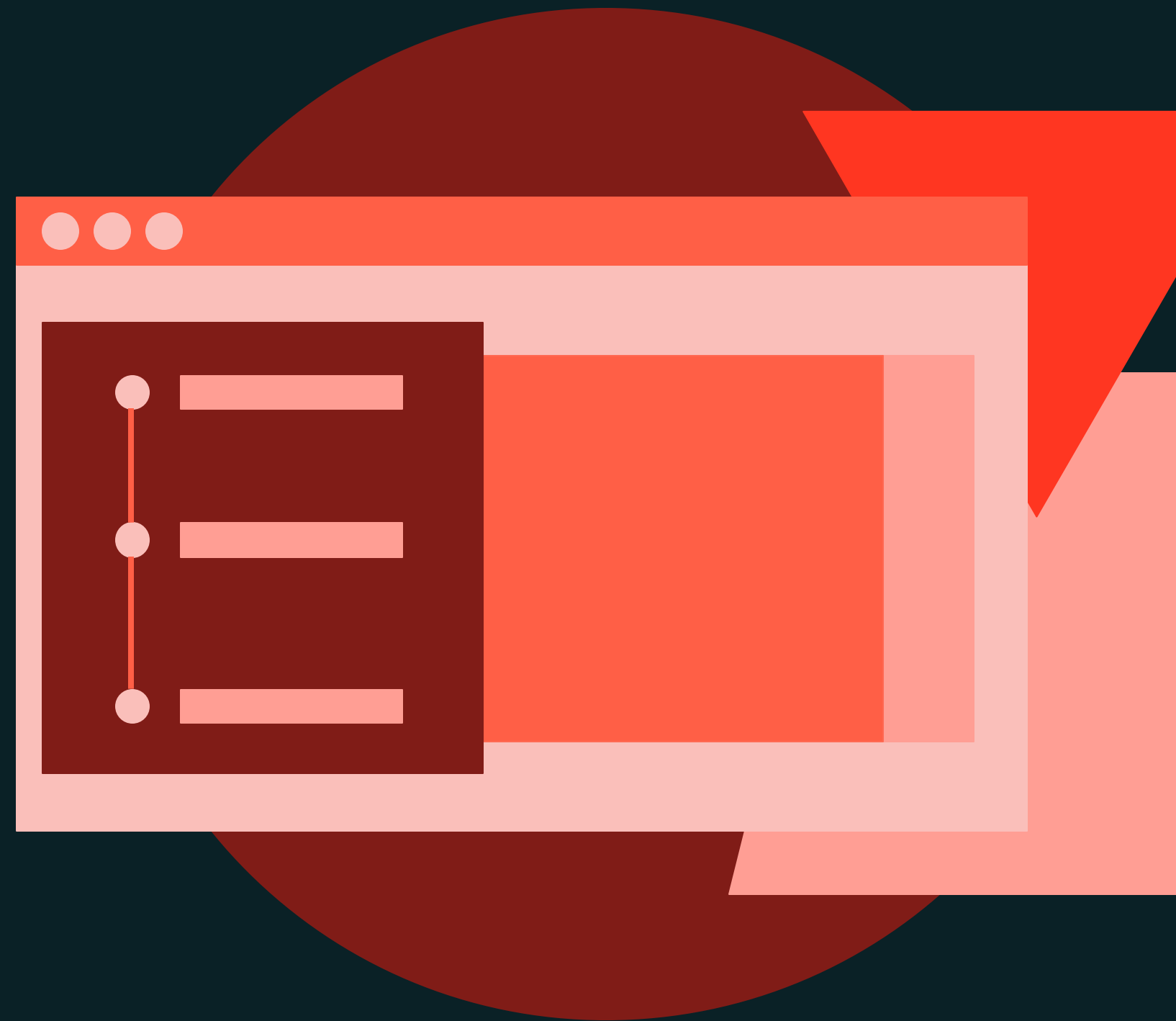




Data Ingestion and Transformation

DEMONSTRATION

Exploring Data Transformation in Databricks



High Level Steps

- Create a multi-hop transformation pipeline using the Medallion Architecture.
- Use Streaming Tables and Materialized Views to transform raw data into meaningful insights.
- Demonstrate how to move data from the Bronze layer to the Silver and Gold layers while maintaining data quality.





Data Orchestration and Querying Capabilities

Get Started with Databricks for Data Warehousing



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Objectives

- Explain how Databricks support data orchestration needs within the platform.



Agenda

Data Orchestration and Querying Capabilities	Time	Lecture	Demo	Lab
Orchestration in Databricks	5 mins	✓		
Setting Up and Managing Serverless Lakeflow Jobs	15 mins		✓	
Databricks Querying Capabilities	5 mins	✓		





Data Orchestration and Querying Capabilities

LECTURE

Orchestration in Databricks

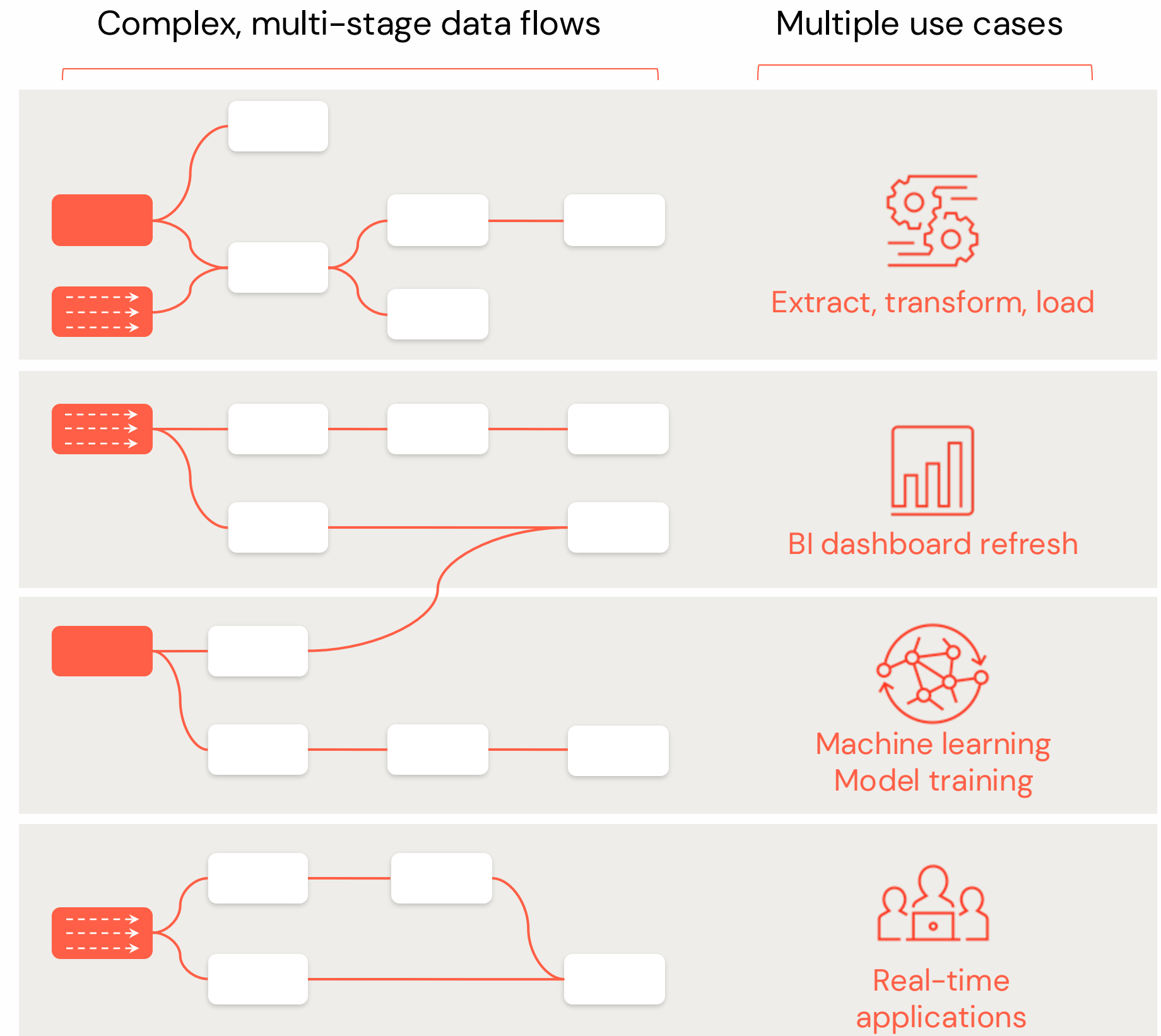


Modern Data Warehousing Requires Modern Orchestration

Orchestrating processes across all data, analytics and AI use cases is business critical

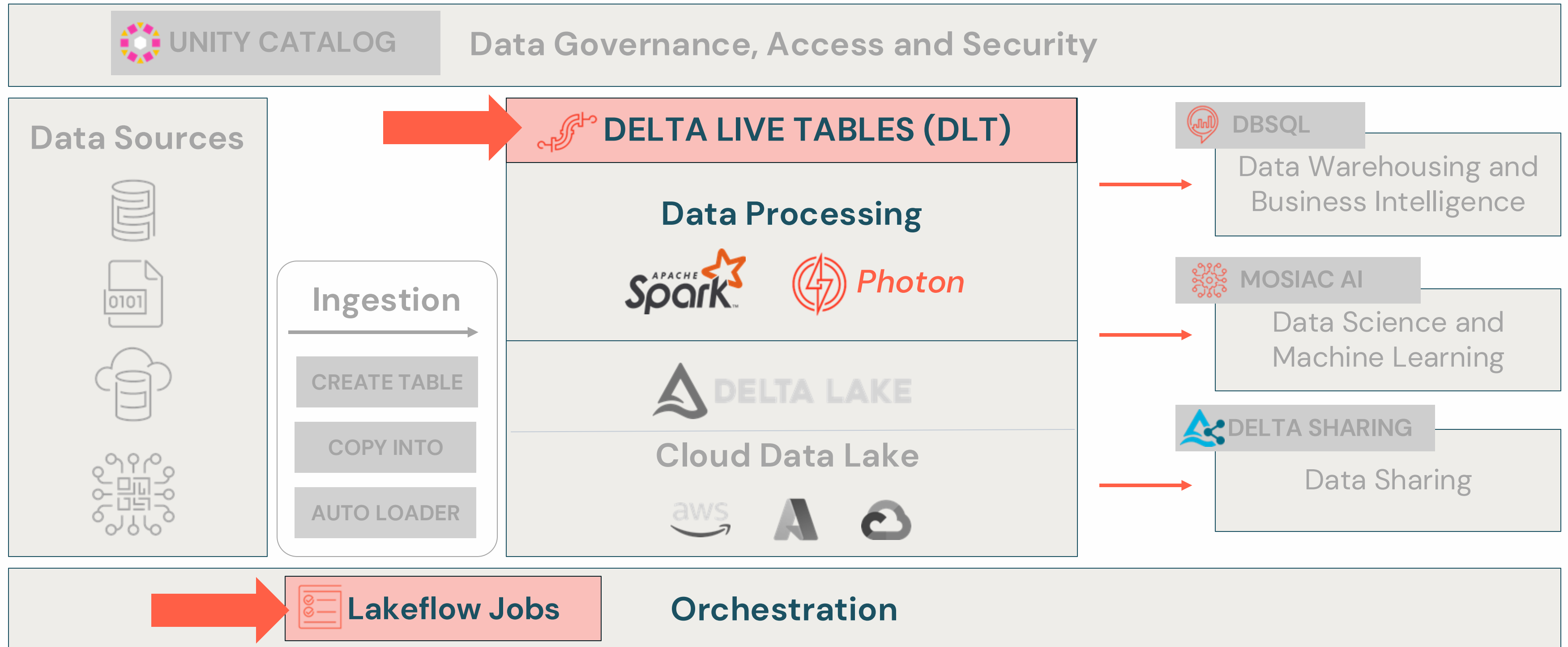
- Handles complex, multi-stage data flows.
- Manages multiple dependencies in Lakeflow Jobs.
- Supports diverse use cases, including:
 - Extract, transform, load (ETL).
 - BI dashboard refresh.
 - Machine learning model training.
 - Real-time applications.

Multiple data sources & triggers



Unified Orchestration Using Databricks

Overview



Databricks Lakeflow Jobs

Reliable orchestration for data, analytics, and AI



**Orchestrate
Anything
Anywhere**



**Simple
Lakeflow Jobs
Authoring**



**Deep Platform
Integration**



**Proven
Reliability**

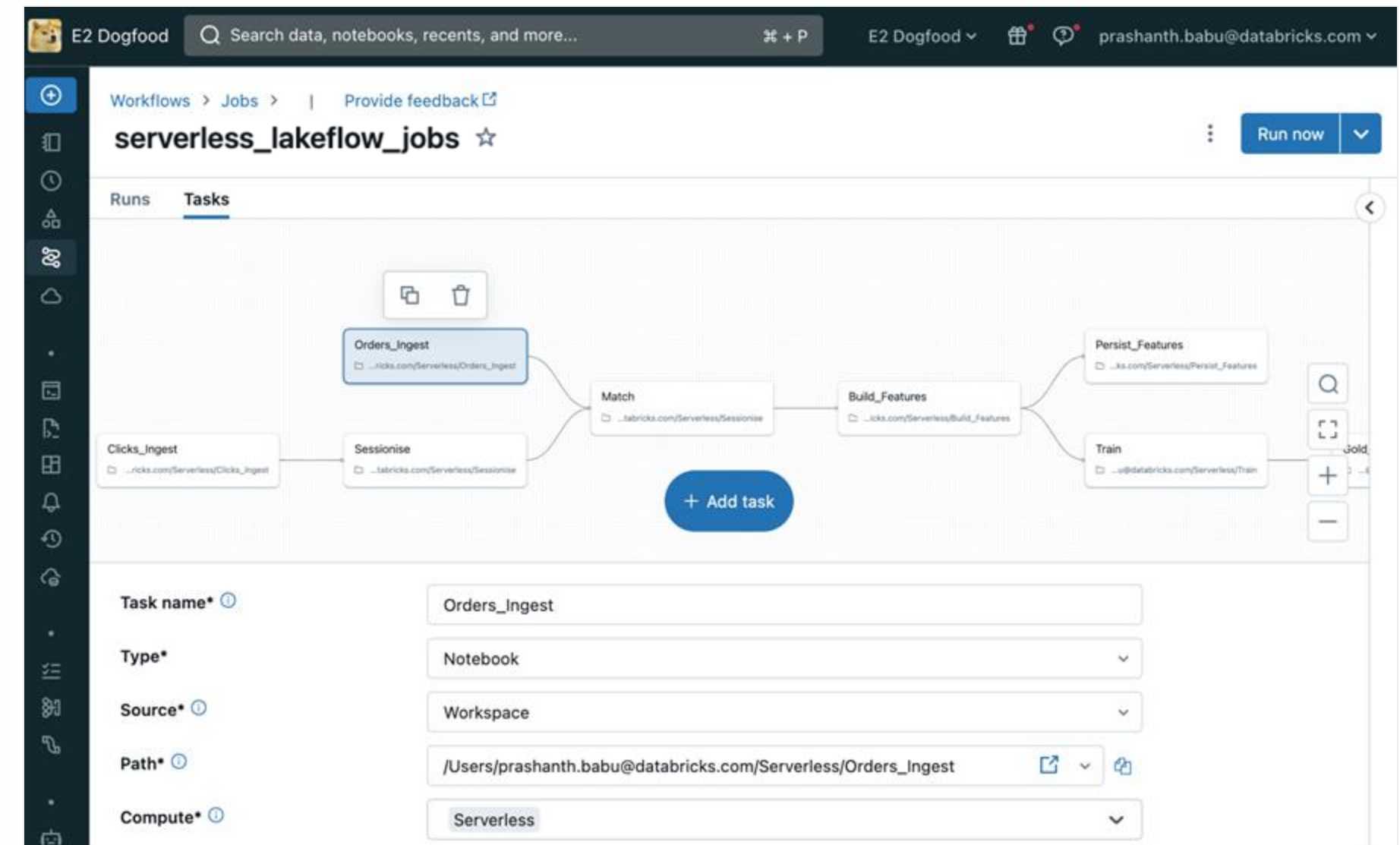


**Fully
Managed**



Serverless Lakeflow Jobs

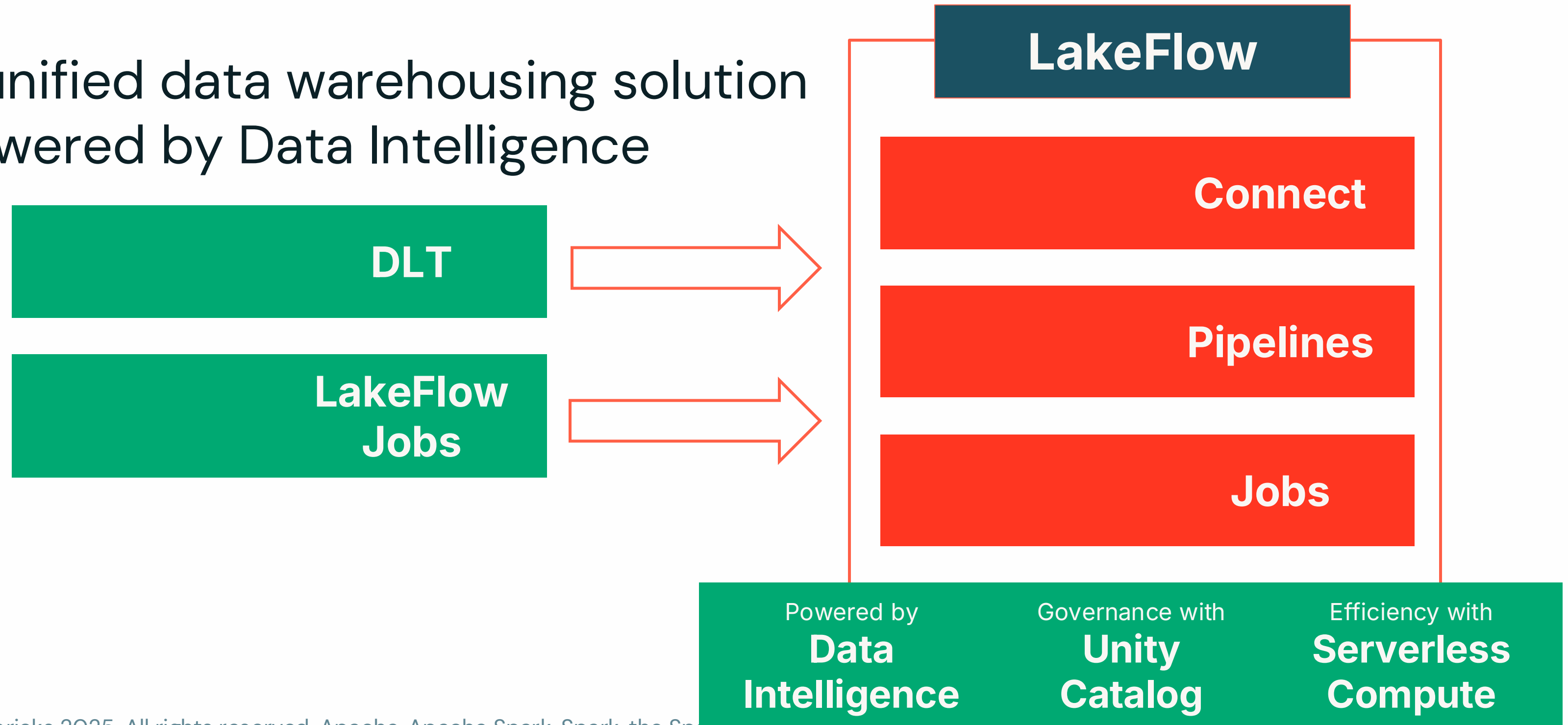
- Fully **Managed Infrastructure**, eliminates the human efforts to handle compute, providing fast analysis & cost savings.
- Advanced **Autoscaler** ensures efficient resource management
- **Elastic scaling** up and down is achieved using a warm pool of machines.



LakeFlow

The evolution of data warehousing on Databricks

A unified data warehousing solution powered by Data Intelligence





Data Orchestration and Querying Capabilities

DEMONSTRATION

Setting Up and Managing Serverless Lakeflow Jobs



High Level Steps

- Create a Lakeflow Job with SQL tasks for a Medallion Architecture pipeline.
- Set up task dependencies and implement conditional logic for lakeflow jobs control.
- Use the Lakeflow Job UI for monitoring and data lineage visualization to trace data transformations and dependencies.
- Configure error handling and retries for tasks.
- Schedule Lakeflow Job using manual triggers.
- Set up notifications for monitoring and analyze the execution history





Data Orchestration and Querying Capabilities

LECTURE

Databricks Querying Capabilities



Serverless Compute for Notebooks

With *Environments* for dependency management

01-IngestData Peo...

mainPython☆

Run allConnectedScheduleShare

FileEditViewRunHelpLa...New cell UI: ON

08:55 AM (4s)2Python

1df = spark.read.format("delta").load("dbfs:/databricks-datasets/learning-spark-v2/people/people-10m.delta")

2

3# Write the data to a Delta table (Delta format is default)

4df.write.format("delta").mode("overwrite").saveAsTable("people")

Hide performance

View all in query history

	Spark state...	Started At	Duration	Rows read	Bytes read	Bytes written
✓ L4	df.w...	2024-04-16 08...	3 s 718 ms	10,000,018	211.00 MB	210.95 MB

Fast and easy to use

- Starts in seconds

Portable, reusable environments

Automatically cached

Base Python and package versions

Customizable by users or admins

Shared Autos...ScheduleShare

EnvironmentsEdit

Channel: Current

Client: 1

Base EnvironmentTeam AWorkspace default

Packages (150)

Filter

Type: Notebook

Name	Version
Adjusttext	0.8
pip	22.3.1
dice	1.0.1
boto	4.5.6
foo	88.1.5

Environment Variables

Key	Value
workspace_name	prod
job	entitleme...

Properties

Key	Value
Timezone	PST
ANSI	Enabled
Catalog	users

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Query Insights

Extended to SQL Queries & DataFrames on Serverless Compute

- **Query History**

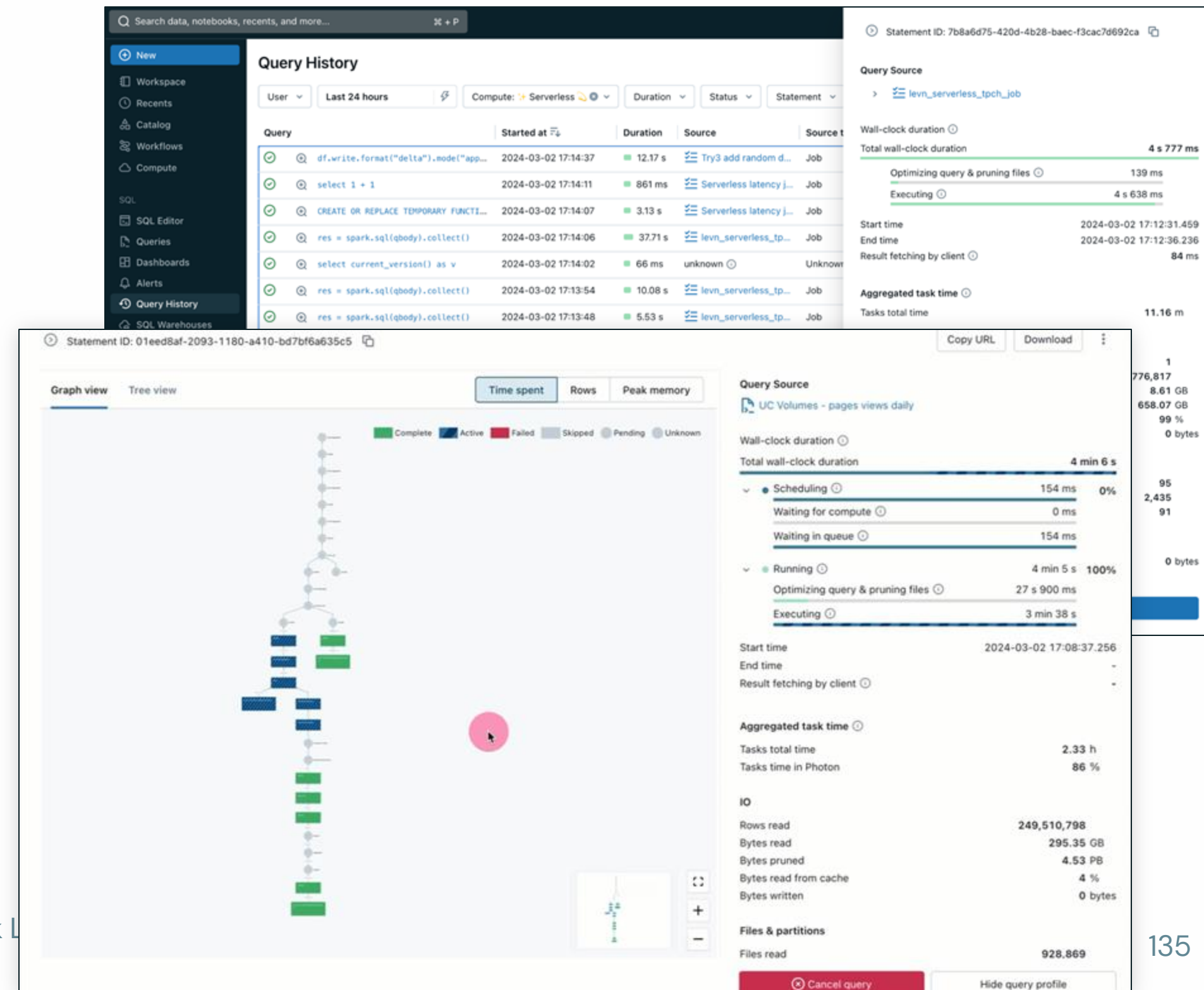
- Searchable and filterable overview of all current and past statements + cross-linking to source object that generated the query

- **Query Profile**

- Deep-dive into execution details to identify performance bottlenecks and optimization opportunities

- **Live Query Profile – coming soon**

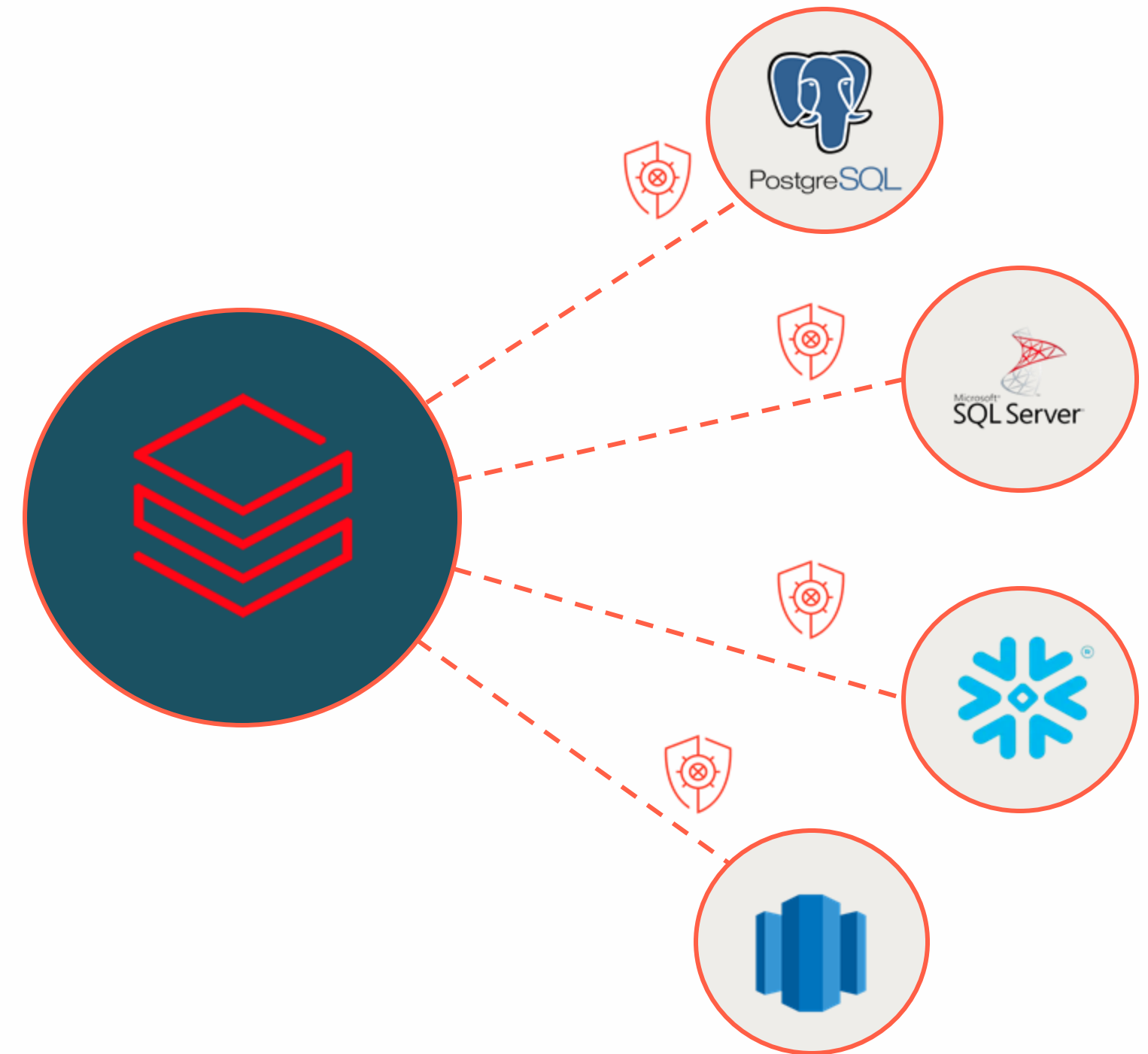
- Performance insights updated live as the query executes to enable immediate action



Query Federation

Discover, query, and govern your data wherever it lives

- Connect data sources to Unity Catalog and set policies across them
 - Table, Row, Col, tags, etc.
- Efficient optimization & caching across sources



AI Functions for DBSQL

Access Proprietary and Open Source LLMs directly within DBSQL

- Enable analysts to classify data with LLMs
- Extract actionable insights using AI language models.
- Generate product descriptions, summaries, and more by simply adding the ai_query function.

```
SELECT
  sku_id,
  product_name,
  ai_query(
    "my-external-openai-chat",
    "You are a marketing expert for a winter holiday promotion targeting GenZ. Generate a promotional text in 30 words mentioning a 50% discount for product: " || product_name
  )
FROM
  uc_catalog.schema.retail_products
WHERE
  inventory > 2 * forecasted_sales
```



Autocomplete Code or Queries

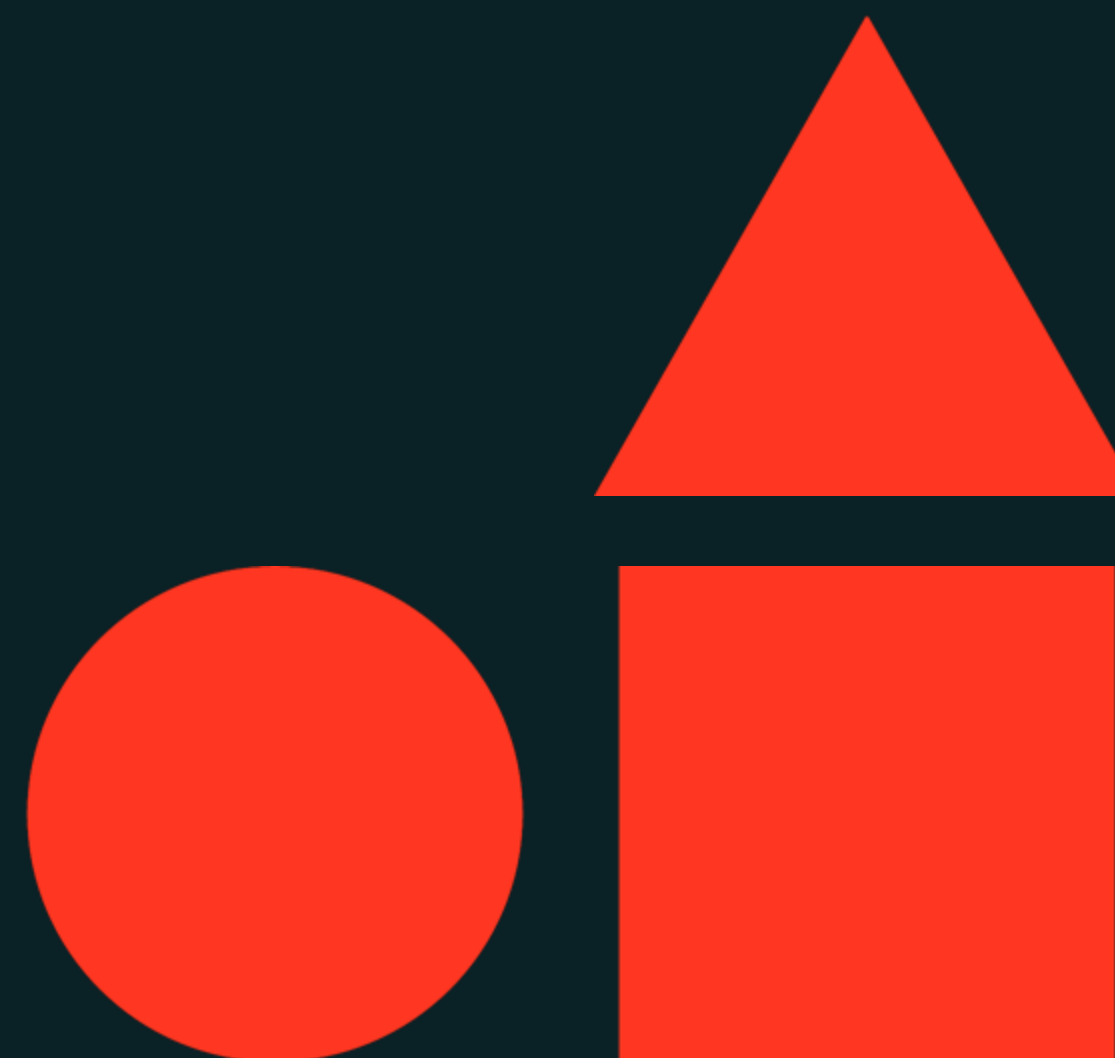
- Generate code from comments inside a notebook cell or SQL editor





Data Presentation

Get Started with Databricks for Data Warehousing



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Objectives

- Describe the impact of data warehousing work on Databricks AI/BI needs.
- Identify the AI supported features in the Databricks platform.



Agenda

Data Presentation	Time	Lecture	Demo	Lab
Introduction to AI/BI	5 mins	✓		
Creating a Dashboard in Databricks	5 mins		✓	
Creating Genie Spaces	10 mins		✓	
Data Warehousing Comprehensive Lab	30 mins			✓





Data Presentation

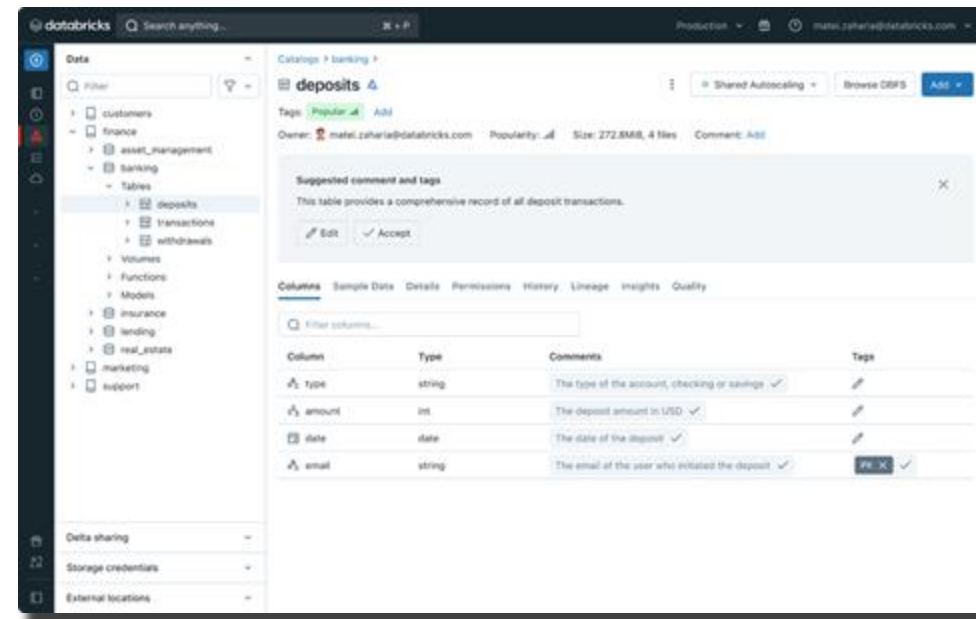
LECTURE

Introduction to AI/BI

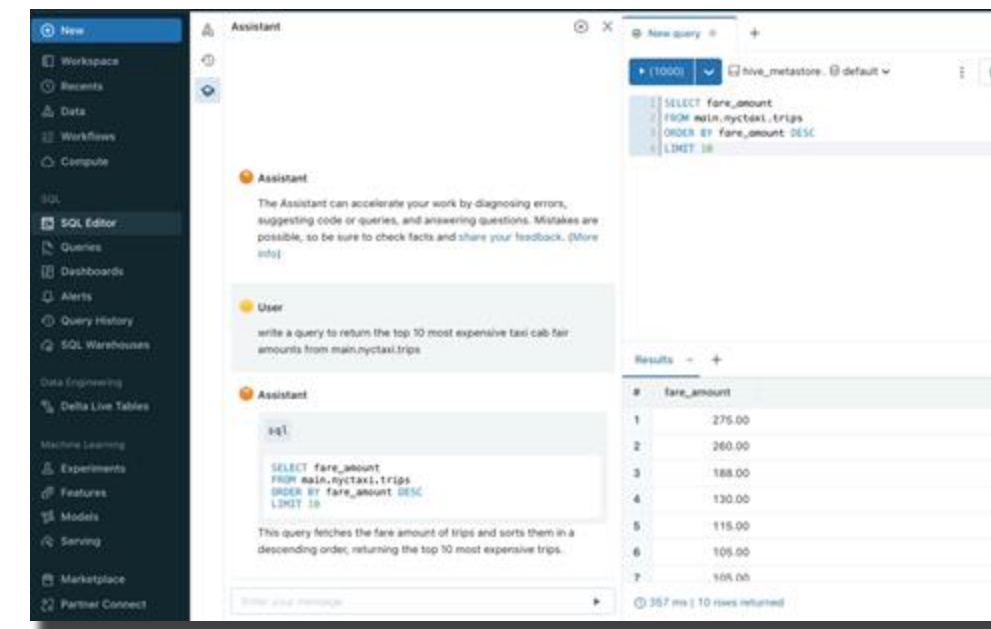


Infusing AI in our user experience

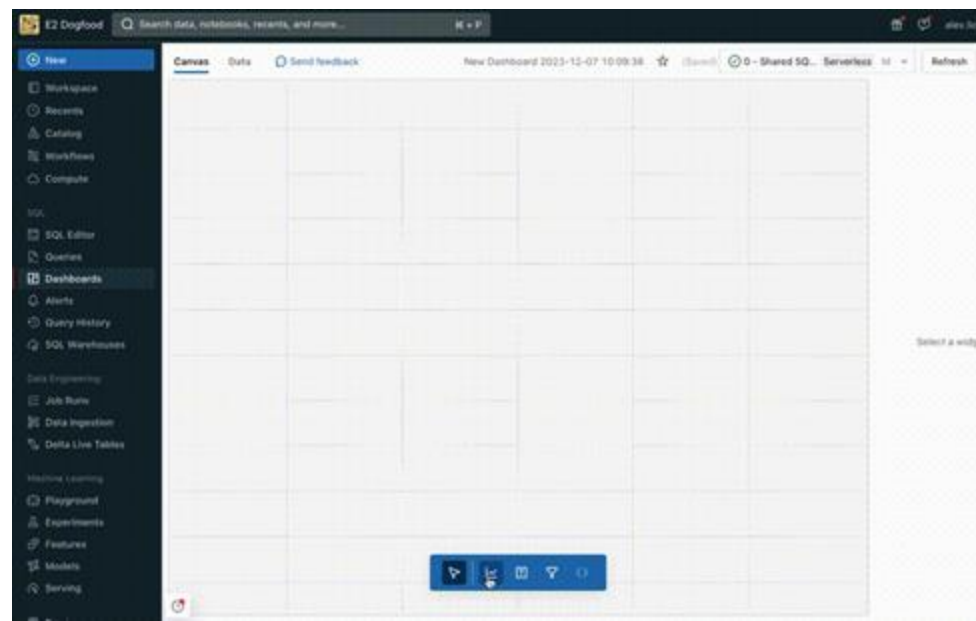
AI-generated
docs for easier
search and
discoverability



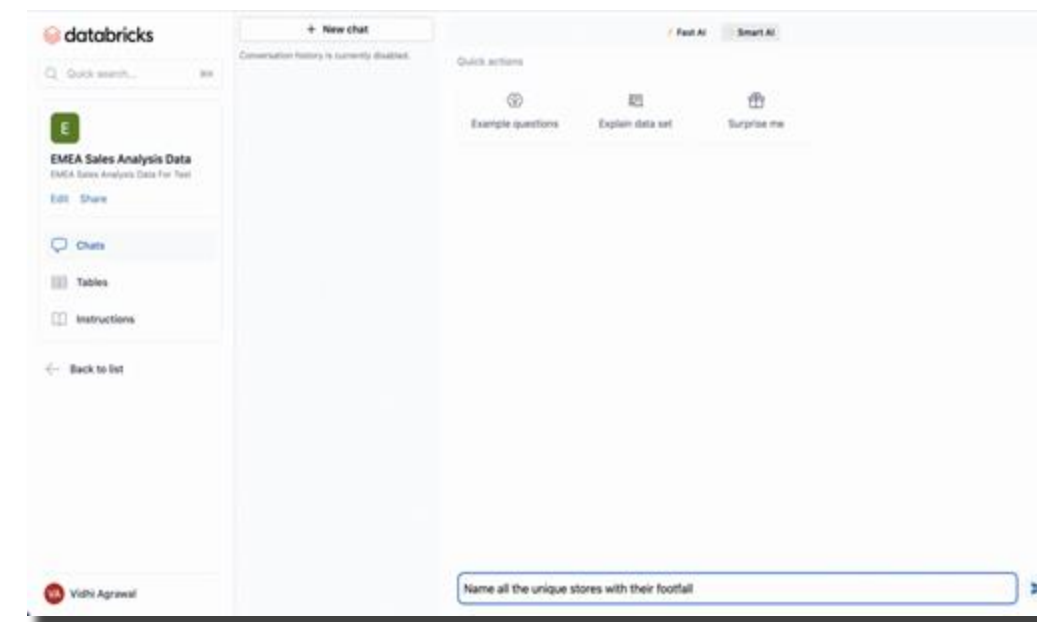
AI Assistant for
quick code fixes,
queries and
descriptions



AI Assistant
for quick
creation of
visualizations



AI-powered
Genie Spaces to
quickly
understand your
data using
natural language



Databrick's AI/BI Offerings

 Workspace

 Recents

 Catalog

 Workflows

 Compute

SQL

 SQL Editor

 Queries

 Dashboards

 Genie

 Alerts

 Query History

 SQL Warehouses

Located under the SQL header in the navigation:

AI/BI Dashboards

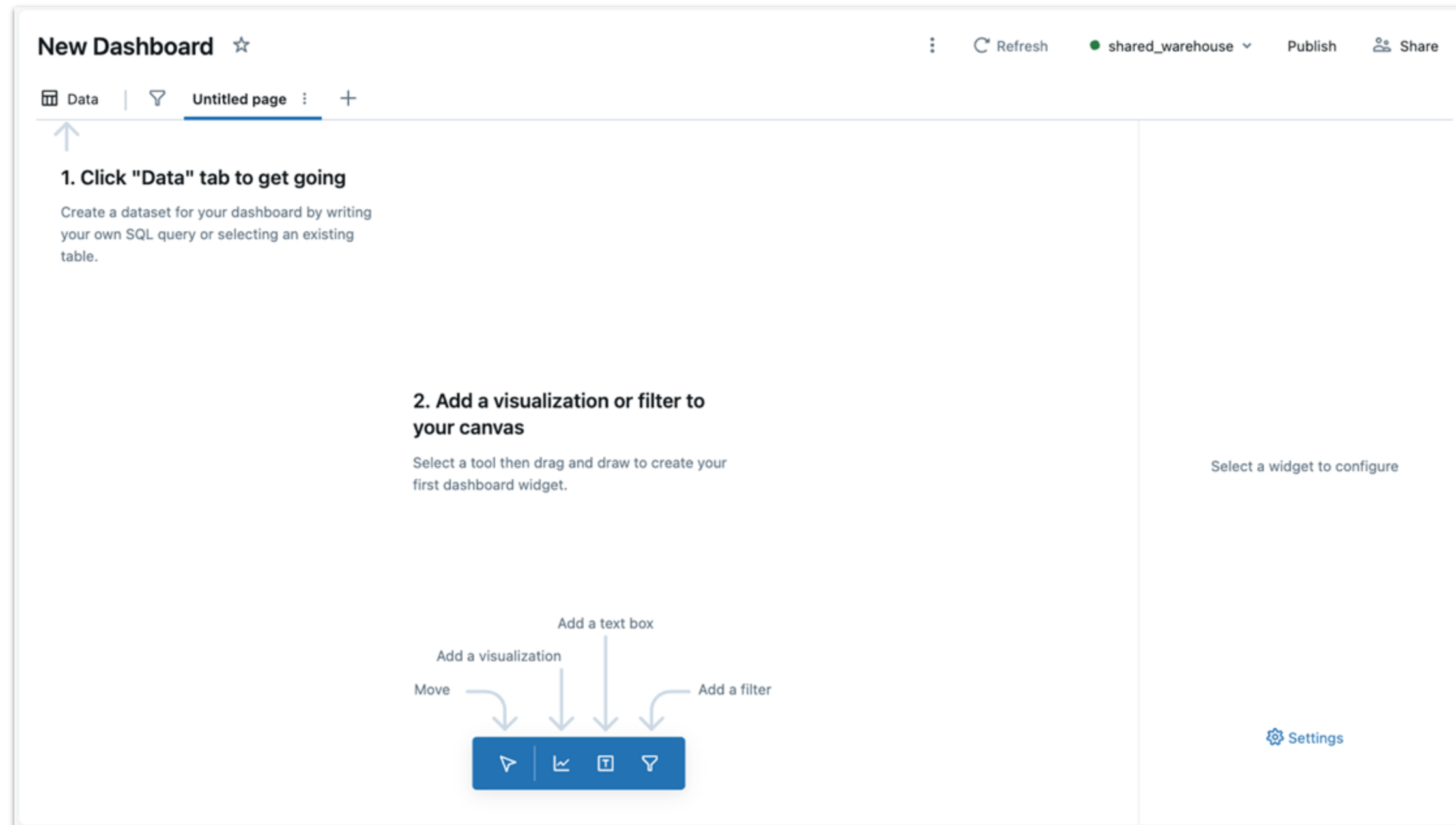
- Presentation environment for visualizations, titles, filters, and other features traditionally found in presentation software.

AI/BI Genie

- Unique space associated to a specific curated data set for asking questions using natural language to gain further insight or alternative visualizations without needing a data analyst to oversee the development.



AI/BI Dashboards

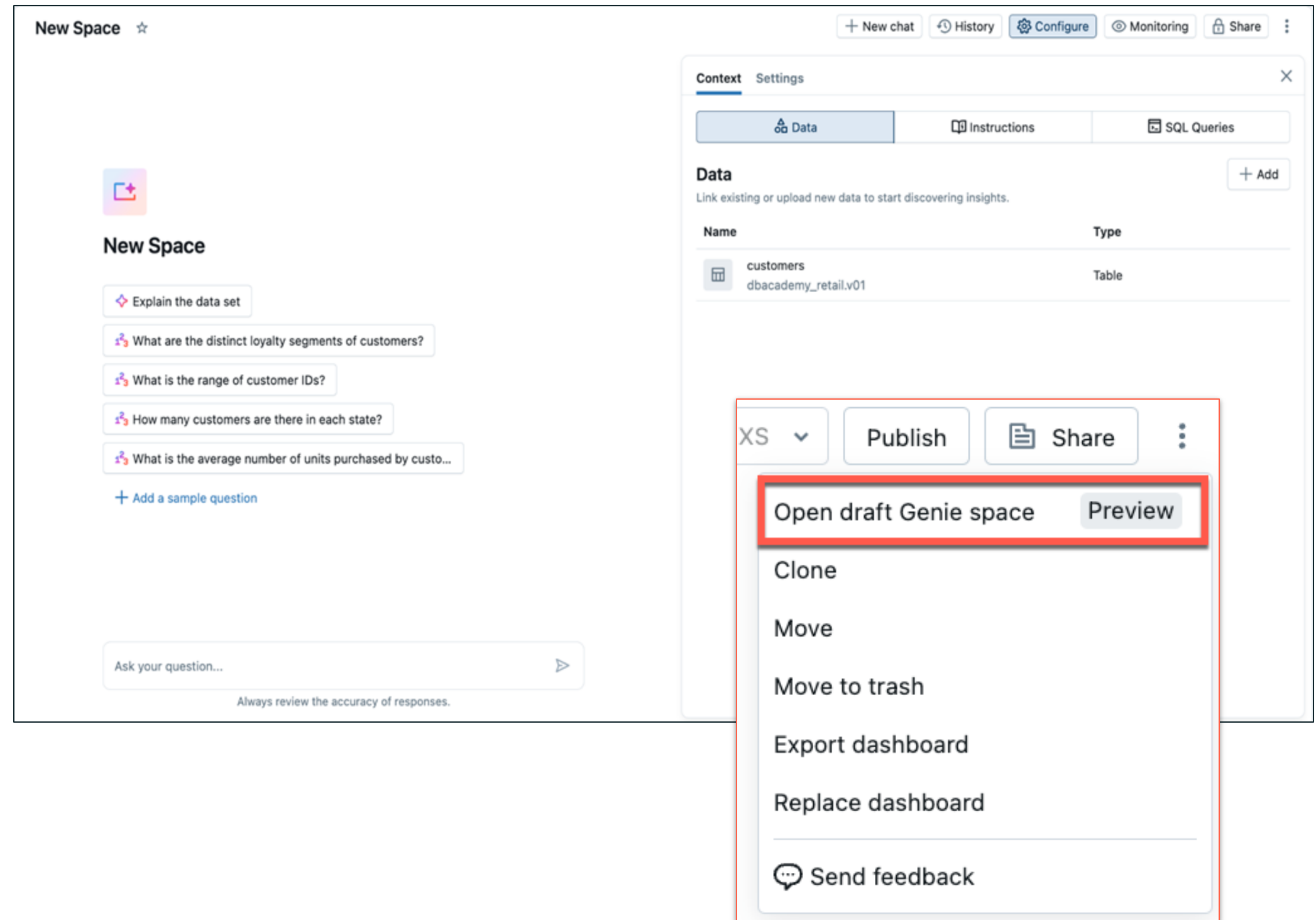


- Support for technical and non-technical users.
- Simple, easy to use interface.
- All-in-one sharing manages the permissions for the data, queries, and dashboard all from one interface.



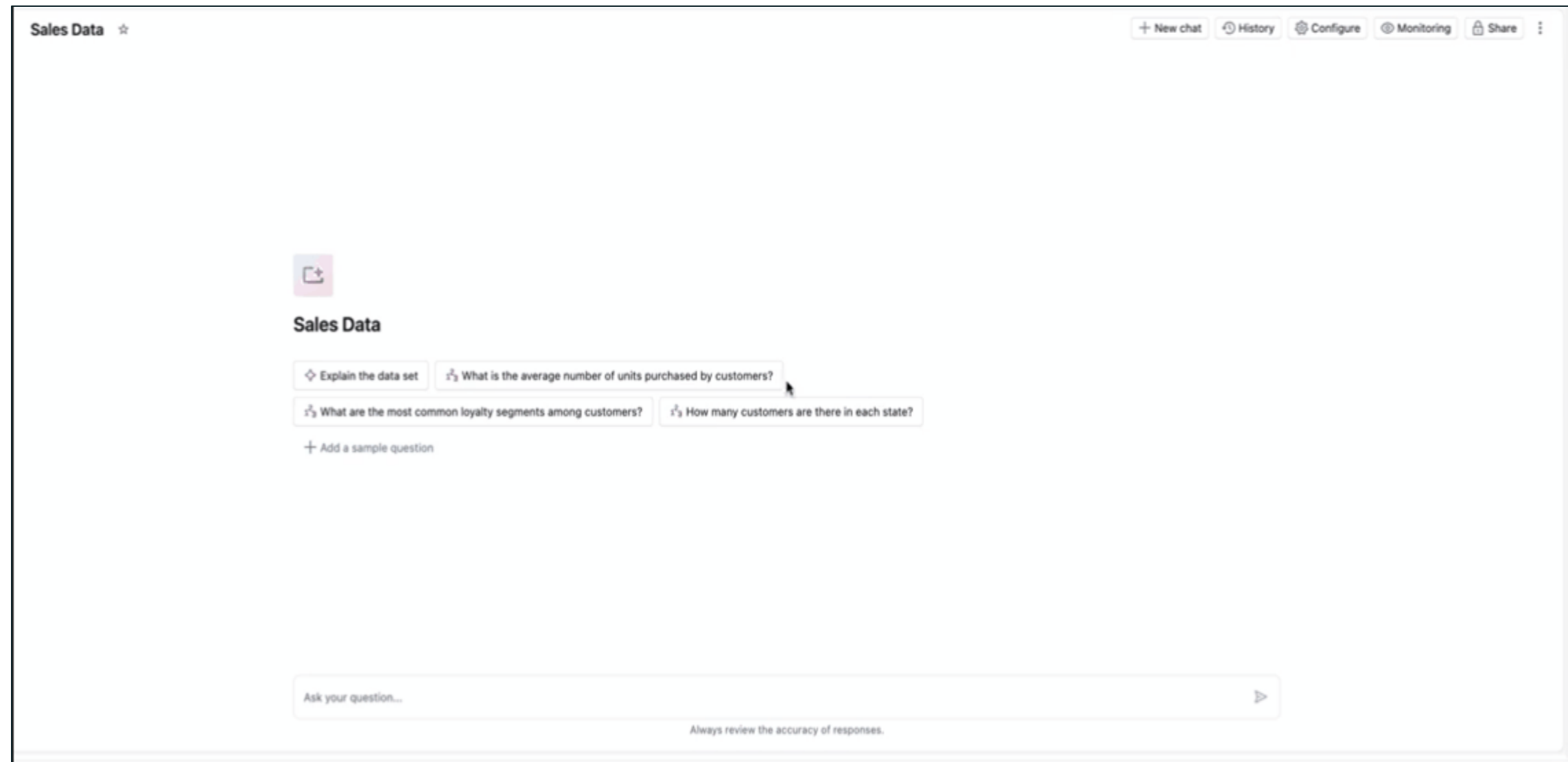
AI/BI Genie

- Easily created straight from a Dashboard.
- Simple UI for creation outside of Dashboards.
- Supports follow-up discussions directly with the data.



AI/BI Genie Spaces

Ask questions of your data in natural language using your terms



- Supports non technical users with a no-code solution to gaining data insights using natural language.
- Provides a self-service access point to the data for follow-up questions without needing to involve a data analytics specialist.

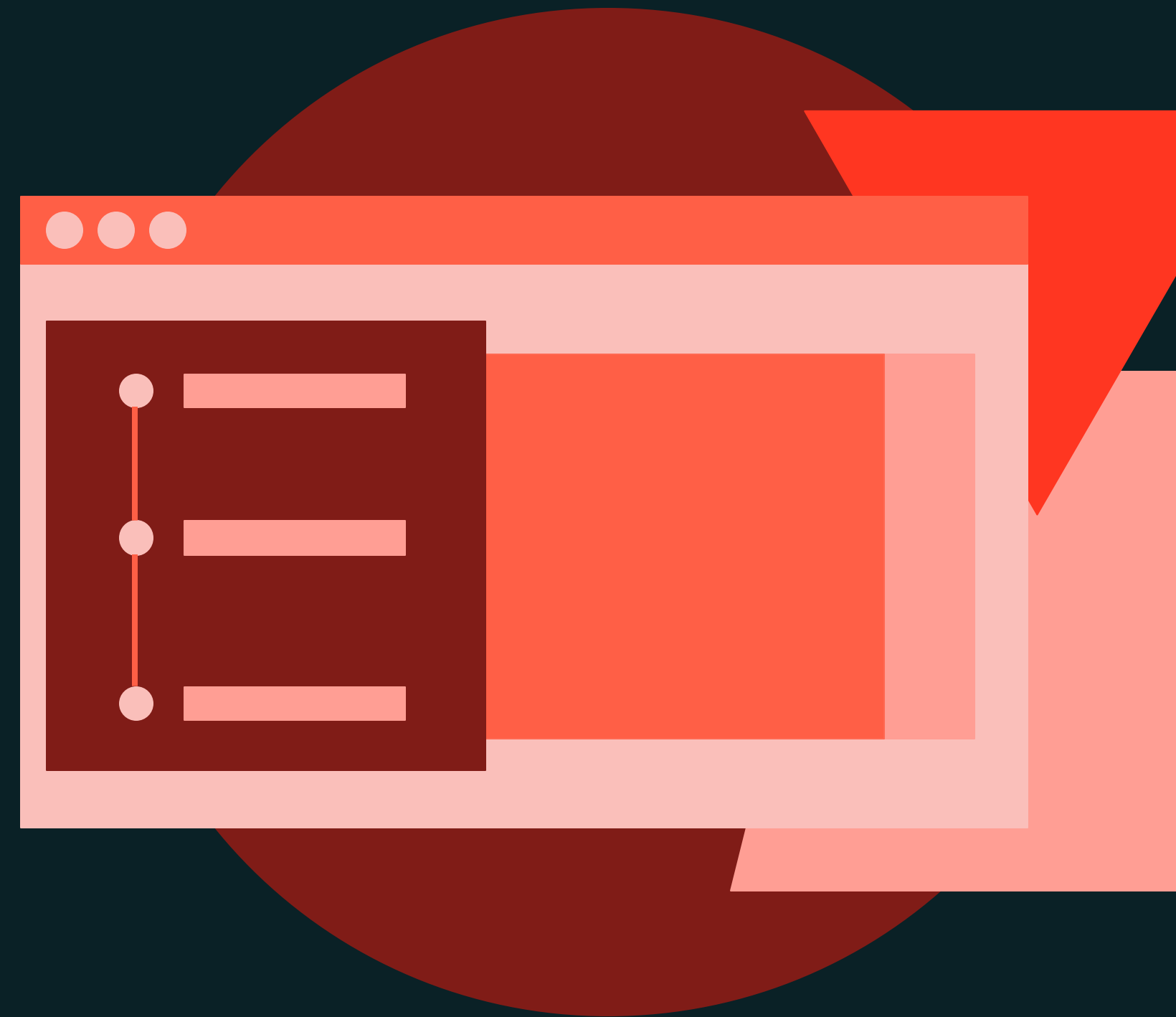




Data Presentation

DEMONSTRATION

Creating a Dashboard in Databricks



Follow-along Instructions

Estimated Time: 5 minutes

For this demonstration, the instructor will guide you through the Dashboards interface, including adding data sets both with the UI and through a query, creating visualizations, and adding features like filters to the dashboard. This demo closes with a brief introduction to sharing your dashboards with others in your business. If you have access to the Vocareum lab environment, feel free to follow-along to explore the workspace and get oriented with where future labs will be completed.

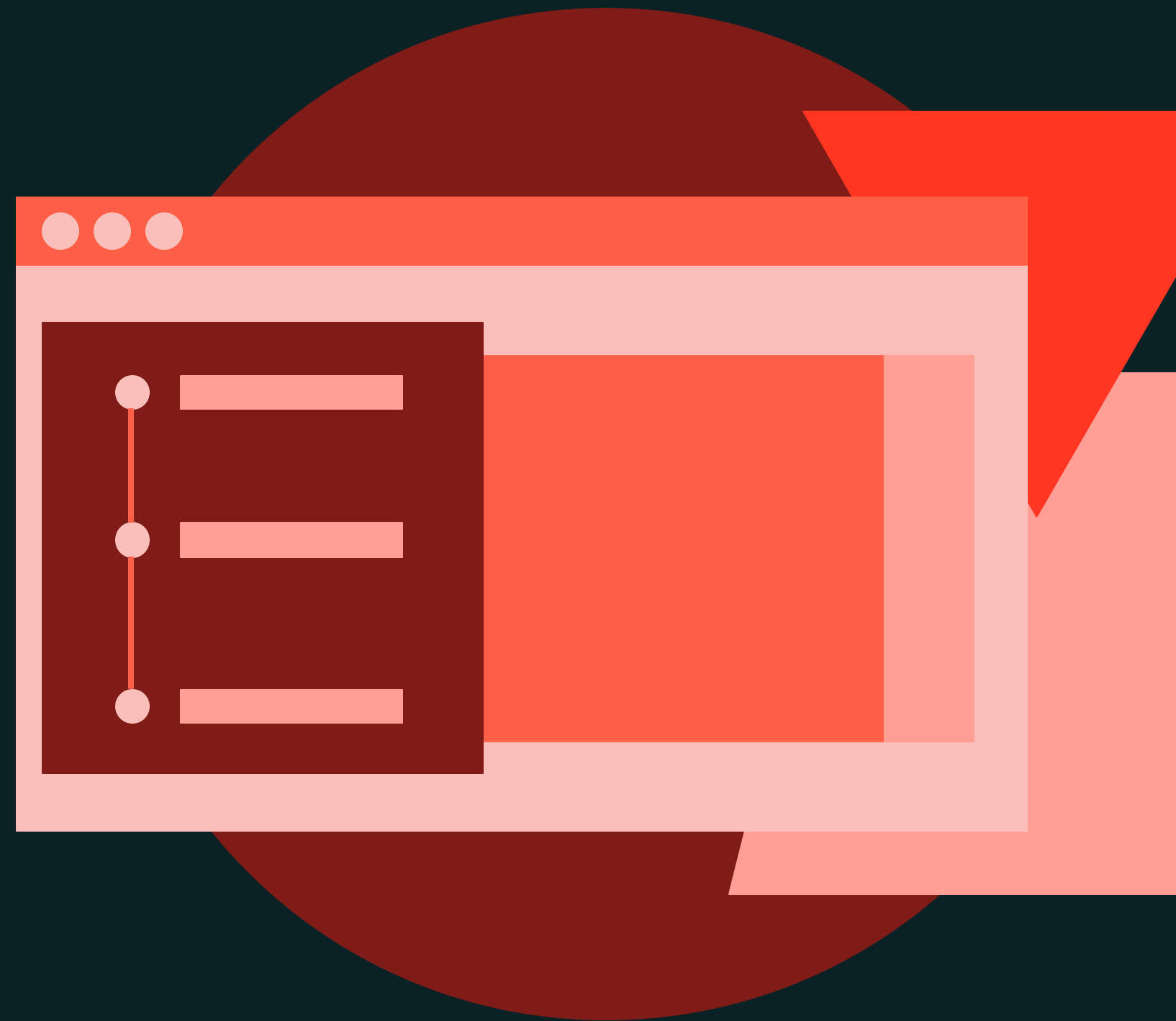




Data Presentation

DEMONSTRATION

Creating AI/BI Genie Spaces



Demonstration Details

Estimated Time: 5 minutes

For this demonstration, the instructor will walk you through creating Genie Spaces both from a dashboard but also from scratch and discuss the capabilities of this tool.



High Level Steps

- Understand Databricks AI/BI Genie Spaces:
- Create Genie Spaces:
- Leverage Genie Space Features:
- Create Genie Spaces from Dashboards:
- Understand Genie's Monitoring and Preview Features:

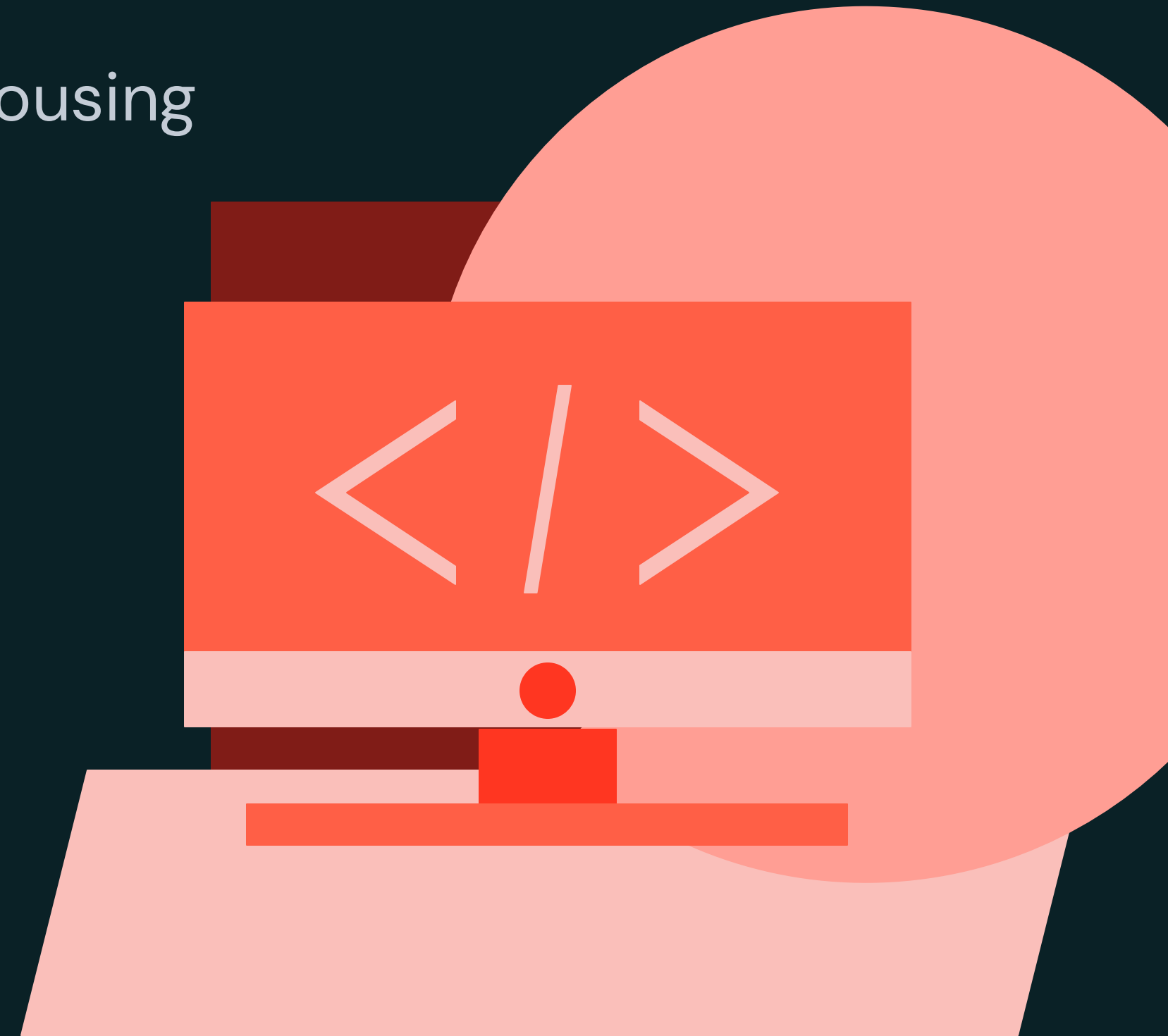




Get Started with Databricks for Data Warehousing

LAB EXERCISE

Data Warehousing Comprehensive Lab



Lab Exercise Goals

- What the learner is to accomplish within the lab environment as part of completing the lab.
- Explain how it relates to the lecture or demonstration preceding the exercise.
- Announce whether the lab will be completed with the instructor or if it's an independent exercise to be completed in the following X minutes.



Summary and Next Steps

Get Started with Databricks for Data Warehousing



Course Learning Objective Recap

- Describe the available compute options for workloads performed on the Databricks Data Intelligence Platform.
- List the products and features Databricks offers for different data-centric needs within the Databricks Platform.
- Navigate the Databricks Workspace UI.
- Use Databricks to complete common data warehousing tasks.
- Explain the purpose of Delta Lake for data warehousing purposes.
- Describe and apply various techniques for ingesting and transforming data in Databricks.
- Explain how Databricks support data orchestration needs within the platform.
- Describe the impact of data warehousing work on Databricks AI/BI needs.
- Identify the AI supported features in the Databricks platform.



